INVITED LECTURE · ITBA · 23 OCTOBER 2026

Machine learning for neural time series

How models learn from EEG, fMRI and spikes, and how to know what they really learned

Gonzalo Uribarri
Stockholm University

ITBA
Instituto Tecnológico de Buenos Aires

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About me: from dynamical systems in Buenos Aires to machine learning for brain data in Stockholm

PhD · Buenos Aires

PhD in Physics, Universidad de Buenos Aires, 2022, with Gabriel Mindlin: dynamical systems.

Postdoc · Stockholm

KTH Royal Institute of Technology, 2022–2025, with Erik Fransén: machine learning for biomedical data (Parkinson's, Alzheimer's).

Now · Stockholm University

Associate Senior Lecturer, Department of Computer and Systems Sciences (DSV), since September 2025: machine learning for time series in biomedicine and neuroscience (EEG, MEG, eye tracking).

Today you will see three of my projects, marked **OWN RESEARCH**: Detach-ROCKET · Parkinson's from eye movements · decoding raw MEG and EEG (Parts 2–3).

@ gonzalo.uribarri@dsv.su.se

𝕏 @gonzauri

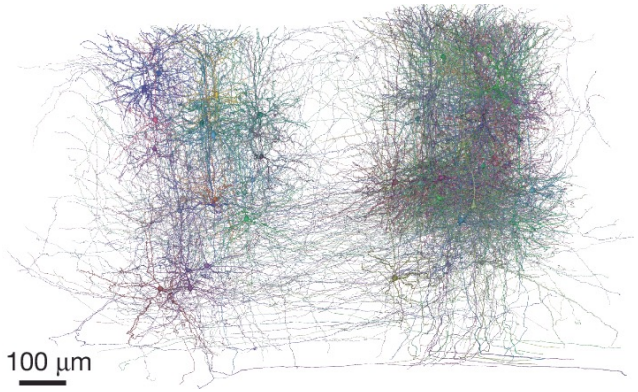
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The brain is extremely complex, and we only see it through partial, noisy windows

A VERY COMPLEX SYSTEM



one neuron



78 neurons in a piece of mouse cortex

Left: S. Ramón y Cajal, c. 1900 · Instituto Cajal (CSIC) · public domain, via Wikimedia Commons.
Right: MICrONS Consortium 2025, Fig. 5c · CC BY 4.0 · cropped, panel letter removed

- **86 billion** neurons, **~150 trillion** synapses in the neocortex alone
- A model of one cortical neuron needs a **5–8-layer** deep network
- How the brain computes is still **an open question**

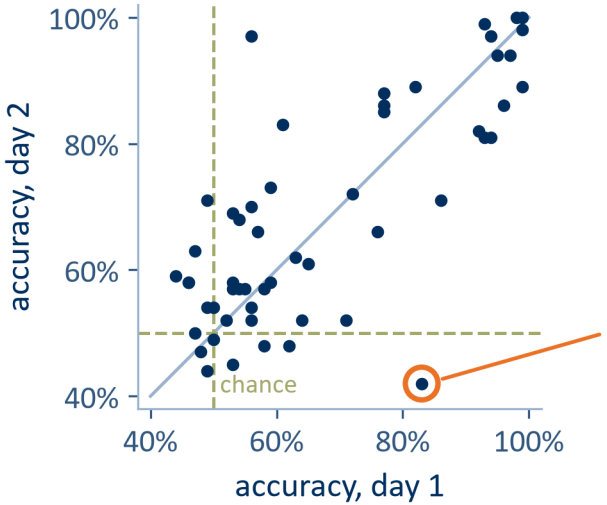
PARTIAL, NOISY, VARIABLE MEASUREMENTS

EEG: in one study, no epileptic spike smaller than **6 cm²** of cortex reached the scalp

fMRI: one voxel blends **~1–3 million** neurons

Same brain–computer interface task, 54 people, two days

each dot = one person · spread = people differ · off diagonal = days differ



same person, next day:

83% → 42%

Data: Lee et al. 2019, Table 4 · CC BY 4.0 · Figure: own

Azevedo 2009 · Pakkenberg 2003 · Beniaguev 2021 · Adolphs 2015 · Tao 2005 · Logothetis 2008 · Lee 2019

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Recordings are getting denser, cheaper and portable: we have more data than we know how to use

DENSER: NEUROPIXELS PROBE



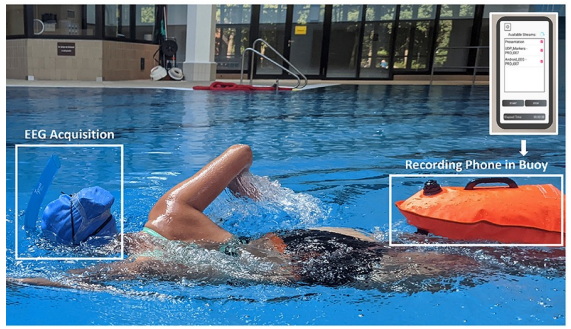
384
channels recorded at once

30,000
samples per second, each

Steinmetz et al. 2018, Fig. 1c · CC BY 4.0 · cropped

- Two probes: **> 700** neurons at once in one mouse

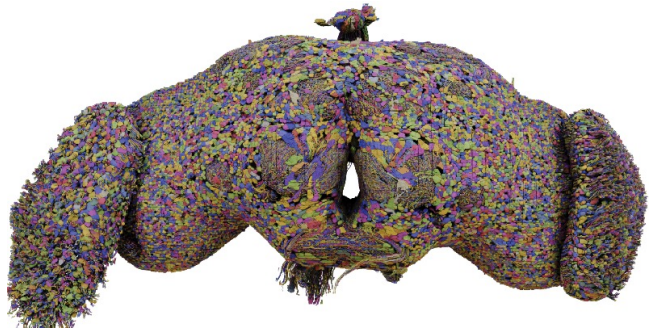
PORTABLE, CHEAPER: WIRELESS EEG



Klapprott & Debener 2024, Fig. 1 · CC BY 4.0 · cropped

- A **32**-channel wireless amplifier under a swim cap, streaming to a phone
- Consumer EEG from **≈ $400**; lab systems: tens of thousands of dollars

COMPLETE WIRING MAPS: FLY CONNECTOME



Dorkenwald et al. 2024, Fig. 1a · CC BY 4.0 · cropped, letter removed

- Whole fly brain: **139,255** neurons, **~50 million** synapses (2024)
- 2026: brain + nerve cord of a male fly, **~167,000** neurons
- Explore it: codex.flywire.ai

KEY IDEA

This is why ML is gaining ground in neuroscience: it makes sense of large, noisy data and powers both analysis and applications.

Jun et al. 2017 · mBrainTrain specs · Sugden et al. 2023 · Dorkenwald et al. 2024 · Berg et al. 2026

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Brain signals can now be decoded into words, and decoders can also fool us

DECODED INTO WORDS

electrodes in speech motor cortex

↓

spike activity in short time bins

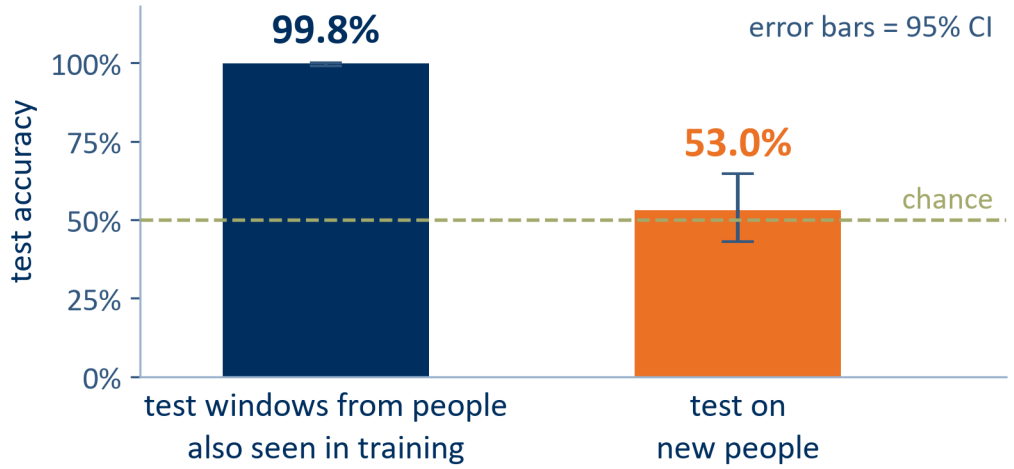
↓

neural network → speech sounds (phonemes)

↓

language model → text

…OR FOOLING US



- **97.5%** of words correct over 8.4 months: a man with ALS, 256 electrodes
- A second participant: **62** words per minute

Same EEG data, same network: Alzheimer's vs subjective cognitive impairment, 49 + 49 people. Only 17 of 63 deep-learning EEG papers clearly avoided this leak.

Today's question: what can a machine learn from brain signals that unfold in time, and how do we know it really learned it?

Card et al. 2024 NEJM · Willett et al. 2023 Nature · Brookshire et al. 2024 Front Neurosci · Figures: own

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Today

1 · ML for time series	2 · EEG: one feature, a fair test	Break	3 · EEG/MEG models	4 · fMRI	5 · Spikes	Notebook	Wrap-up

After each part: a short stop for your questions.

BY THE END YOU CAN

O1 **Explain** why a brain recording is a multivariate time series, and why that is hard for generic ML.

O2 **Explain** how a model learns from data, and recognise the two tasks: predict (classify or regress) and represent (embed or cluster).

O3 **Place** a model between hand-made features and deep learning (ROCKET halfway), and pick a sensible first one.

O4 **Judge** whether a model learned what we think it learned (six trust checks), and **run** an EEG + fMRI analysis in Colab.

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Today we ask two kinds of questions of three kinds of brain data

	PREDICT classify or regress	**REPRESENT** embed or cluster (compress)
EEG/MEG Q1 · eyes closed?	eyes open or closed? from one feature to deep models	a map of 1,200 windows; sleep epochs grouped without labels
fMRI Q2 · child or adult?	the connectome as features → child or adult? age in years	PCA of 155 connectomes
SPIKES Q3 · where is the rat?	where is the rat? (position)	a curve that follows the track; spike sorting = clustering

SIX TRUST CHECKS, COLLECTED ALONG THE WAY

BASELINE | **SPLIT** | **CONFOUND** | **UNCERTAINTY** | **METRIC** | **TIME**

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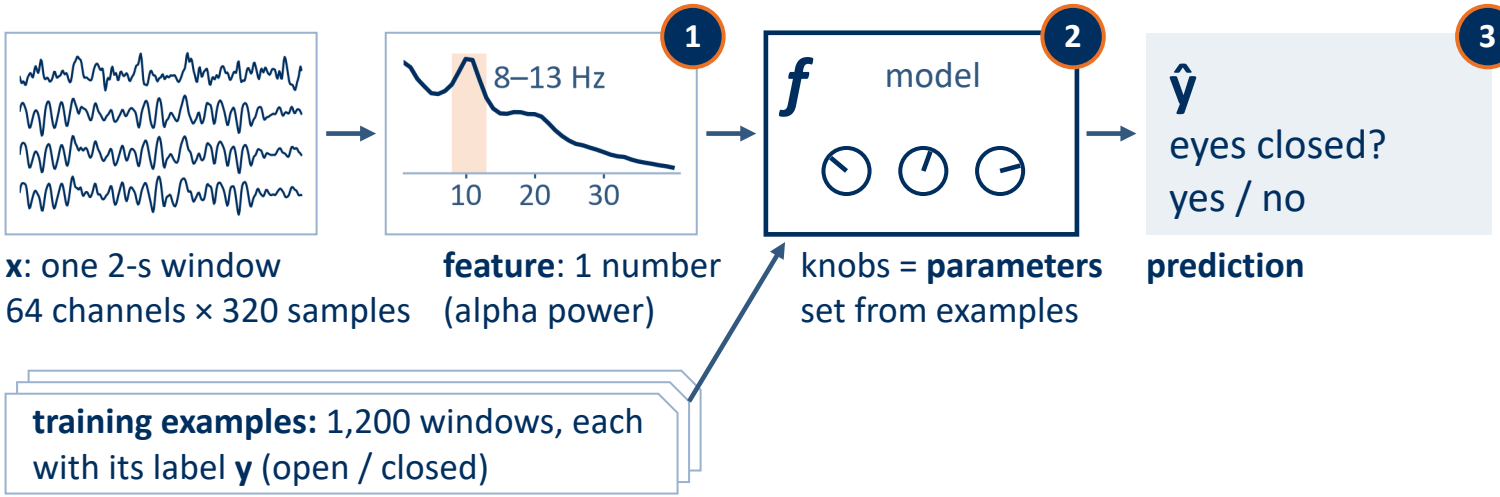
PART 1 OF 5

Machine learning for time series

What is a time series, and how can a machine learn from it?

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A model is a function learned from examples: it outputs a category (classification) or a number (regression)



Classification
the answer is a category

- EEG: eyes open or closed? (Q1)
- fMRI: child or adult? (Q2)

4

Regression
the answer is a number

- fMRI: age in years
- Spikes: where is the rat on the track, in cm? (Q3)

1. **Example** x: what the model sees (one window). **Feature**: a number computed from it
2. **Model** f: a rule with knobs (**parameters**) set from labelled examples
3. **Label** y: the right answer. **Prediction** ŷ = f(x): the model's answer
4. Category → **classification**; number → **regression**. Both: the task PREDICT

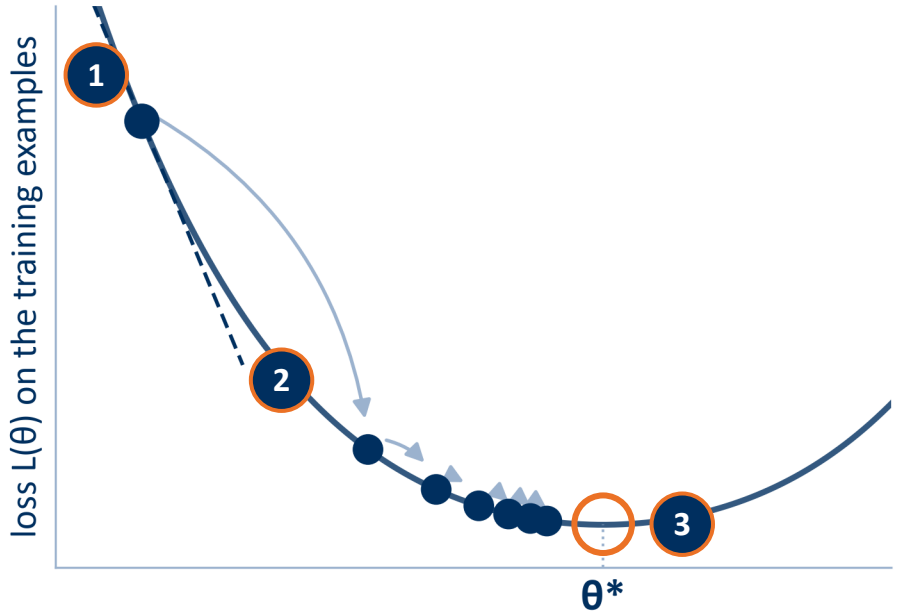
Figure: own, after Uribarri, UNaB talk · Pereira et al. 2009 · data: PhysioNet EEGBCI, ODC-By 1.0

→ slides 26, 40 in depth

Notebook §2

A model learns by turning its knobs (parameters), step by step, to lower its loss on the training examples

predict the training examples → measure the loss → turn the knobs a little downhill → repeat



1. start: knobs set arbitrarily (e.g. at zero): high loss
2. the slope says which way is downhill: **gradient descent** steps against it
3. θ*: lowest loss = the learned knobs

Σ DEFINITION · LEARNING

Learning = finding the knobs with the smallest loss:

$$\theta^* = \arg\min_{\theta} L(\theta)$$

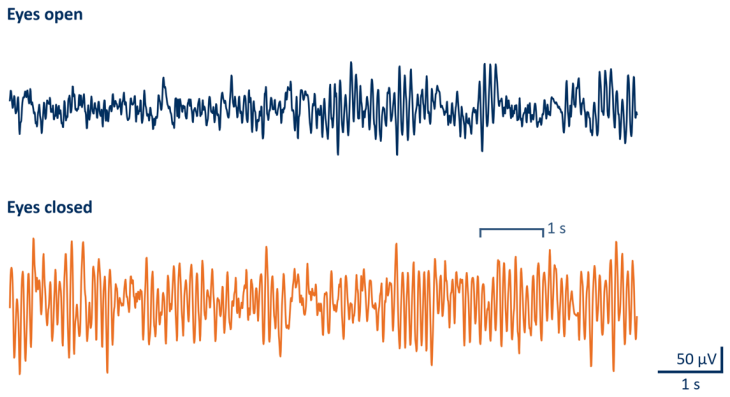
θ	the parameters (knobs): in a linear model, one weight per feature, plus a bias
$L(\theta)$	the loss: how wrong it is on the training examples
$\arg\min_{\theta}$	the setting that makes L smallest
θ^*	the learned knobs

Same recipe for 2 knobs (our alpha detector) or 1.7 billion (an EEG foundation model).

Figure: own · framework: Richards et al. 2019 · NeuroLM: Jiang et al. 2025

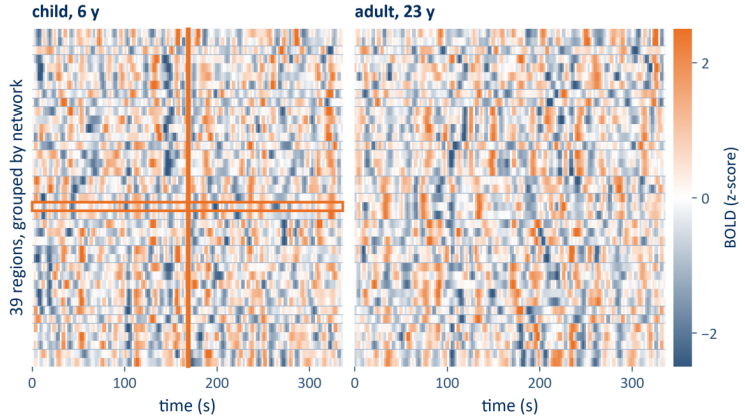
Almost every brain recording is a multivariate time series: many channels, measured in time

EEG (AND MEG): SENSORS × TIME



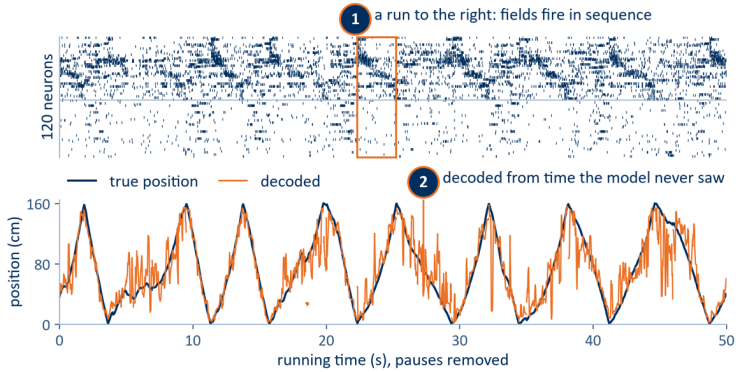
64 electrodes × 160 samples/s

FMRI: BRAIN REGIONS × SCANS



39 regions × 168 scans (one every 2 s)

SPIKES: NEURONS × TIME BINS



120 neurons × 25-ms bins

Data: PhysioNet EEGBCI (ODC-By) · ds000228 / nilearn (CC BY 4.0) · DANDI 000044 (CC BY 4.0) · Figures: own

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EEG and MEG are fast but blurred, fMRI is sharp but slow, and spikes need electrodes inside the brain

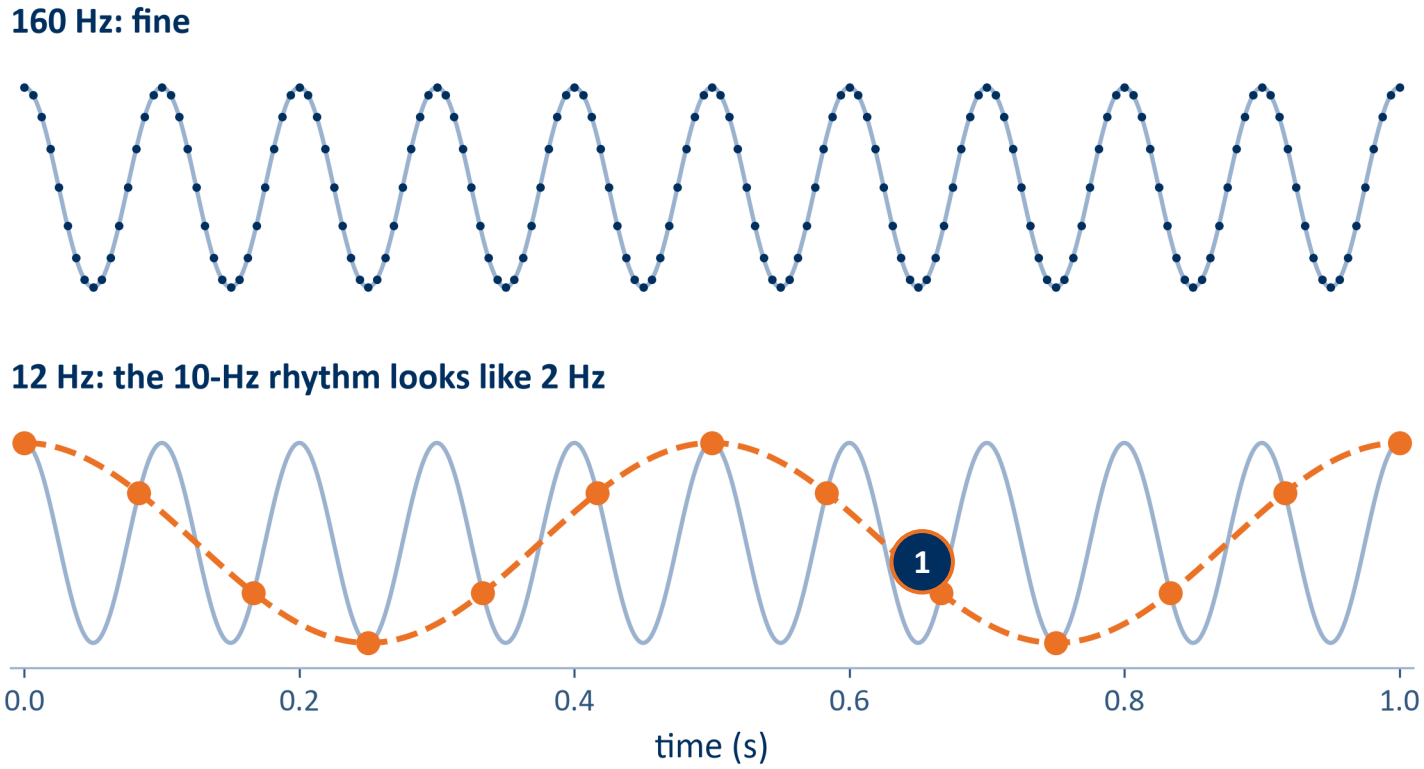
	What it measures	One sample every	Spatial detail	Our time series today
EEG	scalp voltage from many neurons firing in sync	6.25 ms (160 Hz)	cm: only large patches of cortex in sync reach the scalp (slide 3)	64 electrodes × 160 samples/s (Parts 1–3, notebook)
MEG	magnetic field of the same currents	≈ 1–2 ms (600–1,100 Hz)	mm to cm	306 sensors (slide 39)
fMRI	blood oxygenation, peaking ≈ 5 s after activity	2 s	2–3 mm voxels	39 regions × 168 scans per person (Part 4, notebook)
Spikes	action potentials of single neurons	≈ 0.03–0.05 ms (20–30 kHz), then counted in 25 ms bins	single neurons	120 neurons × 40 bins/s (Part 5)

Same shape every time: channels × time. What changes is the speed, the sharpness and the noise.

Tao et al. 2005 · Hämäläinen et al. 1993 · SPM HRF · Jun et al. 2017 · data: EEGBCI, Richardson et al. 2018, Grosmark & Buzsáki 2016

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The sampling rate sets the fastest rhythm you can see: half the number of samples per second



EEGBCI: 160 Hz → up to 80 Hz
fMRI: one scan every 2 s → up to 0.25 Hz (slide 46)

Σ DEFINITION

$$f_{\max} = \frac{f_s}{2}$$

f_s	sampling rate: samples per second (Hz)
$f_{\max}$	fastest frequency the samples can show (the Nyquist frequency)

1. Faster rhythms fold back to slower ones: **aliasing**

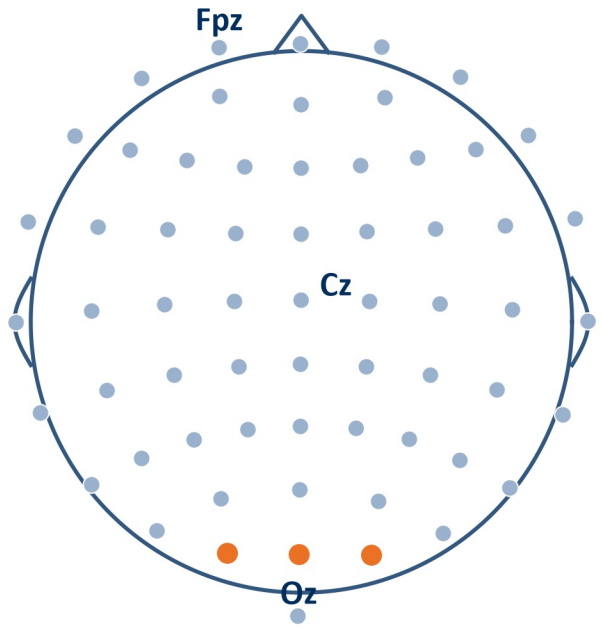
Shannon (1949) · Figure: own (figures/make_figs_p1.py)

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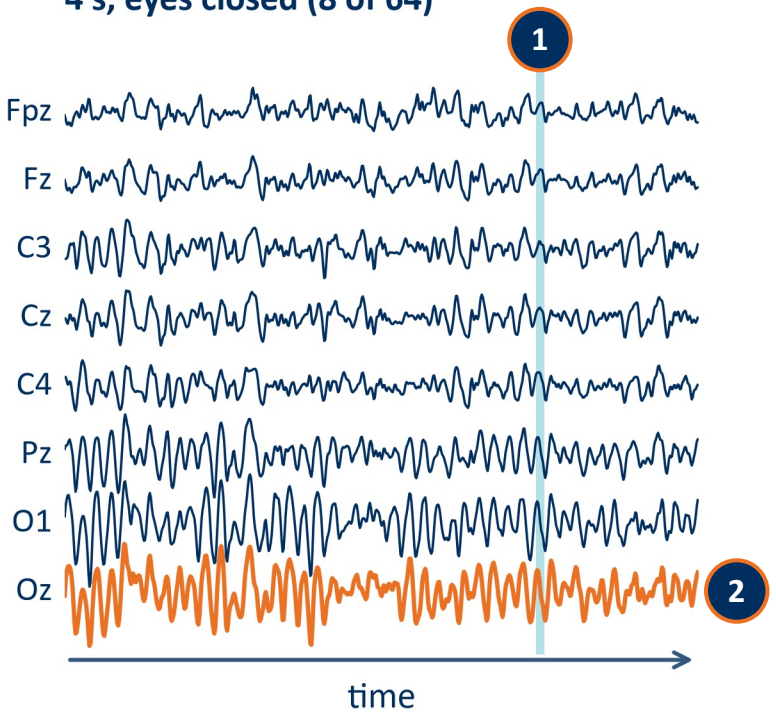
Notebook §0

An EEG recording is a matrix: 64 electrodes, each measured 160 times per second, in a fixed order

64 electrodes (top view)



4 s, eyes closed (8 of 64)



Σ DEFINITION

Time series: measurements taken one after another at a fixed rate (here 160 per second).

Multivariate: several channels measured at the same moments.

1. One moment = 64 numbers, one per electrode
2. One electrode = 160 numbers every second

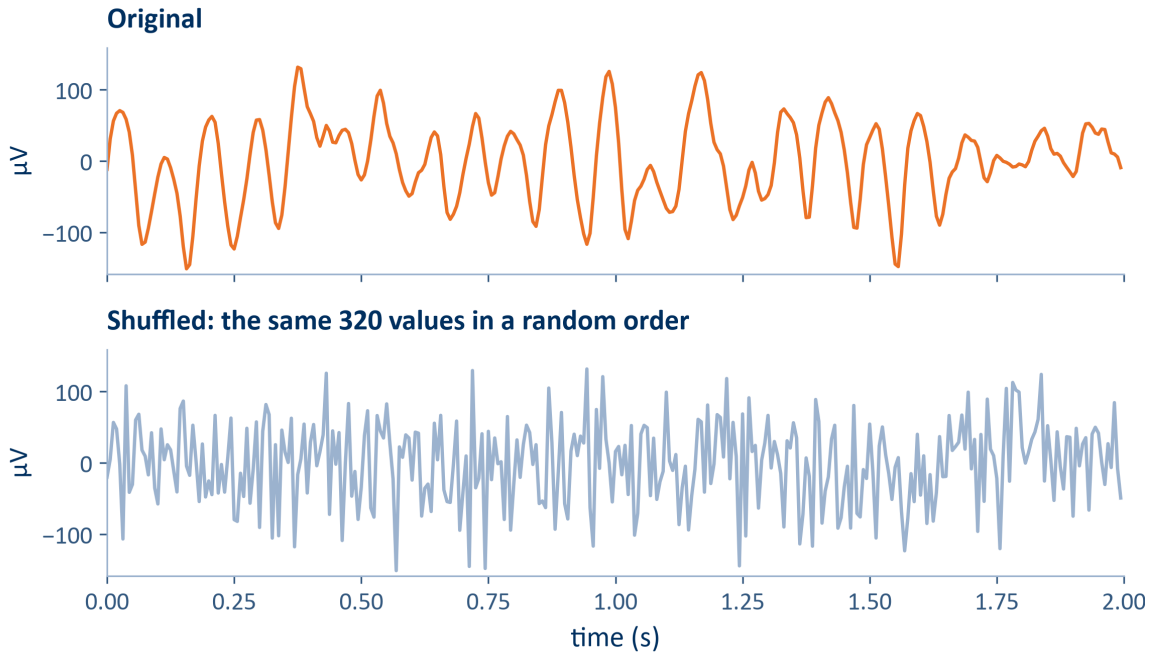
1 min = 64 × 9,600 = 614,400 numbers. Unlike the rows of a table, they cannot be reordered.

Data: PhysioNet EEGBCI (Schalk et al. 2004), ODC-By 1.0 · MNE standard_1005 montage · Figure: own (figures/make_figs_p1.py)

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In a time series the order carries information; in a table, the order of the columns does not

A TIME SERIES AND THE SAME VALUES SHUFFLED



A TABLE: COLUMN ORDER IS IRRELEVANT

	age	sex	IQ
A	34	F	102
B	51	M	97

= same information

	IQ	age	sex
A	102	34	F
B	97	51	M

KEY IDEA

One EEG window is channels × time: 64 × 320 = 20,480 ordered numbers per example.

Data: PhysioNet EEGBCI (Schalk et al. 2004), ODC-By 1.0 · Figure: own (make_figs_p1.py)

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Two ways to do ML on time series: turn them into features, or let a deep network learn from raw data

High-dimensional series are hard for generic ML, so we use methods that exploit the temporal structure.

FEATURIZATION

series → a few meaningful numbers (band power, spectrum, catch22, connectome)
→ a traditional ML model (e.g. logistic regression)

DEEP LEARNING ON RAW DATA

raw channels × time → a network that learns its own features (CNNs, foundation models)

ROCKET: halfway
random features + linear model

less data needed
more bias (prior knowledge)

more data needed
less bias (more freedom for the model)

Overview after Uribarri, EIVIA 2025 course (own) · Dempster et al. (2020) ROCKET

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PART 2 OF 5

EEG: from one feature to a fair test

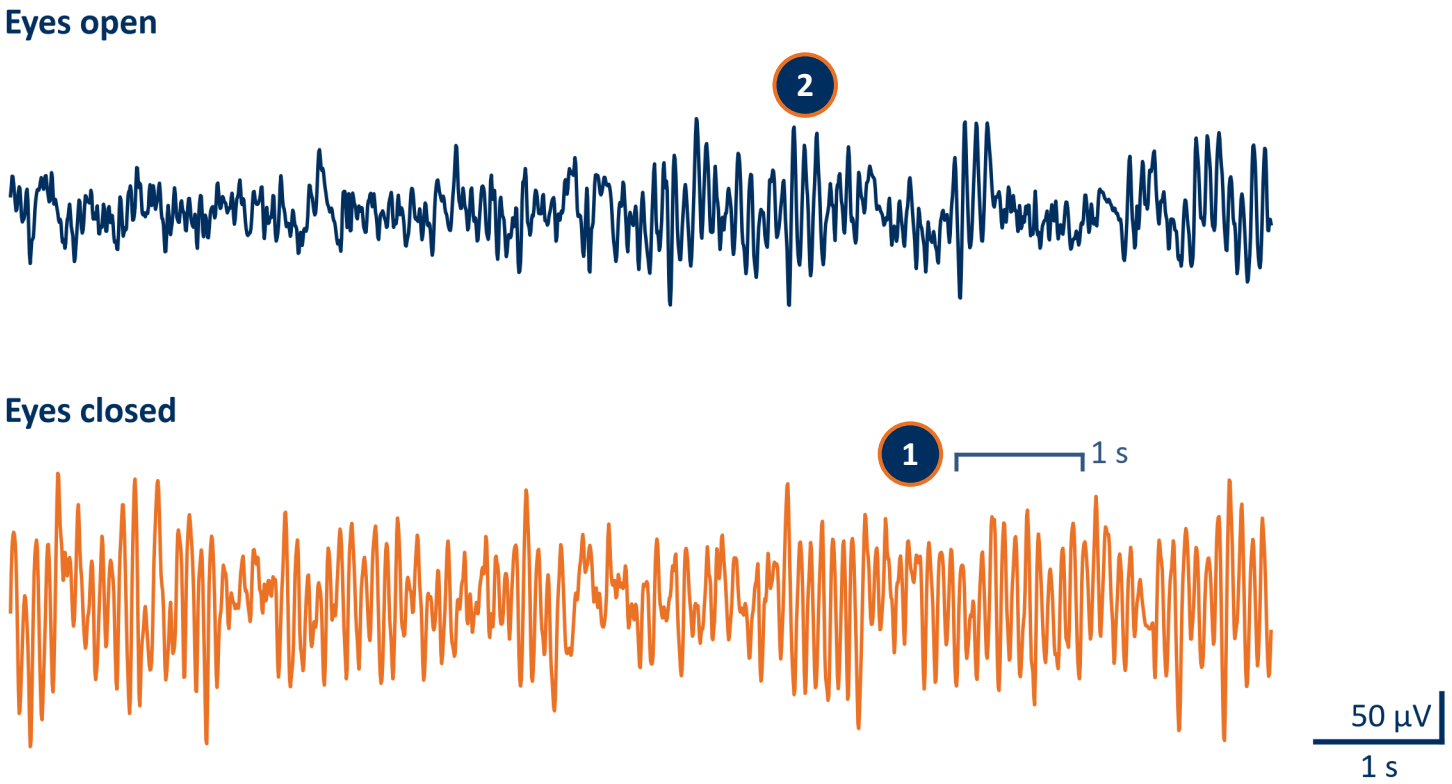
Can a model tell from the EEG whether the eyes are closed?

Notebook §1–§2

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EEG/MEG | Notebook §1

With the eyes closed, the EEG over the visual cortex fills with a rhythm of about 10 waves per second



RUNNING EXAMPLE

PhysioNet EEGBCI: 109 people, 64 electrodes, 160 samples per second. We use 20 people: 1 min with eyes open, 1 min with eyes closed.

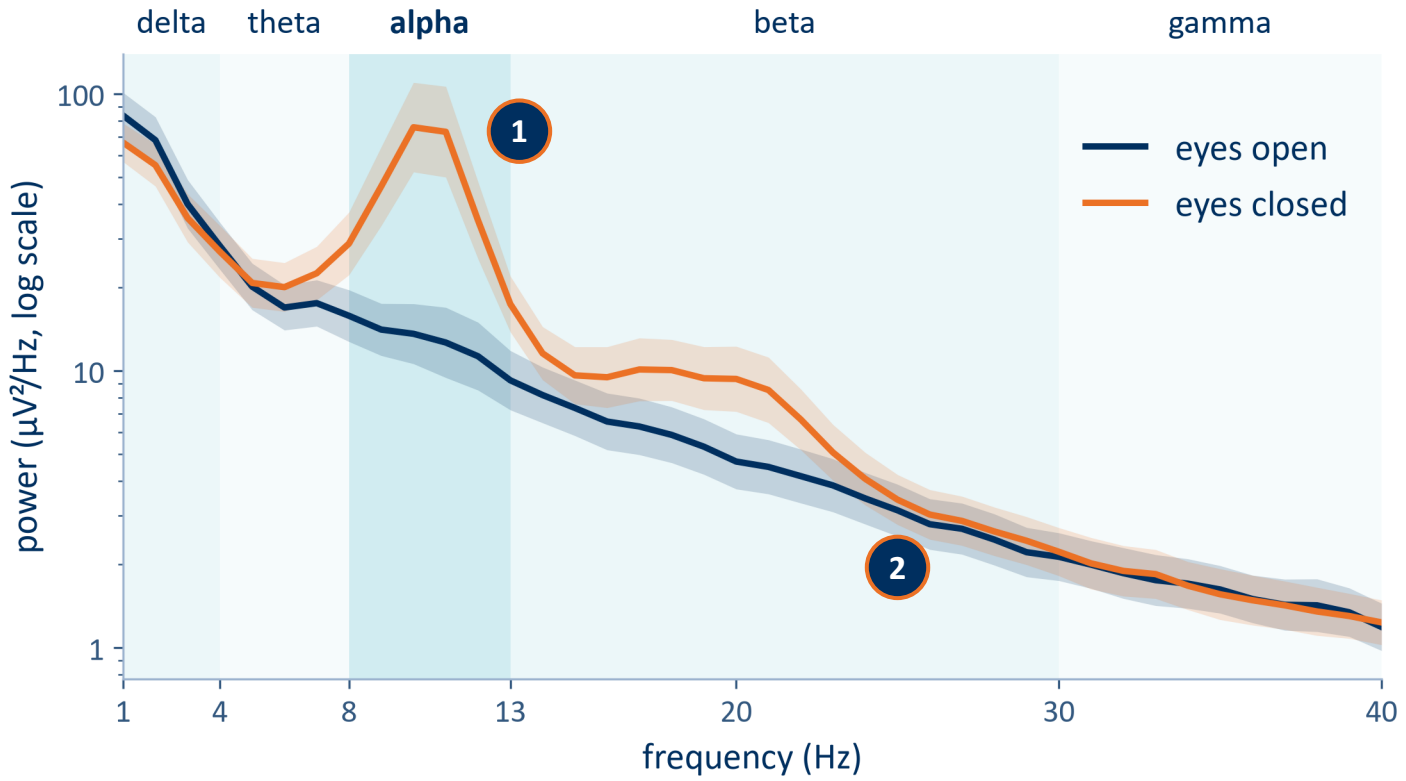
1. ≈ 10 waves in 1 s: the **alpha rhythm**
2. Eyes open: smaller and less regular

Q1 · Can a model tell whether the eyes are closed, in a person it has never seen?

Data: PhysioNet EEGBCI (Schalk et al. 2004), ODC-By 1.0 · Figure: own (figures/make_figs_p1.py)

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Seen as frequencies, closing the eyes adds a peak near 10 Hz: the alpha rhythm



1. Eyes closed: a peak near 10 Hz. The median person has **3.4×** more alpha power; it rises in all 20.

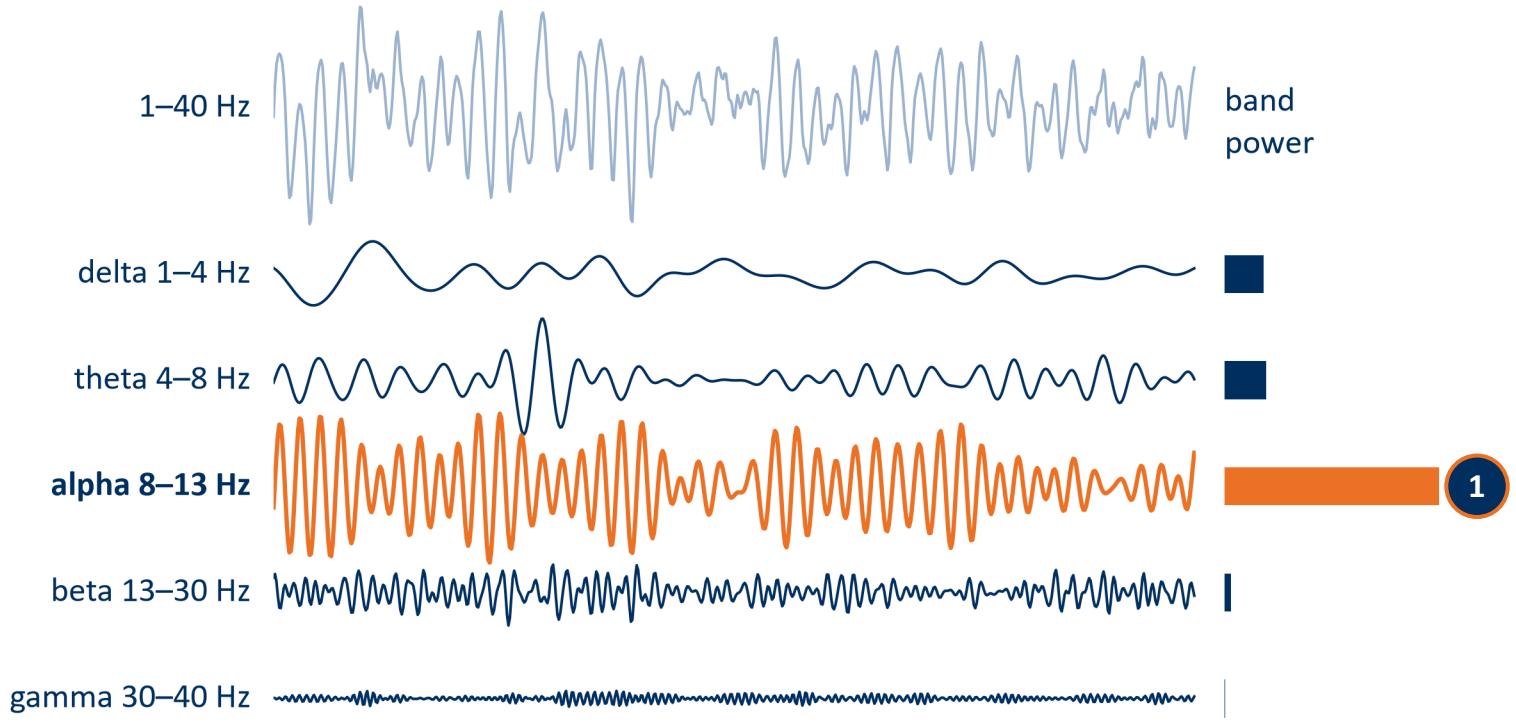
2. The background (often called 1/f) is similar: most of the difference sits in one band.

Spectrum: how much of each frequency the signal holds, averaged over 1-s pieces of each 2-s window.

Welch (1967) · Data: PhysioNet EEGBCI (Schalk et al. 2004), ODC-By 1.0 · Figure: own (figures/make_figs_p1.py)

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A band decomposition splits the EEG into rhythms, and band power measures how strong each one is



1 Alpha: the strongest rhythm with eyes closed (biggest row, longest bar)

Band edges vary between papers. The five rows add back up to (almost) the top row.

∑ DEFINITION · BAND POWER

How strong one band is: the **average height** of the power spectrum between two frequencies (µV²/Hz).

$$\overline{P}_{[f_1,f_2]} = \frac{1}{K} \sum_{f_1 \le f_k \le f_2} S(f_k)$$

$S(f_k)$	power spectrum at step k (µV²/Hz)
f_1, f_2	band edges (alpha: 8–13 Hz)
K	number of 1-Hz steps in the band (alpha: 6)

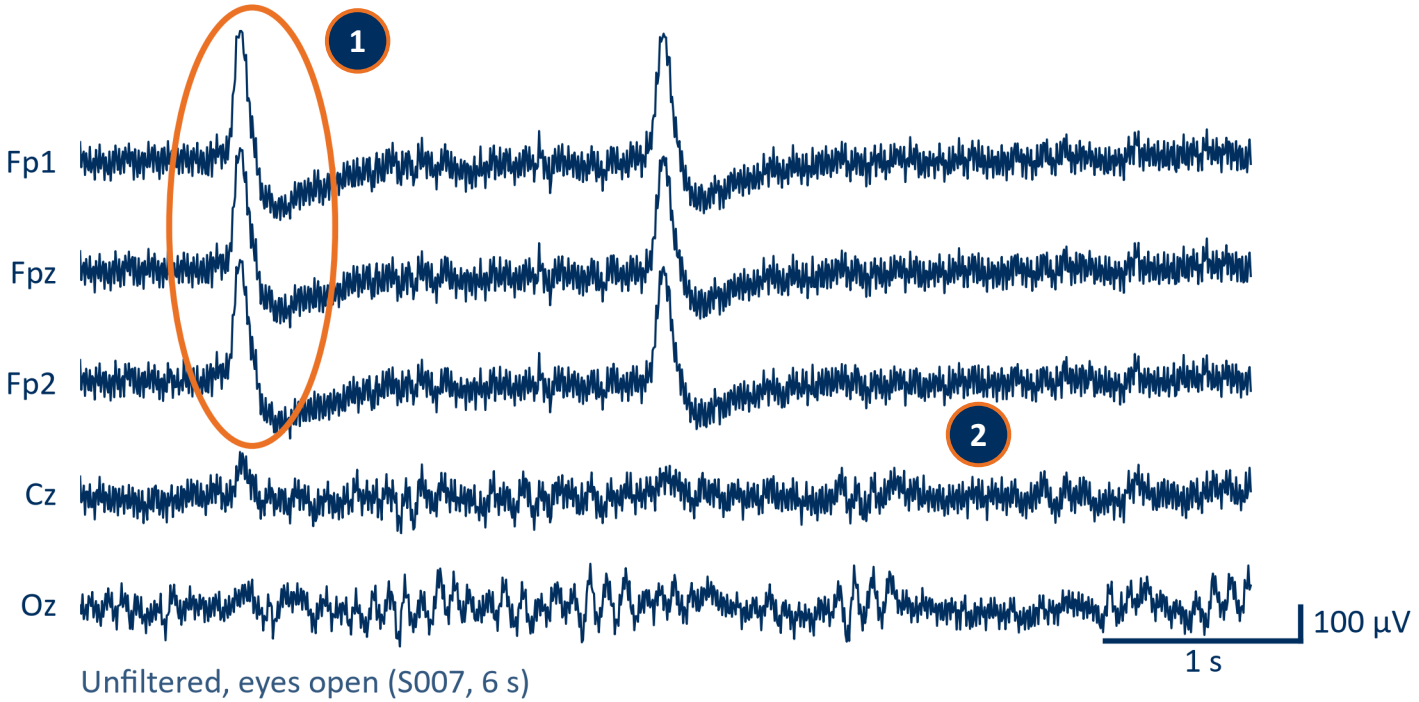
Teplan (2002) · Donoghue et al. (2020) · Data: PhysioNet EEGBCI, ODC-By 1.0 · Figure: own (figures/make_figs_p1.py)

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→ slide 34 · CONFOUND

Notebook §1

Artefacts can be bigger than the brain: eyes, muscles, drifts and the mains all leave their mark



Unfiltered, eyes open (S007, 6 s)

1. **Eyes**: blinks and movements; big, slow, frontal.
2. **Mains**: the fast fuzz is 60-Hz hum (US data; ≈ 14× its neighbours in the spectrum). In Argentina: 50 Hz.

Drift: slow wandering (electrodes, sweat).

Muscle: above 20 Hz, much of scalp EEG is muscle (Whitham et al. 2007).

⚠ **PITFALL** **Not:** if a model decodes it, the brain produced it. **But:** artefacts can differ between the conditions you compare. **Do:** look at the raw data, and ask what else differs.

Whitham et al. (2007), n = 2 · Data: PhysioNet EEGBCI, ODC-By 1.0 · Figure: own (figures/make_figs_p1.py)

EEG/MEG · ← slide 21 · artefacts · Notebook §1

Before any model, the recording is filtered, cleaned of eye and heart artefacts and cut into epochs

1 Look

Plot the raw traces and mark broken channels.

2 Filter

Keep e.g. 1–40 Hz (drops slow drift and fast noise); remove mains noise (50 Hz in Argentina, 60 Hz in our US data).

3 Re-reference

EEG only: express every electrode relative to, e.g., the average of all electrodes.

4 ICA

Remove components that look like blinks, eye movements or heartbeats.

skipped: our data has no eye channel (slide 34)

5 Epochs

Cut windows around events (or fixed 2-s windows) and drop the bad ones.

∑ DEFINITION

ICA (independent component analysis) splits the channels into source-like components. You inspect them, delete the eye and heart ones and rebuild the signal.

⚠ PITFALL

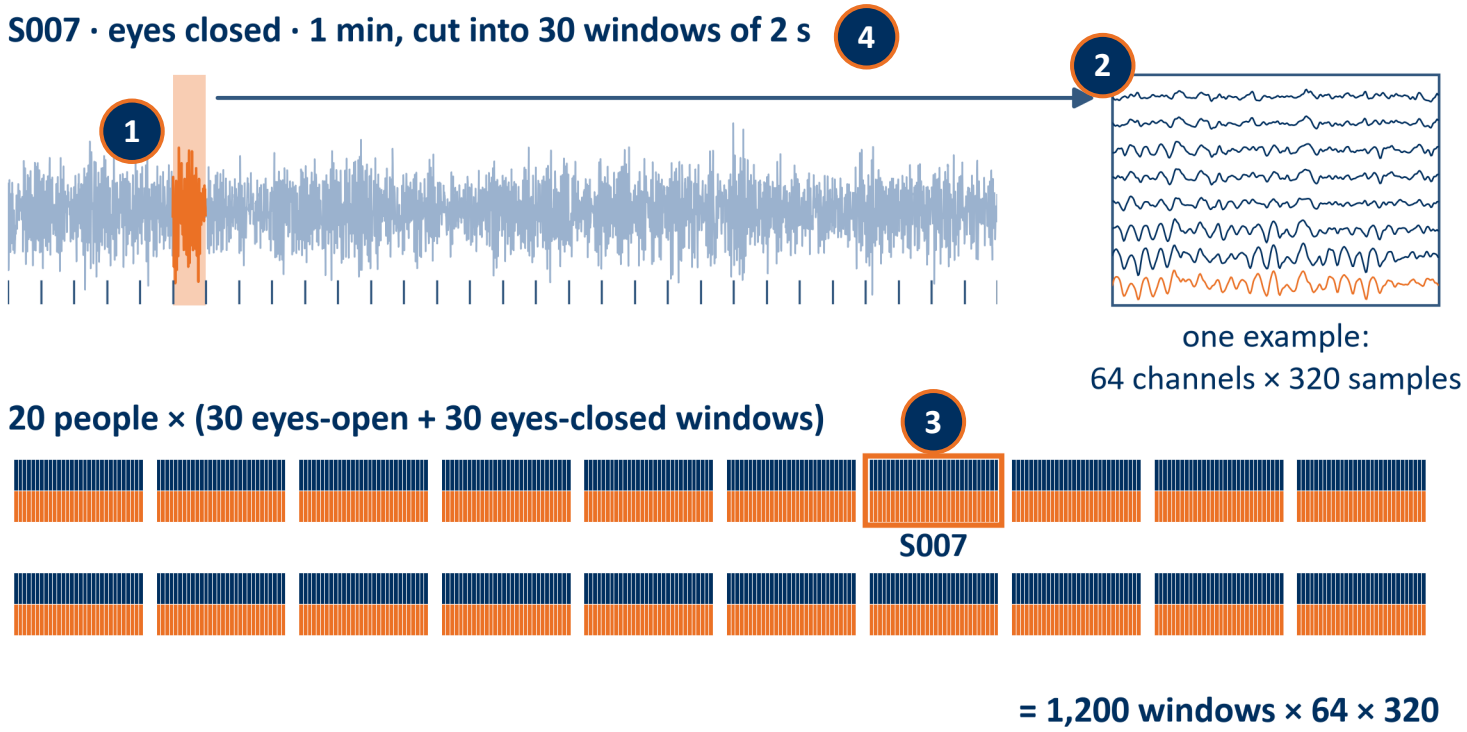
Not: cleaning cannot leak.
But: any step that learns from the data (scaling, CSP, feature selection) can carry the test set into training.
Do: put every learned step inside the cross-validation loop.

Gramfort et al. (2013) MNE-Python · MNE tutorial “Repairing artifacts with ICA”

→ slide 29 · SPLIT

Notebook §2

To get examples, we cut each 1-min recording into 2-s windows: 1,200 windows, but only 20 people



1. 2 s = 320 samples per channel.
2. One example = 64 × 320 = 20,480 numbers.
3. 30 eyes-open + 30 eyes-closed windows per person.
4. The **label** (the answer to learn: open or closed) belongs to the whole 1-min recording.

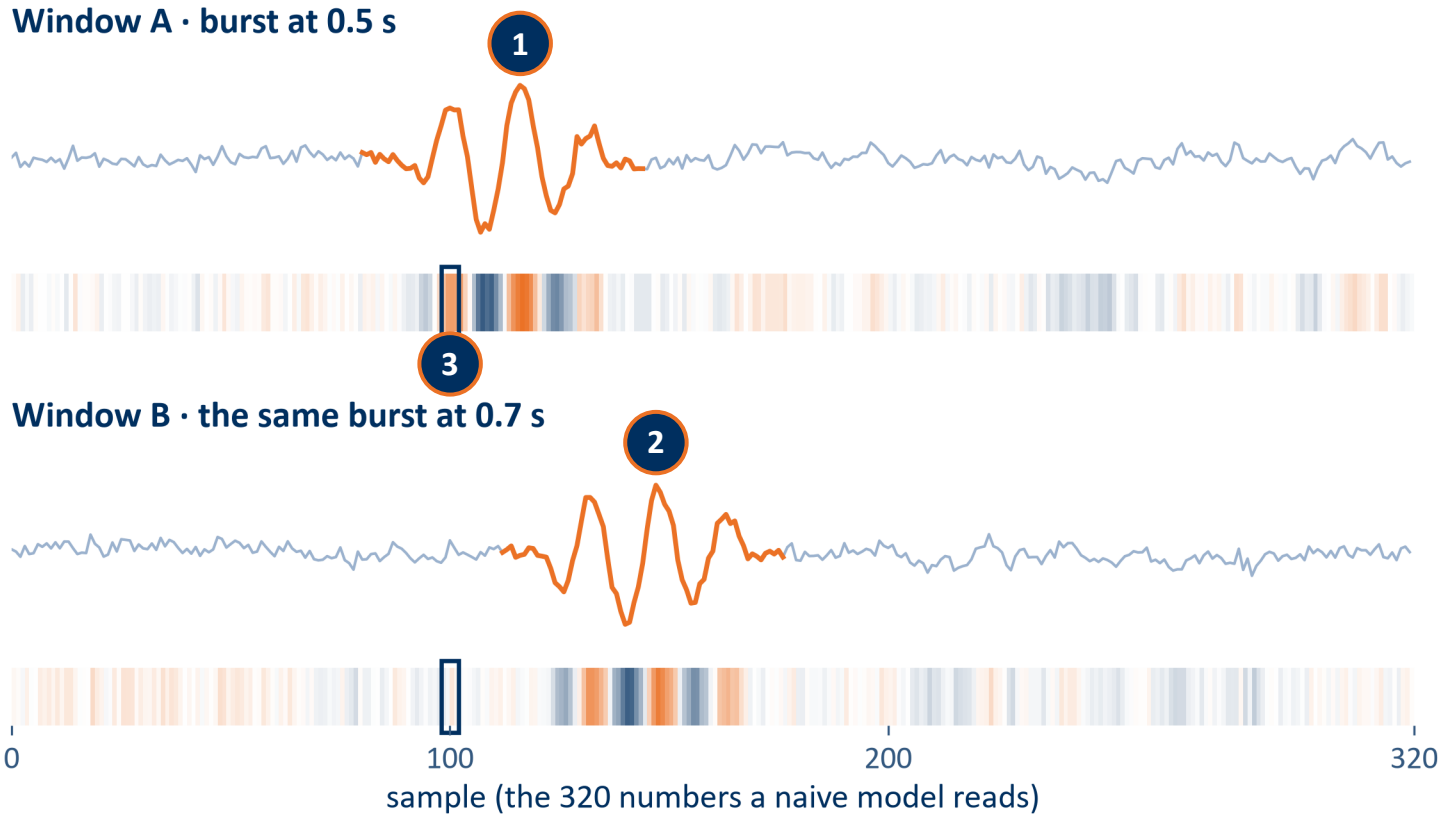
How many independent examples: 1,200 or 20? → Part 2

Σ DEFINITION **Sample** = one measurement in time (1/160 s). **Example** = one window shown to the model.

Data: PhysioNet EEGBCI (Schalk et al. 2004), ODC-By 1.0 · Figure: own (figures/make_figs_p1.py)

→ slide 36 · sliding filters

Using every sample as a feature is fragile: the same burst, 200 ms later, lands on different numbers



Feature = one number the model reads about an example.

1 2 3 With samples as features, A and B look like two different windows.

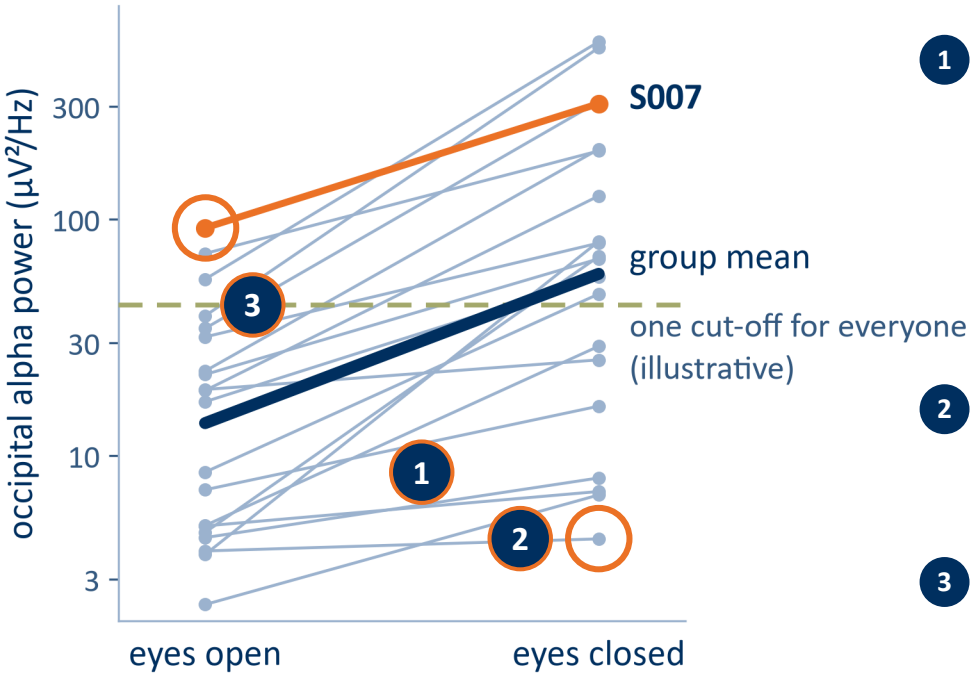
20,480 numbers per example, only 1,200 examples: the model can memorise the noise of its training windows (overfitting, slide 33).

Fix: summaries that ignore when (spectrum, band power), or filters that slide along time (ROCKET, CNNs: Part 3).

Figure: own (figures/make_figs_p1.py), after Uribarri, EIVIA 2025 course

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Alpha power at the back of the head alone gets 64% right on new people, where chance is 50%



1. every line goes up: 20 of 20 people
2. one person's eyes-open level is above another's eyes-closed level
3. so any one cut-off gets someone wrong: here 10 of the 40 person averages

Accuracy on new people (5 rounds)

Model	Accuracy
Always the same answer	50%
1 number (alpha power, O1/Oz/O2)	64%
Whole spectrum (64 channels × 40 frequencies)	75%

Accuracy = share of test windows labelled correctly.

KEY IDEA · TRUST CHECK · BASELINE

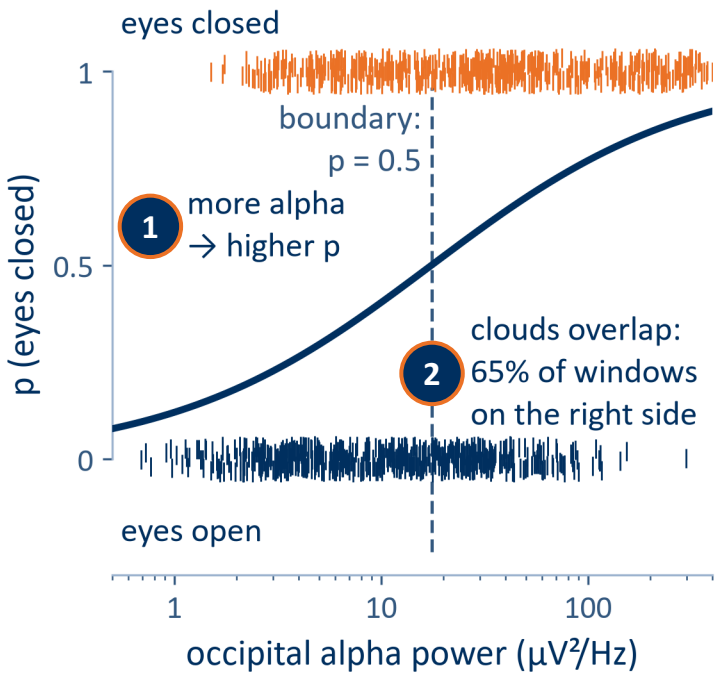
Always report what a trivial model scores. Here, always giving the same answer scores 50%.

Figure: own · data: PhysioNet EEGBCI, S001–S020 (Schalk et al. 2004), ODC-By 1.0 · scikit-learn

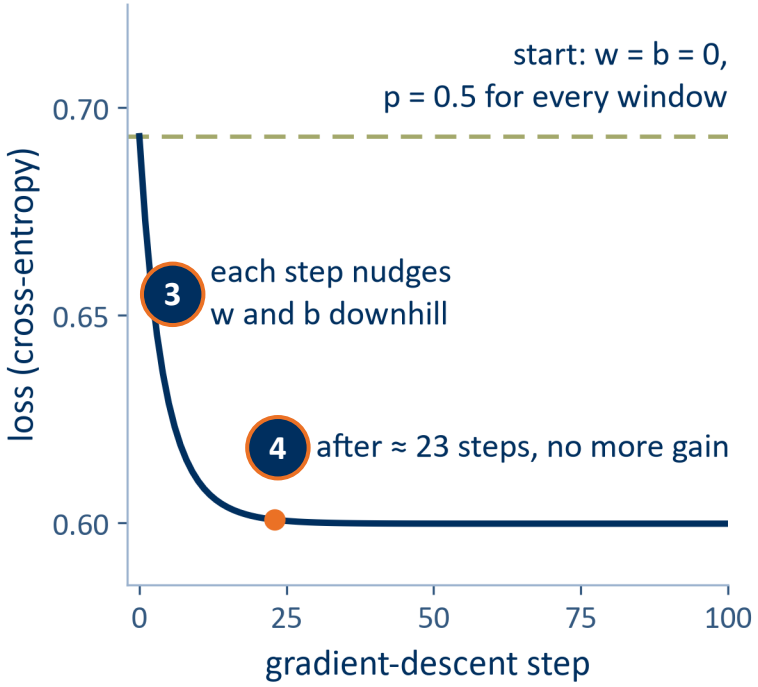
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Logistic regression is a single artificial neuron: it turns alpha power into a probability of “eyes closed”

What it learned



How it learned



Σ LOGISTIC REGRESSION = ONE NE



$$p = \sigma(w \cdot x + b)$$

x	alpha power (standardized)
w	weight
b	bias
σ	squashes any number into 0–1
p	probability of “eyes closed”

Learned: w = 0.97, b = 0.01
65% on training, 64% on new people

Parameters w and b, the knobs

Loss cross-entropy; confident mistakes cost most

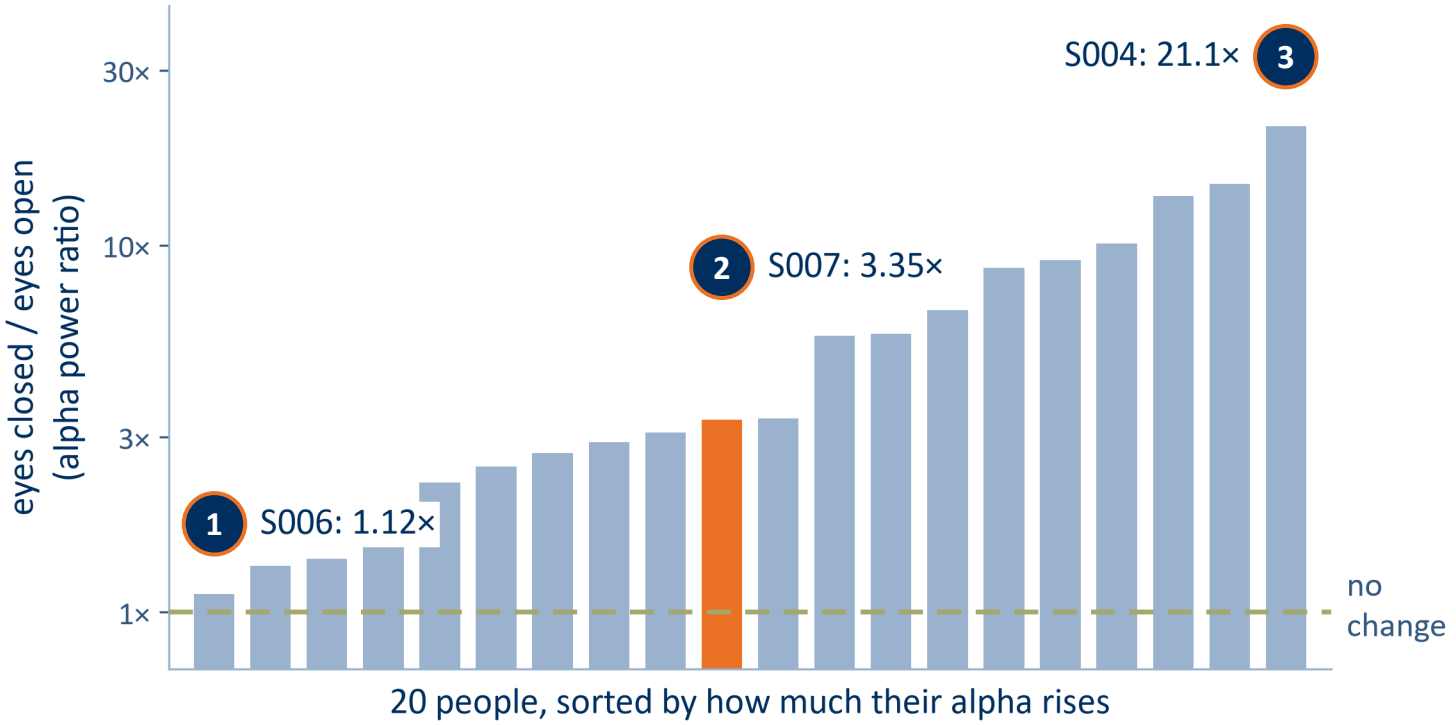
Learning gradient descent: step downhill, repeat

Regularization a penalty that keeps w small (slide 33)

Figure: own (make_figs_p3.py) · data: PhysioNet EEGBCI (Schalk et al. 2004), ODC-By 1.0 · scikit-learn

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Every person's alpha rises when the eyes close, but by 1.1× in one person and 21× in another



- All 20 people: more alpha with eyes closed
- The size of the change: 1.12× to 21×, median 3.4×
- So accuracy also varies by person: 50% to 100% (median 73%) for the whole-spectrum model

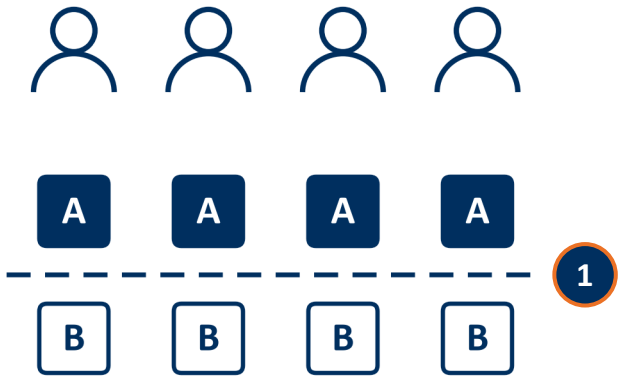
Read: one bar = one person · O1/Oz/O2, 8–13 Hz · log scale

Figure: own · data: PhysioNet EEGBCI, S001–S020 (Schalk et al. 2004), ODC-By 1.0

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When each person carries one label, what counts is the number of people, not of trials

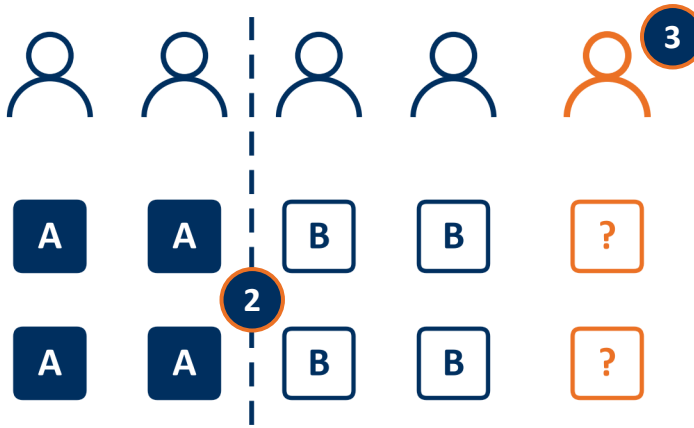
Each person gives both conditions (our eyes data)



1 the model must learn **WHAT** changes

A = eyes open · B = eyes closed

Each person has one label (a diagnosis)



2 the model can learn **WHO** is who

3 a new person: which label?

A = control · B = Parkinson's

EXAMPLE · PARKINSON'S FROM EYE MOVEMENTS

1,584 short eye-tracking recordings (trials) from 84 people: 54 with Parkinson's, 30 controls. The final test: 440 trials, but only 24 people.

PITFALL

Not: 1,584 trials make a big dataset.
But: a diagnosis must work on a new person, and we have 84 people.
Do: count people, and split by person.

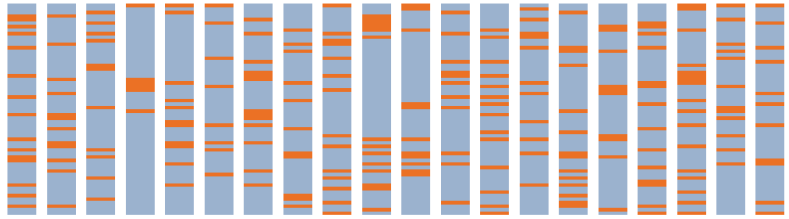
Uribarri et al. 2026, Int J Med Inform 220:106626 · data not public · Figure: own

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PREDICT ← slide 23 · 1,200 or 20? Notebook §2

Split by the unit you want to generalize to: for a new patient, that unit is the person

Random window split



both: 80% of windows for training

1 the test person's other windows are in training: the model can recognise the person

Split by person



20 people (one column each, 60 windows per person)

2 test people never seen

training test

99.8% → 53.0%

same deep network, Alzheimer's vs subjective cognitive impairment, 49 + 49 people: window split vs person split (the 53% interval includes chance)

- Only 17 of 63 deep-learning EEG papers clearly avoided this leak
- Why it works: 1-s MEG windows tell which of 124 people they came from with > 99% accuracy

KEY IDEA · TRUST CHECK · SPLIT Test on the unit you will meet in use: a new person, session, site, or a later time.

Brookshire et al. 2024 · Kampel et al. 2023 (both Front. Neurosci.) · Figure: own

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BREAK

Break



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PART 3 OF 5

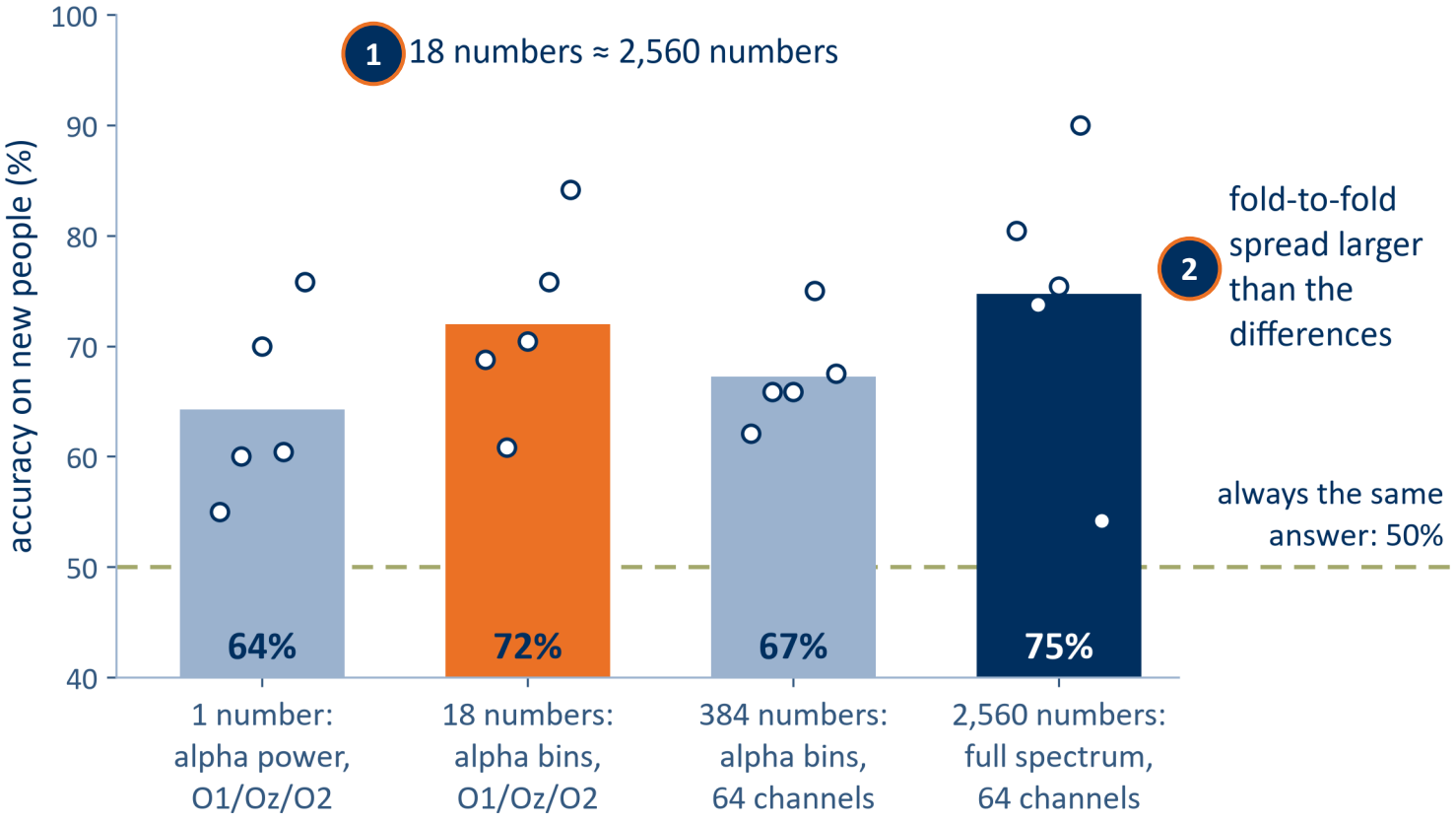
EEG/MEG: from hand-made features to deep models

Who designs the features: you, chance, or the model?

Notebook §2–§3

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Hand-crafted features are cheap and readable: 18 alpha numbers reach 72%, the whole spectrum 75%



- **Band power: the average height of the spectrum in one band and channel, log scale (slide 20).**
- Fast, no tuning, readable: “alpha over occipital cortex”.

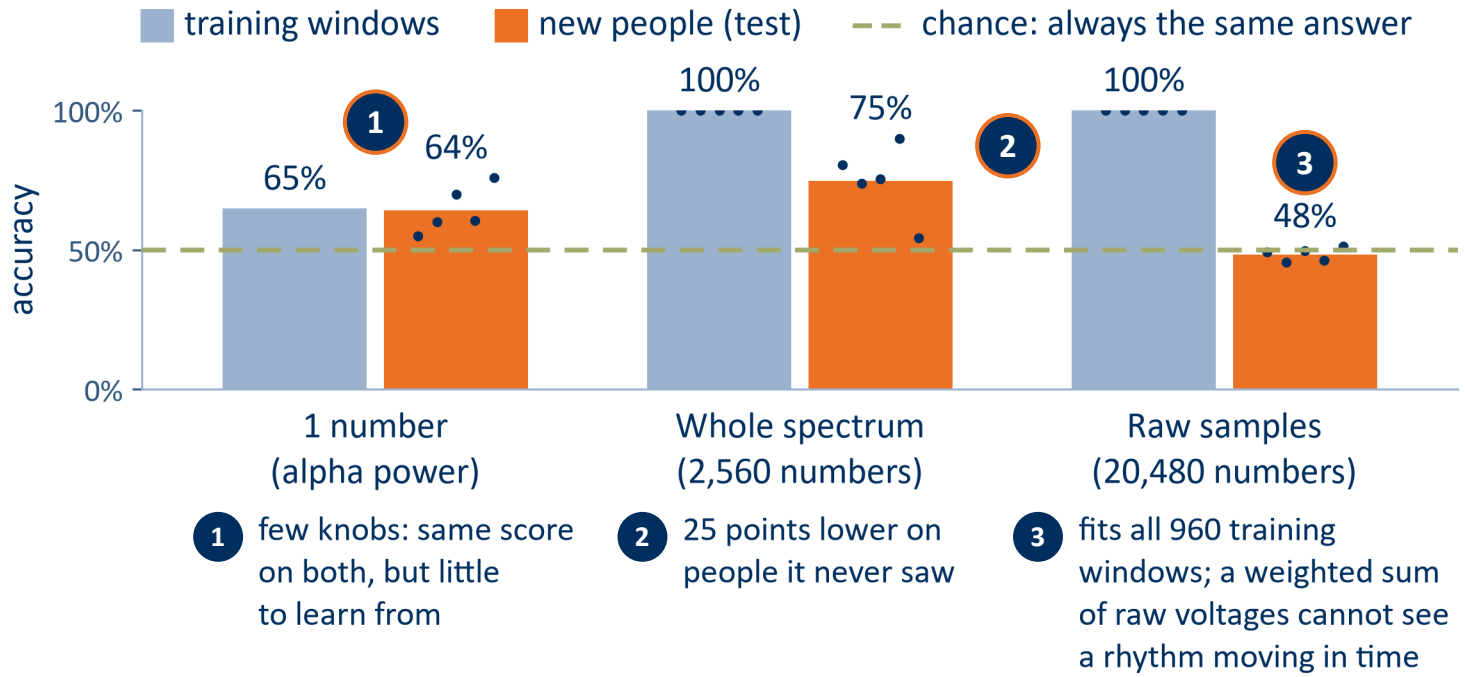
EXAMPLE · GENERIC FEATURES

catch22: 22 readable statistics of any time series, chosen from 4,791 candidates: about 1000× faster than all 4,791, for ≈ 7% lower accuracy on 93 datasets.

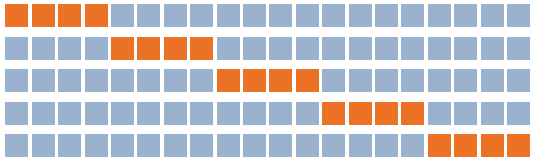
Lubba et al. (2019) DMKD 33:1821–1852 · Welch (1967) · Figure: own · data: EEGBCI (Schalk 2004), ODC-By 1.0

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Learning is not memorizing: perfect on its training windows, a model can score 75%, or chance, on new people



Cross-validation: split the 20 people into 5 groups; train on 4 groups, test on the 5th; rotate; average the 5 scores.



20 people × 5 rounds → average of 5 scores

Regularization: a penalty, added to the loss, for large weights. Here it narrows the gap; new people stay at 74–77%.

Read: test = 4 people never seen, 5 rounds · dots = the 5 rounds

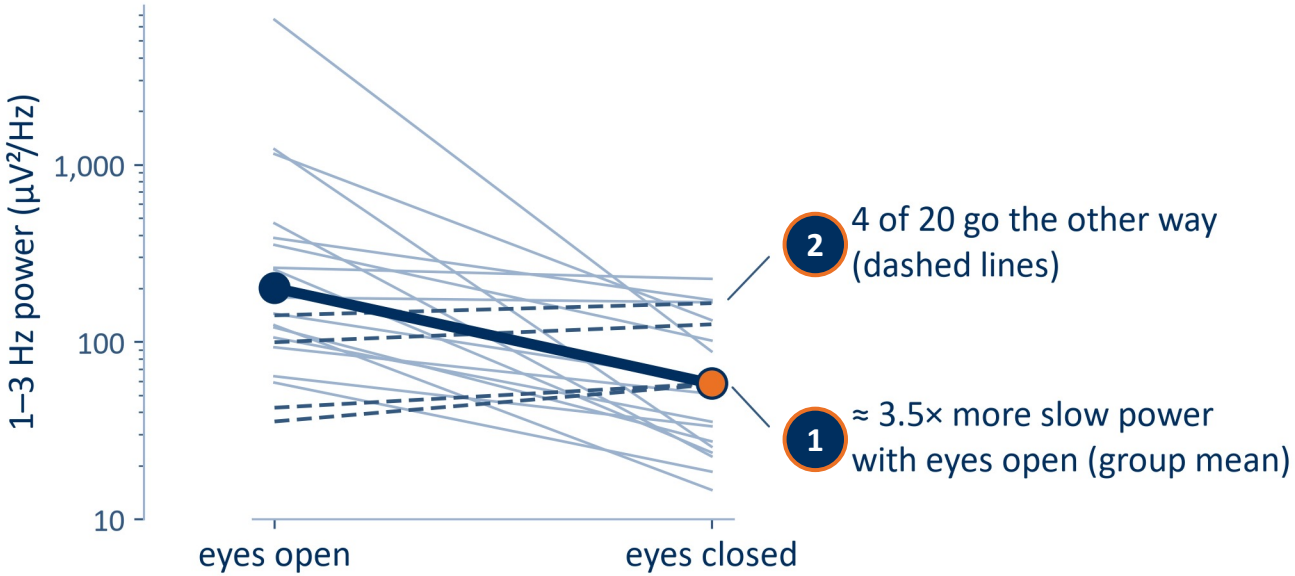
Σ **DEFINITION · TRAINING AND TEST**

The model learns on the **training set** and is scored on a **test set** it never saw. **Overfitting**: good on the first, worse on the second.

Poldrack et al. 2020 · Varoquaux et al. 2017 · Figure: own · data: PhysioNet EEGBCI, ODC-By 1.0

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Forehead electrodes likely pick up blinks and eye movements: slow frontal power is 3.5× higher with eyes open



Each line = one person · 1–3 Hz power at the three forehead electrodes

Eyes open or closed?
Accuracy on new people

Features	Accuracy
All 64 channels, 1–40 Hz	75%
Fp1/Fpz/Fp2 only	**75%**
Fp1/Fpz/Fp2, no 8–13 Hz	71%
Fp1/Fpz/Fp2, 1–3 Hz power	61%
Always the same answer	50%

No eye channel was recorded, so “the eyes” is an inference.

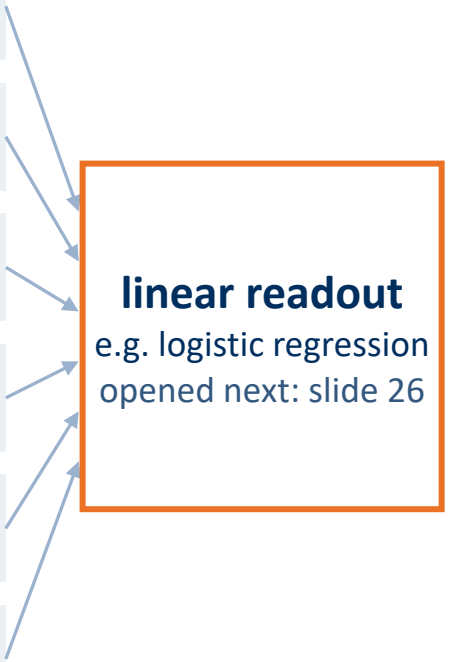
⚠ **TRUST CHECK · CONFOUND**

Not: a high score shows that the model found visual cortex.
But: a model uses whatever separates the classes, and open eyes blink and move.
Do: test a confound-only feature: in one EEG study, a single 1-ms sample “decoded” 40 image classes at ≈ 92% (Li 2021).

EEGBCI (Schalk et al. 2004), ODC-By 1.0 · Figure: own (make_figs_p3.py) · Li et al. (2021) IEEE TPAMI 43:316–333

Every decoder is a representation plus a simple readout; higher rungs learn more and need more data

rule of thumb, not a measurement	slide	labelled data needed	easy to read
Pretrained foundation model	41	few, in principle	○○○
Learned filters (CNN)	40	●●●	●○○
Random kernels (ROCKET)	36–39	●●○	●○○
Covariance → tangent space	73	●○○	●○○
Spatial filters (CSP)	72	●○○	●●○
Band power per channel	32	●○○	●●●



bottom → top: learns more, needs more labelled data

Σ DEFINITION

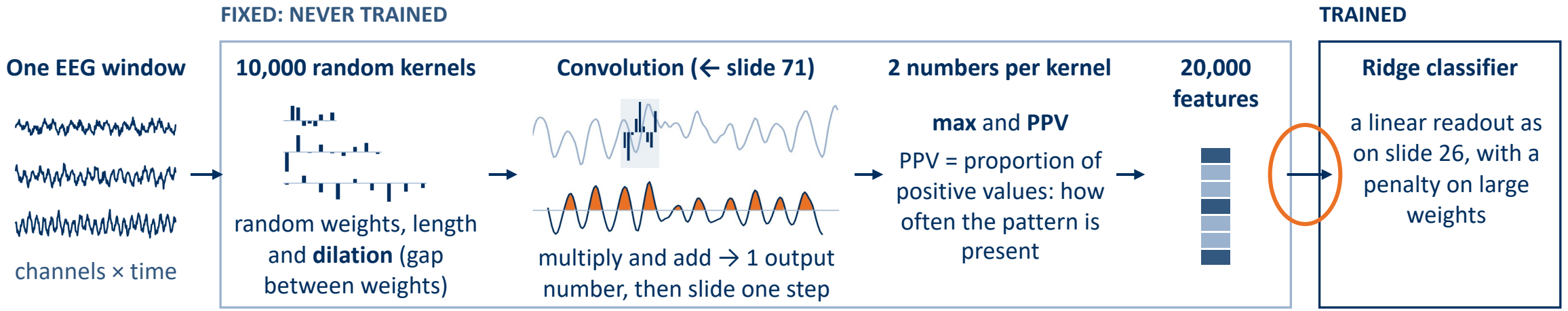
Representation (slide 42): the numbers a pipeline computes from a window.
Readout: a linear model, a weighted sum of those numbers plus a bias, turned into a class.

Most of the gain comes from the representation. Which rung to use depends on how much labelled data you have.

After Uribarri, EIVIA 2025 (Days 2–4) · Lotte et al. (2018) J Neural Eng · Schirrmeister et al. (2017)

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ROCKET: thousands of random filters and one linear model, trained in seconds on a laptop



EXAMPLE · EEG SLOW CORTICAL POTENTIALS

One person, 6 channels, 3.5-s trials, fixed split (268 training / 293 test trials). Always the same answer: 50% · spectrum features + logistic regression: 63% · MiniRocket: **92%**, about 4 s to train and test on a laptop. The information is in the slow waveform shape, not in a rhythm.

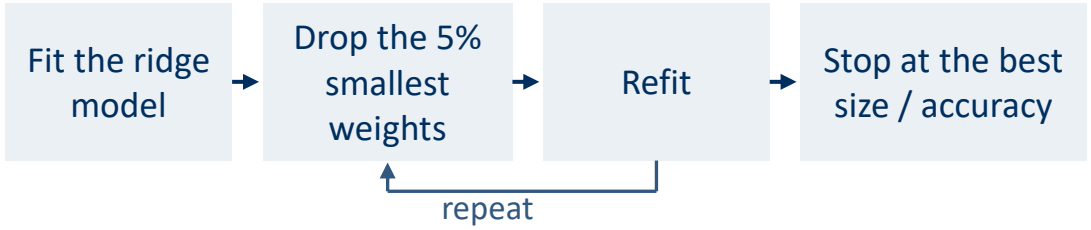
The ROCKET family is fast and near the top of the benchmarks (univariate).

Dempster et al. (2020) DMKD; (2021) KDD · Ruiz et al. (2021) · data: UEA SelfRegulationSCP1, CC BY 4.0 · Figure: own

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Detach-ROCKET keeps 10% of ROCKET's features at equal accuracy; its multichannel version points to channels

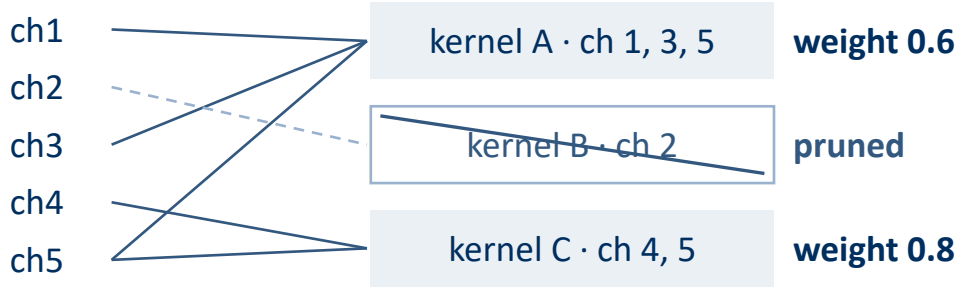
1 · Prune



UCR ARCHIVE AVERAGE · UNIVARIATE, NOT NEURAL

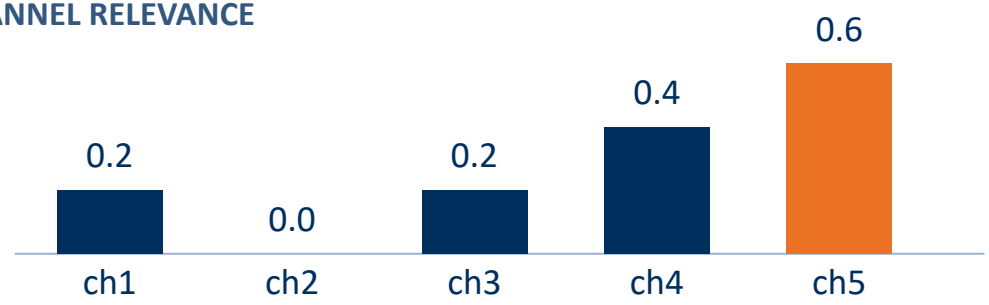
Model	Features kept	Test accuracy
ROCKET	100%	84.74 ± 0.62%
Detach-ROCKET	10%	**85.15 ± 0.78%**
Detach-ROCKET, automatic stop	2.06%	82.90 ± 2.74%

2 · Point to channels (multichannel)



Each kept weight is split among its channels, summed per channel.

CHANNEL RELEVANCE



Next: pruning on a clinical problem (slide 38) and channel maps on real MEG/EEG (slide 39).

Uribarri et al. (2024) DMKD 38:3922–3947 · Solana et al. (2025) AALTD, LNCS · github.com/gon-uri/detach_rocket

Fixational eye movements classify 95% of 24 new people correctly; always answering “Parkinson’s” scores 58%

1 new person

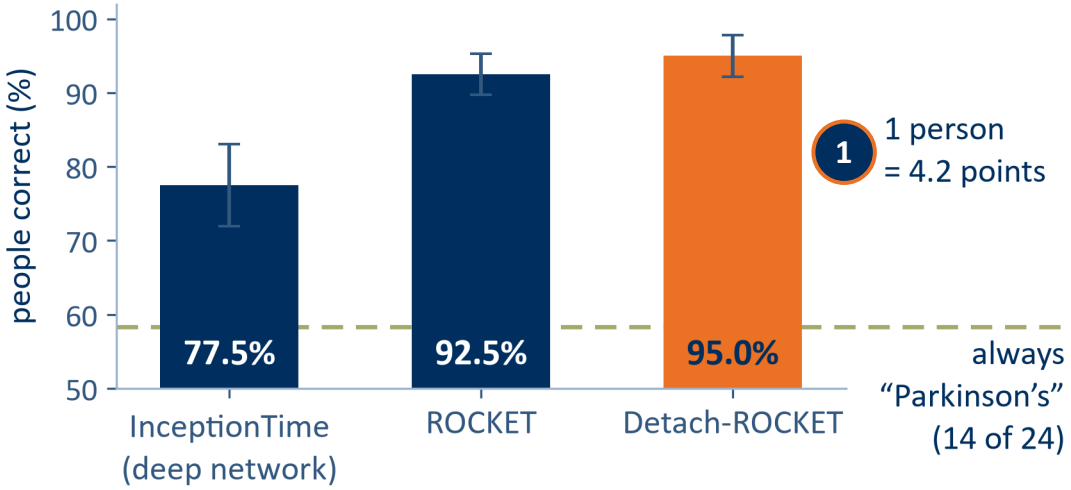
its trials: 1.5-s fixations, gaze x, y + velocities

Detach-ROCKET scores every trial

average the scores (**soft vote**)

Parkinson’s or healthy

Eye tracking: not brain data, but a neuro-clinical time series



- 84 people: 54 with Parkinson’s, 30 healthy. 24 locked away until the end.
- The model’s confidence weakly tracks disease duration and motor score (UPDRS; ρ ≈ 0.2), not age.
- Test groups age-matched (the whole cohort is not).

⚠ **PITFALL** **Not:** “a diagnostic test” · **But:** 24 test people, one clinic, one device; 95% and ± are over 5 training runs, not over new people · **Do:** replicate in other clinics first.

Uribarri et al. (2026) Int J Med Inform 220:106626, CC BY 4.0 · left: after its Fig. 3 · right: its Table 1

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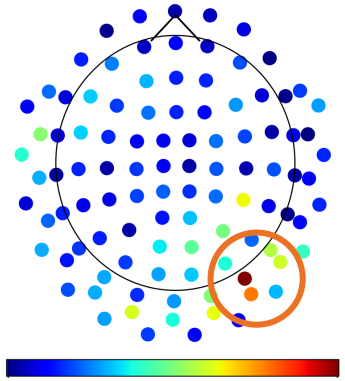
On raw MEG and EEG of new people, our pruned ensemble beats the majority answer and points to the sensors used

Detach-Rocket Ensemble: 25 pruned MiniRocket models vote (an ensemble); channel relevance as on slide 37, median of 25.

Face or scrambled face?

MEG · 306 sensors · raw signals · 16 training people · tested on new people

1 right lateral occipital: a known face-responsive region



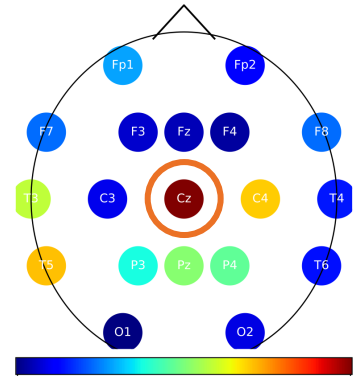
less ← relevance → more

MEG, one decision per trial	Accuracy
Our ensemble, new people	**64.3 ± 0.5%**
Best ROCKET-family rival (Arsenal)	61.5%
Chance (balanced classes)	≈ 50%

Alzheimer's or healthy?

EEG · 19 electrodes · eyes-closed rest · 65 people (36 AD, 29 healthy) · leave one person out

2 strongest at Cz, the recording reference: not interpreted



less ← relevance → more

* DICE-net per person: read by us from their figure; a one-person tie

EEG accuracy	Ours	DICE-net
Per person (of 65)	86% (56)	85% (55)*
Per segment	79.9%	**83.3%**
Always "Alzheimer's", per person	55% (36)	55% (36)

⚠ **PITFALL** **Not:** "the relevant sensor is where the brain does the job" · **But:** each sensor mixes many sources, and model weights are not activation patterns (Haufe 2014) · **Do:** test the map against anatomy or source analysis.

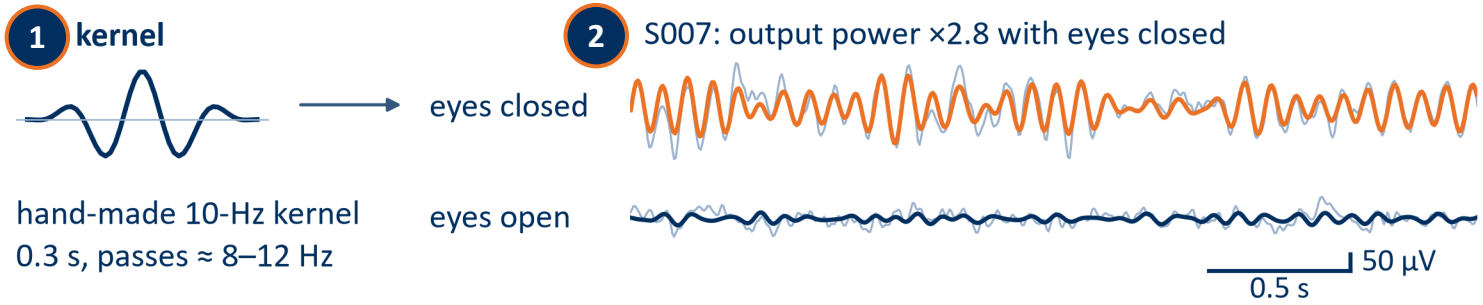
Solana et al. (2025) AALTD, LNCS; maps: its Figs. 6–7, © the authors, reproduced by an author · MEG: Henson et al. (2011), DecMeg2014 · EEG: ds004504

A CNN learns its own filters from the labels: band-pass filters in time, then weighted sums over electrodes

CNN (convolutional neural network): a stack of **layers**, each sliding learned **kernels** over its input. EEGNet:

all weights learned together from the labels: same loss and gradient descent as on slide 26

Input	**Temporal convolution**	**Depthwise spatial conv.**	**Pooling**	**Linear readout**
64 electrodes × 320 samples (2 s)	 kernels slide along time	weights over the 64 electrodes	average over short stretches of time	→ p(eyes closed)
← slide 23 window	≈ band-pass filter (slide 71)	≈ a CSP virtual electrode (slide 72)	exact timing matters less (← slide 24 shift)	= logistic regression (slide 26)



Backpropagation: every kernel weight learns which way lowers the loss.

Evidence: motor decoding, ConvNets 84.0% vs filter-bank CSP 82.1% (Schirrmeister 2017).

All 20 people (medians): raw Oz ×1.5 · this 10-Hz kernel ×2.8 · a 20-Hz kernel ×1.7

⚠ **PITFALL** **Not:** "deep = better" · **But:** deep nets typically need > 150 trials per class (BCI benchmark, preprint); with tens, covariance pipelines win on average · **Do:** beat a cheap baseline first.

Lawhern et al. (2018) · Schirrmeister et al. (2017) · Chevallier et al. (2024, preprint) · Figure: own, EEGBCI

EEG foundation models pretrained on thousands of hours are promising, but the 2026 evidence is mixed

Study	What was compared	Verdict for foundation models
EEG-Arena (2026, preprint)	30 models vs 25 supervised baselines, 57 tasks	**Better on most tasks**; more pretraining data helps, bigger models not always
EEG-FM-Compass (2026, Natl Sci Rev)	12 models, 13 datasets, new people	A linear readout on frozen features (**linear probe**) is often too weak; specialists trained from scratch are competitive
Zare (2026, preprint)	dementia EEG, 1,187 recordings	Classical features beat the foundation models (0.73 vs ≤ 0.68; 0.5 = chance), and every model can tell **which dataset** a recording came from

Before you believe it: compare with a strong cheap baseline, on new people, with the same split.

∑ FOUNDATION MODEL

1 Pretrain a big network **without labels** on thousands of hours from many people (REVE: > 60,000 h, 25,000 people).

↓

2 Hide parts of the signal (**masking**), or hold back what comes next, and learn to predict them.

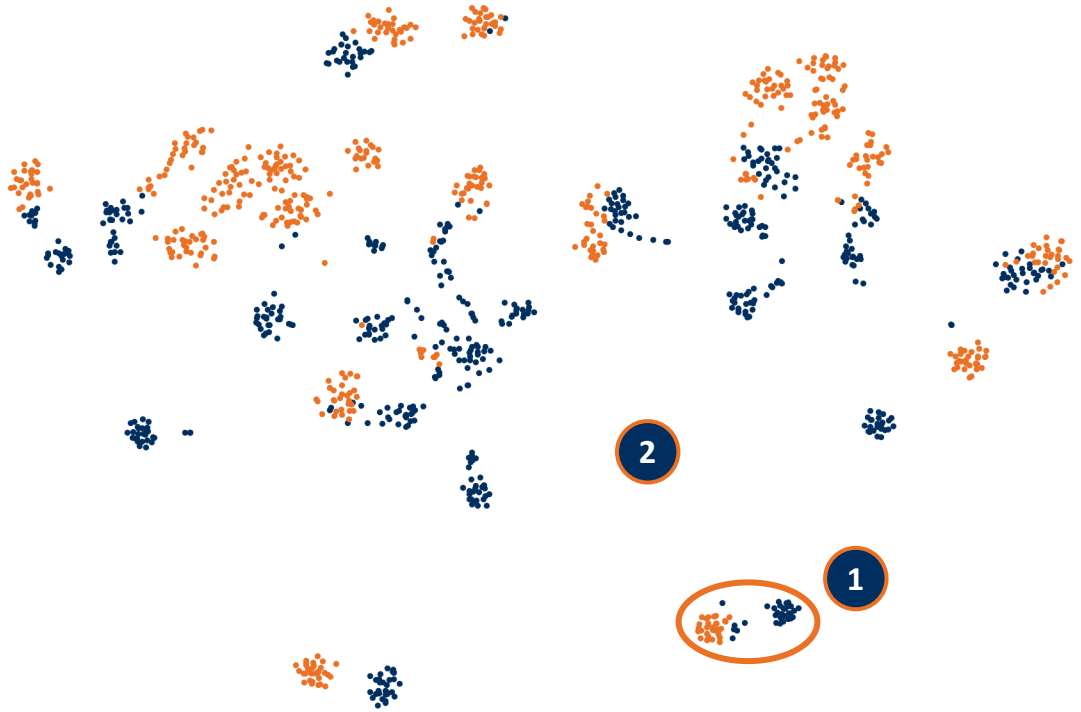


↓

3 The result is a reusable representation (slide 42); fine-tuning adapts it to a task with few labels.

Chen et al. (2026) arXiv:2609.32743 · Liu et al. (2026) Natl Sci Rev · Zare (2026) arXiv:2607.24519

Mapped from 2,560 numbers to 2, the 1,200 windows form about 20 islands, and eye state does not explain them



- eyes open
- eyes closed

1. an island: both colours side by side
2. gaps and island sizes mean nothing in t-SNE

Read: one dot = one 2-s window · 2,560 numbers → 2 · no labels used to draw it

Predict: what else could make these ~20 islands?

Σ DEFINITION · REPRESENTATION (EMBEDDING)

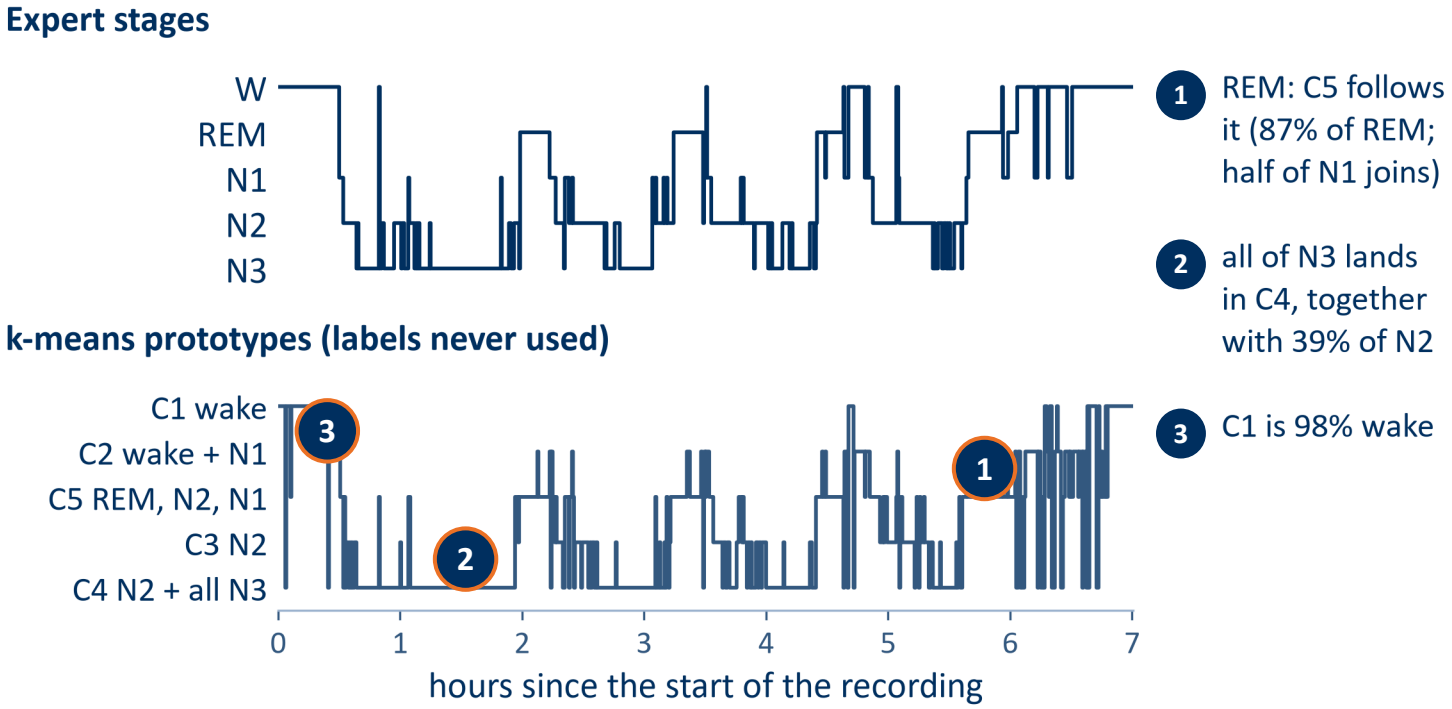
New numbers for each example, so that similar examples land close together. **Representation learning** finds them from data.

FROM HAND-MADE TO LEARNED

- **Hand-crafted**: band power (Part 1)
- **PCA**: directions of most variance (Part 4)
- **t-SNE / UMAP**: 2-D maps (this figure)
- **Learned**: autoencoders, contrastive (Part 5), foundation models (Part 3)

van der Maaten & Hinton 2008 · Wattenberg et al. 2016 · Figure: own · data: PhysioNet EEGBCI, ODC-By 1.0

k-means compresses each 30-s epoch to 1 of 5 prototypes, which partly match the expert sleep stages (ARI 0.26)



Read: W wake · N1 drowsy · N2 light sleep (spindles) · N3 deep sleep (slow waves) · REM dreaming sleep (rapid eye movements) · % = all 8 nights

∑ DEFINITION · K-MEANS

Find k prototypes that make this loss smallest:

$$L = \sum_i \left\| \mathbf{x}_i - \boldsymbol{\mu}_{c(i)} \right\|^2$$

$\mathbf{x}_i$	the 12 numbers of 30-s epoch i (band powers)
$\boldsymbol{\mu}_{c(i)}$	its nearest prototype (k = 5)

How: give each epoch to its nearest prototype; move each prototype to the middle of its group; repeat

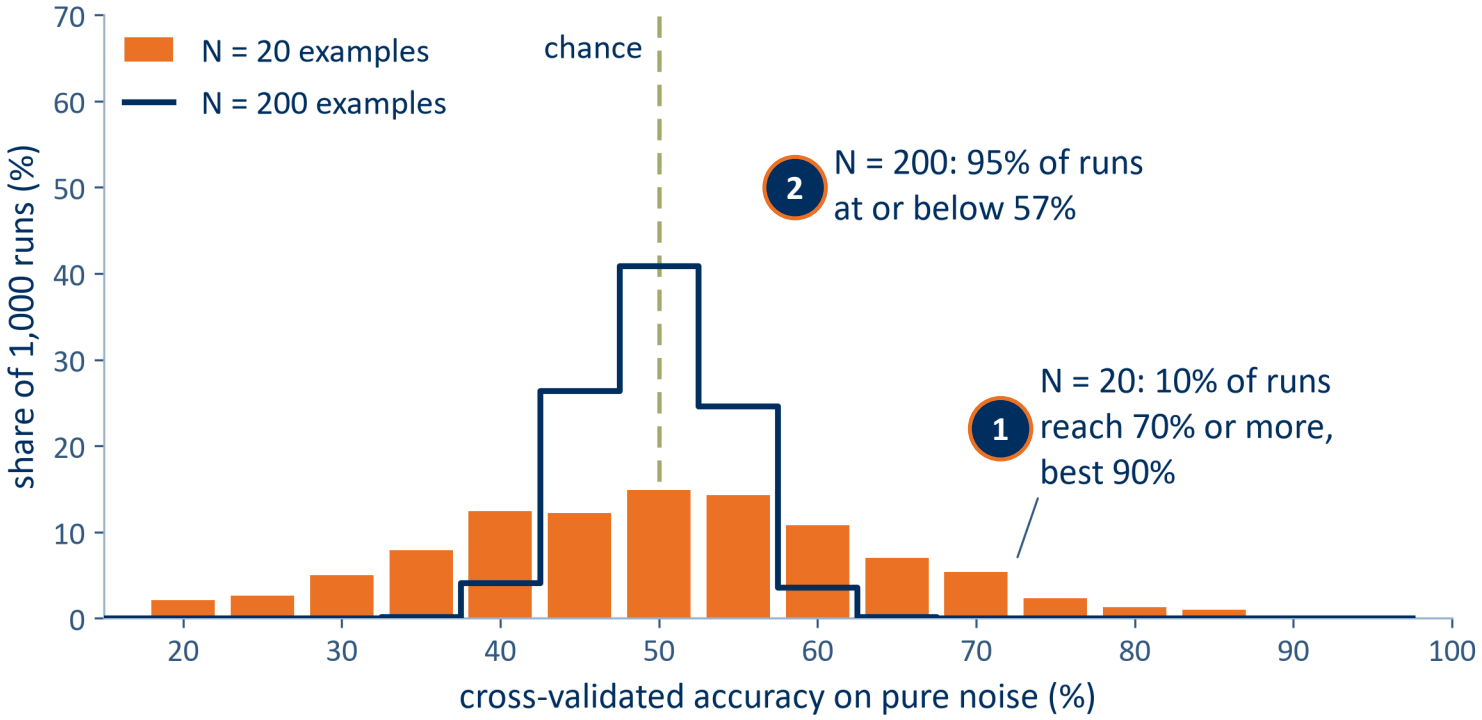
Compression: 12 numbers → 1 of 5 prototypes; k = 5 picked without labels

ARI: 0 = chance, 1 = the expert's groups; shuffled: 0.00

Decat et al. 2022, Sleep Med 98:39–52 · Figure: own · data: Sleep-EDF Expanded (Kemp et al. 2000), ODC-By 1.0

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On pure noise, a classifier with 20 examples scores 70% or more one time in ten



1 Q1, our eyes classifier: 75% overall, but 50% to 100% depending on the person; 95% interval over people: 67–83%.

2 Parkinson's test set: 24 people, so one person moves the score by 4.2 points (slide 38).

3 Permutation test: shuffle the labels, rerun the whole pipeline 1,000 times; your score must beat ~95% of those runs.

KEY IDEA · TRUST CHECK · UNCERTAINTY

Report the N, an interval over people (the unit you generalize to) and a permutation test.

Combrisson & Jerbi (2015) J Neurosci Methods · Varoquaux (2018) NeuroImage · Figure: own, simulated data

PART 4 OF 5

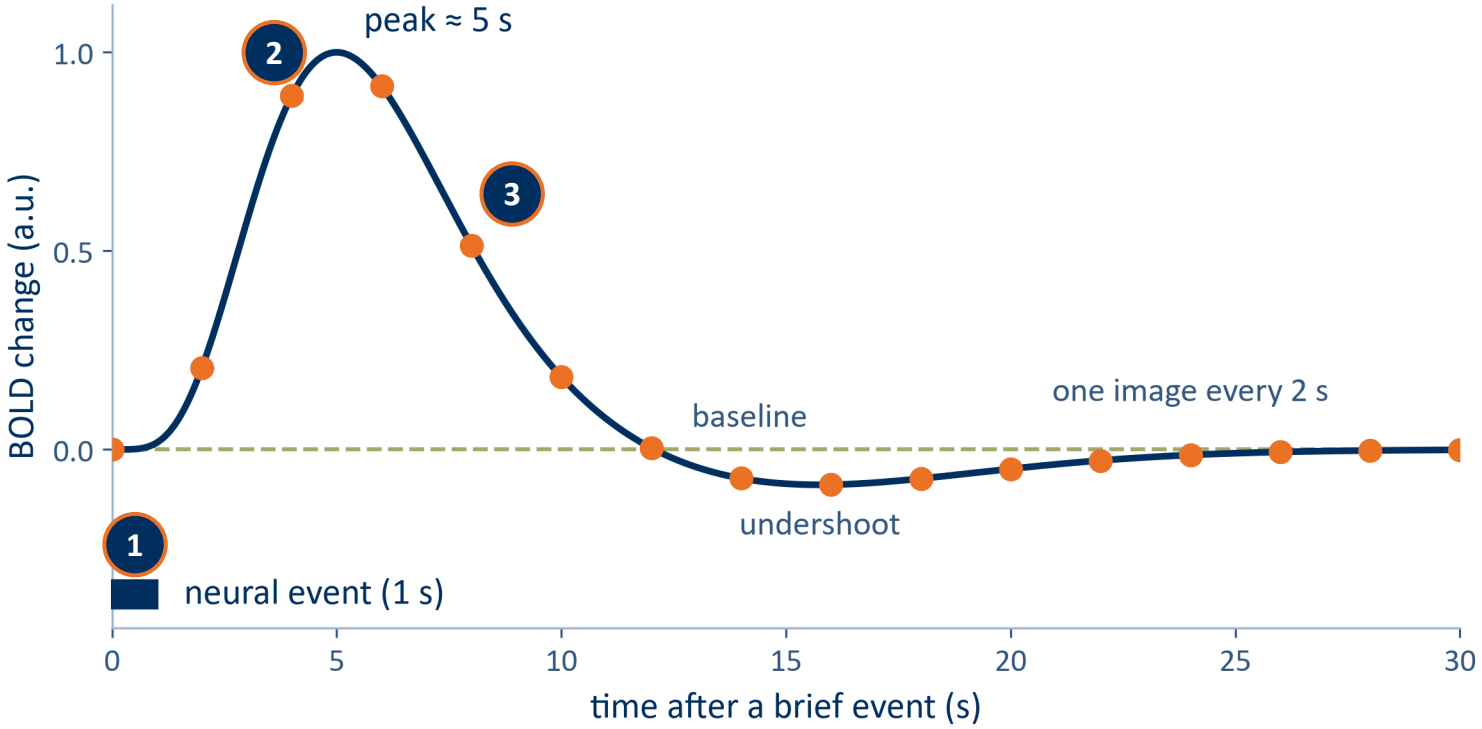
fMRI: connectomes as features

Can a few minutes of fMRI tell a child's brain from an adult's?

Notebook §6–§7

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fMRI measures blood oxygenation, a slow echo of neural activity that peaks about 5 s later



1. **BOLD** (blood-oxygen-level-dependent) signal: blood oxygenation that follows local, mostly synaptic, activity. An indirect, delayed measure
2. Peaks ≈ 5 s after the event and dips below baseline around 15 s
3. **TR** (repetition time) = 2 s here: one whole-brain image every 2 s, so nothing faster than 0.25 Hz is visible

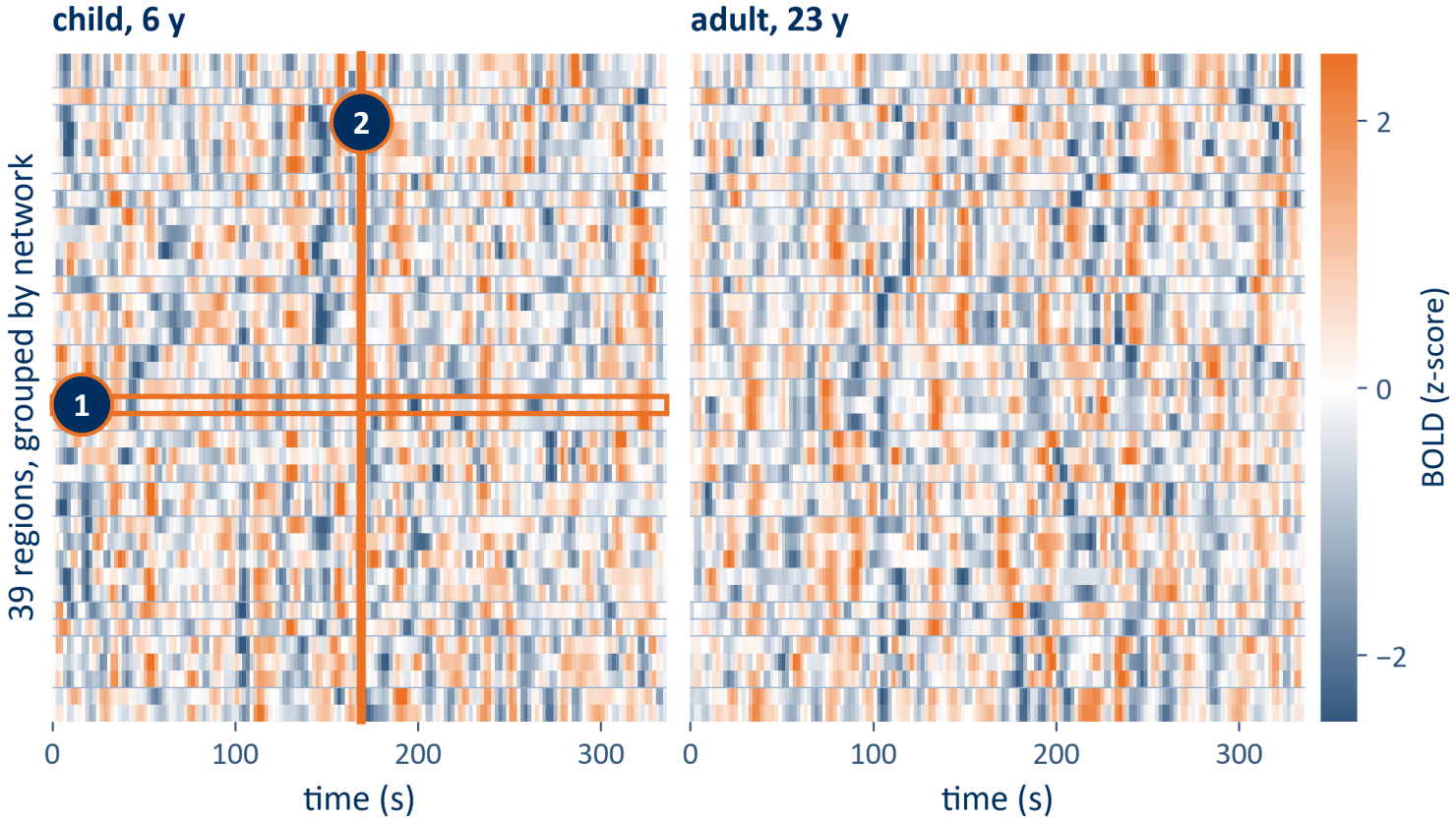
EXAMPLE · ALIASING A heartbeat at 1.1 Hz, sampled every 2 s, shows up as a slow 0.1 Hz wave, inside the 0.01–0.1 Hz band we analyse

Logothetis (2008) · Glover, Li & Ress (2000) · SPM canonical HRF (FIL wiki) · Figure: own (figures/make_figs_p4.py)

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fMRI | Notebook §6

155 people watched the same Pixar short in the scanner: 122 children aged 3–12 and 33 adults



1. one row = one region's signal over time (39 regions, slide 48)
2. one column = one brain image every 2 s; 168 images = 5.6 min
3. no task: everyone simply watched the film

122 children 3–12 y (mean 6.7)
33 adults 18–39 y
open data (CC0)

Richardson et al. (2018) · data: OpenNeuro ds000228 (CC0) · OSF 5hju4 (CC BY 4.0) · Figure: own (figures/make_figs_p4.py)

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An atlas pools the voxels into 39 regions, so each person becomes a time series with 39 channels

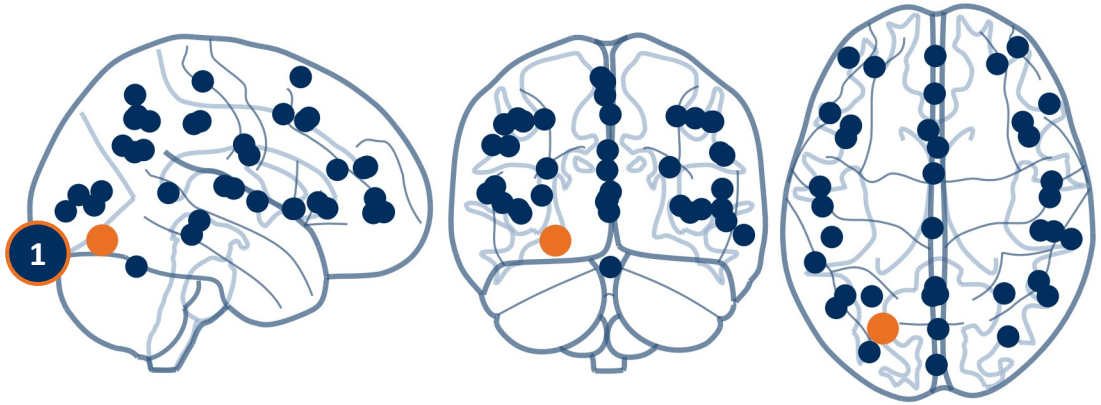
Σ ALGORITHM · FROM SCANNER TO 39 SERIES

1 Correct head motion; align every brain to a common template (fMRIPrep)

2 Take a weighted average of the voxels in each of 39 atlas regions (MSDL)

3 Regress out 15 nuisance signals (head motion, white matter, CSF)

4 Band-pass filter: keep slow waves of 0.01–0.1 Hz (periods of 10–100 s); then z-score each region

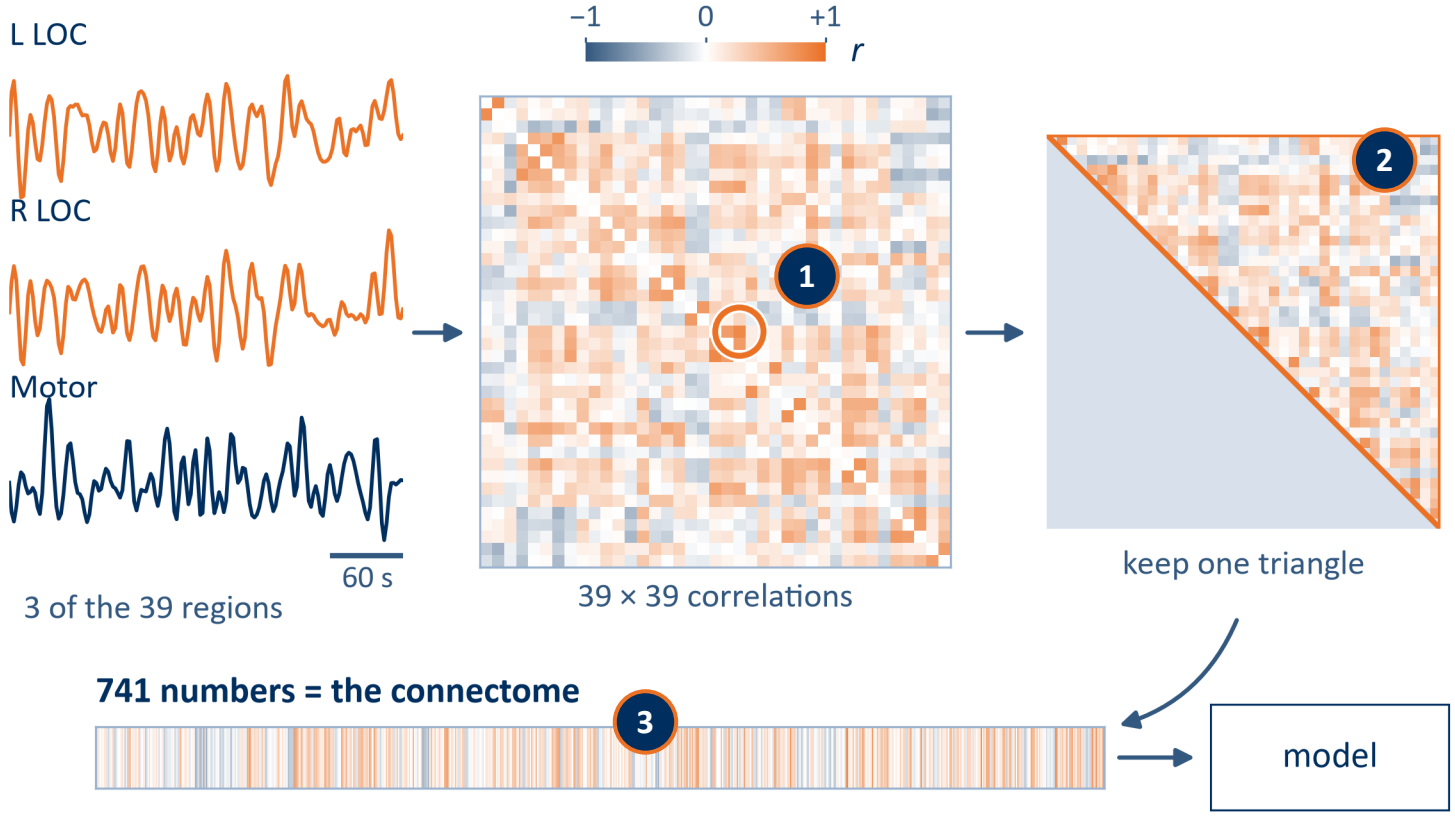


1 one region (orange, “Vis”, a visual region) = one time series

EXAMPLE · CLUSTERING MAKES ATLASES Many atlases cluster voxels whose signals rise and fall together (BASC, Craddock): each voxel is compressed to one region label. MSDL, used here, comes from a cousin method, dictionary learning, with soft, overlapping regions.

Esteban et al. (2019) · Varoquaux et al. (2011) · Bellec et al. (2010) · Craddock et al. (2012) · Figure: own (nilearn glass brain, MSDL centres)

Correlating every pair of regions turns each recording into features: a 741-number connectome

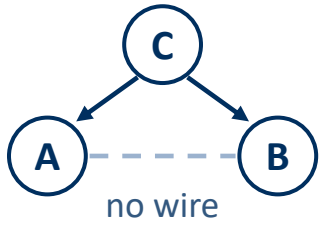


1. each cell = the correlation r of two regions over the 5.6 min (+1 together, 0 unrelated, −1 opposite)
2. the matrix is symmetric: keep one triangle, 39 × 38 / 2 = 741 numbers
3. these 741 numbers = the person's **connectome**: a graph (regions = nodes, correlations = edge weights) and a feature vector for a model

Varoquaux & Craddock (2013) · Figure: own (figures/make_figs_p4.py) · data: ds000228 (CC0) / OSF 5hju4 (CC BY 4.0)

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Partial correlation removes links that other regions explain; a sparse version keeps one edge in four



correlation(A, B) = 0.64

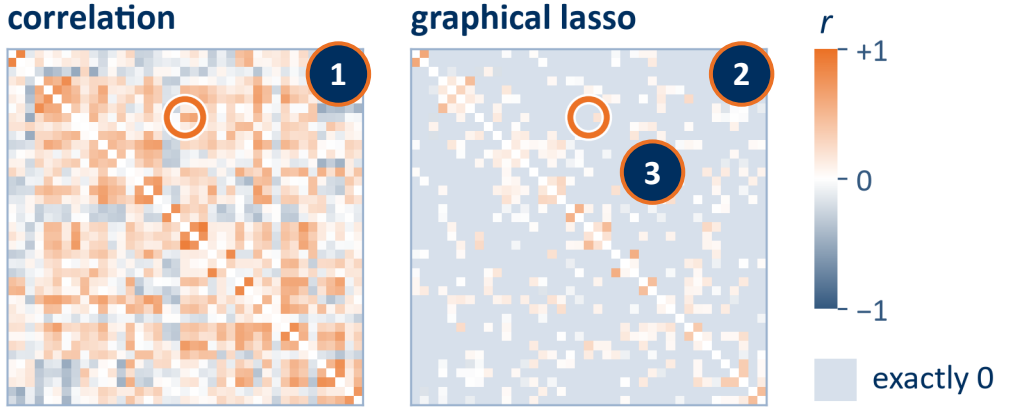
partial correlation = **0.00**

Σ DEFINITION · PARTIAL CORRELATION

What two regions still share without the others:

$$\rho_{AB\cdot\text{rest}} = \text{corr}\left(A - \hat{A}, B - \hat{B}\right)$$

A, B	the time series of two regions
$\hat{A}$	A predicted from all other regions
$A - \hat{A}$	the part of A they cannot explain



1. **plain correlation:** all pairs, direct or not (mean |r| 0.23)
2. **graphical lasso:** all partial correlations at once, plus a penalty that sets weak links to exactly 0: 555 of 741
3. ringed: one of 72 strong pairs (|r| ≥ 0.3) set to 0

⚠ PITFALL

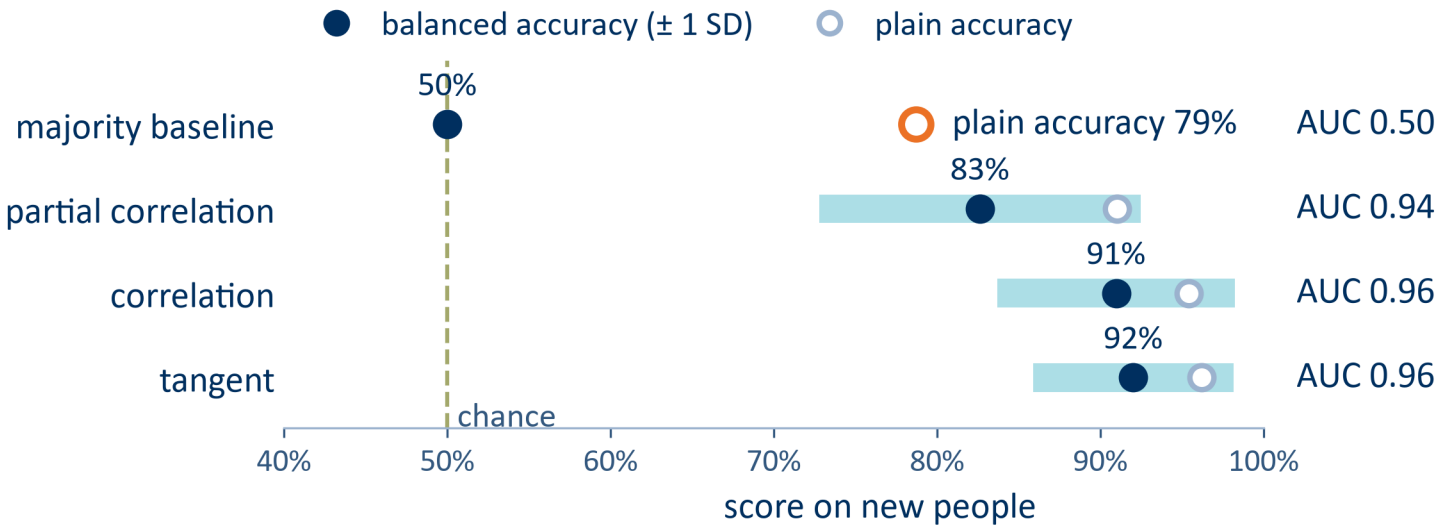
Not: a missing edge means the two regions are not connected.
But: "direct" only among the 39 regions we recorded, and how many edges survive depends on how much data there is.
Do: read a missing edge as "not detected", not as "absent".

Friedman et al. (2008) · Smith et al. (2011) · Figure: own (figures/make_figs_p4.py) · data: ds000228 (CC0) / OSF 5hju4 (CC BY 4.0)

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PREDICT fMRI Notebook §6

Scored fairly, connectomes still tell children from adults: balanced accuracy 91–92%, AUC 0.96, chance 50%



Read: mean over 50 test folds (5-fold split by person × 10 repeats); band = ± 1 SD

Σ DEFINITION · TWO FAIR SCORES

Balanced accuracy: the average of the hit rate in each group. Chance = 50% whatever the group sizes.

AUC: the chance that a random adult gets a higher "adult" score than a random child; 0.5 = chance, 1 = perfect.

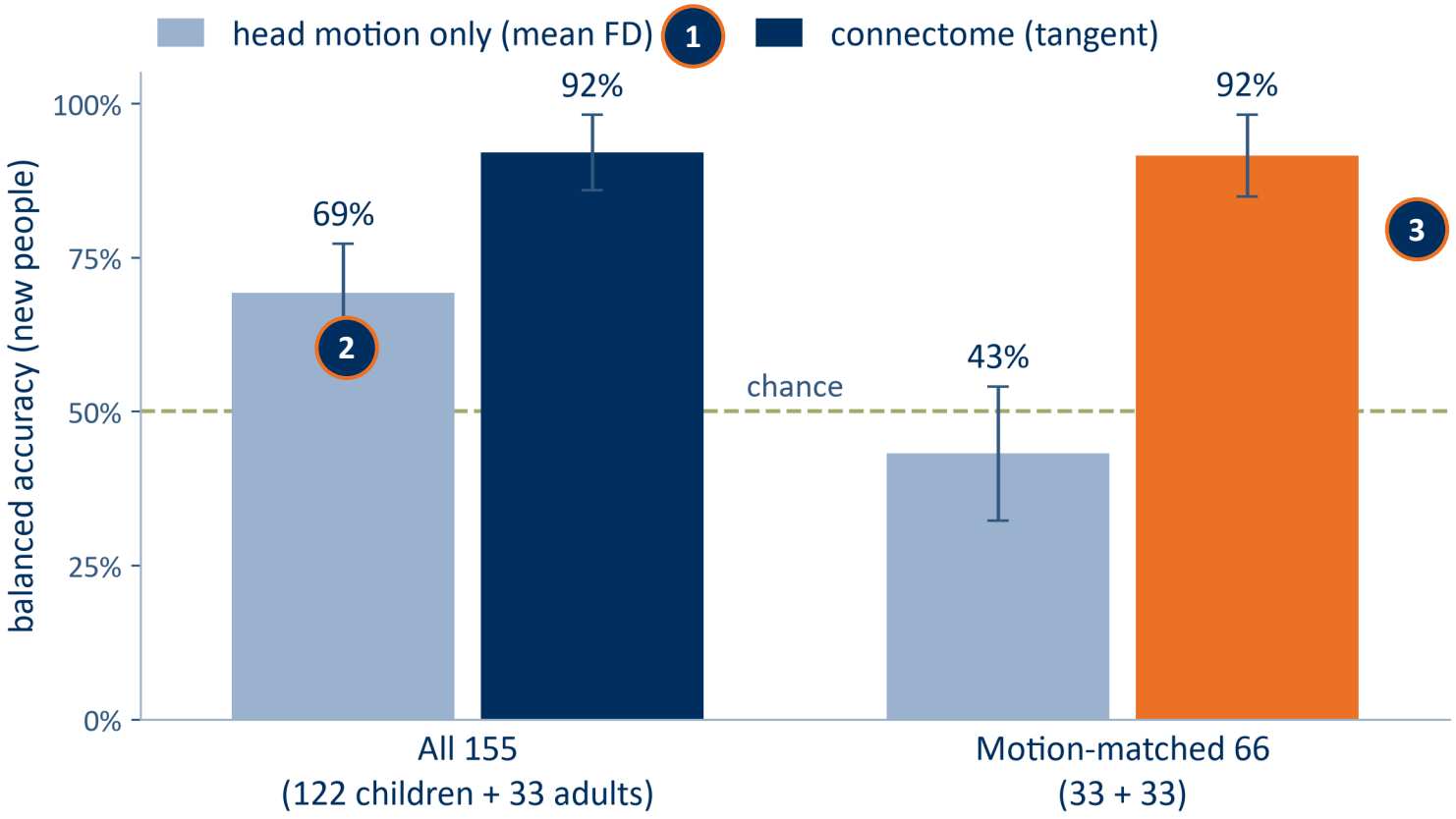
KEY IDEA · TRUST CHECK · METRIC

When groups are unequal, plain accuracy flatters. Report balanced accuracy or AUC, next to the majority baseline.

Dadi et al. (2019) · Poldrack et al. (2020) · Figure: own (figures/make_figs_p4.py) · data: ds000228 (CC0) / OSF 5hju4 (CC BY 4.0)

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Children move twice as much in the scanner, yet the connectome signal survives when motion is matched

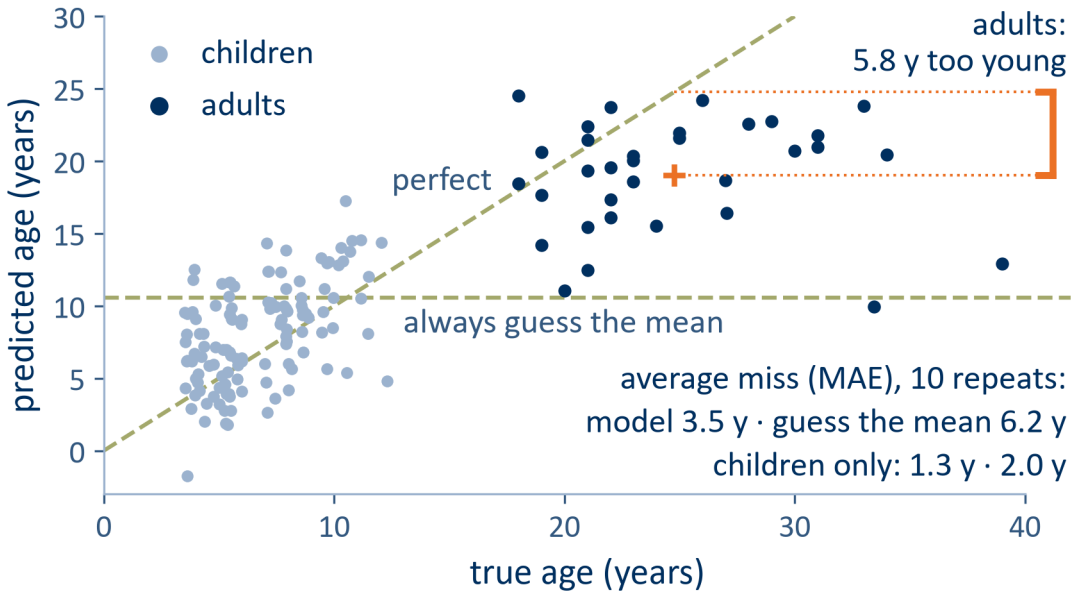


1. **Framewise displacement (FD):** head movement from one image to the next, in mm. Children 0.37 mm on average, adults 0.19 mm
2. Motion alone separates the groups (69% balanced, AUC 0.75), so it is a **confound**: something else that differs between the groups and can carry the label
3. Pair each adult with the child who moved the same amount (0.19 vs 0.19 mm): motion alone no longer predicts (43%); the connectome still does (92%)

Power et al. (2012) · Satterthwaite et al. (2012) · Figure: own (figures/make_figs_p4.py) · data: ds000228 (CC0) / OSF 5hju4 (CC BY 4.0)

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Regression predicts age from the connectome: off by 3.5 years on average, against 6.2 for guessing the mean



Read: points = first repeat, predicted without seeing that person

Σ DEFINITION · RIDGE REGRESSION

Linear regression: logistic regression (slide 26) without the S-curve, so the output is a number:

$$\hat{y} = \mathbf{w} \cdot \mathbf{x} + b$$

$\mathbf{x}$	one person's 741 numbers (slide 49)
$\mathbf{w}, b$	741 weights and a bias, learned
$\hat{y}$	predicted age, in years

Learning: change w and b to lower the squared error $(\text{age} - \hat{y})^2$ on the training people (slide 10). Ridge adds a penalty on large weights (slide 33).

⚠ PITFALL

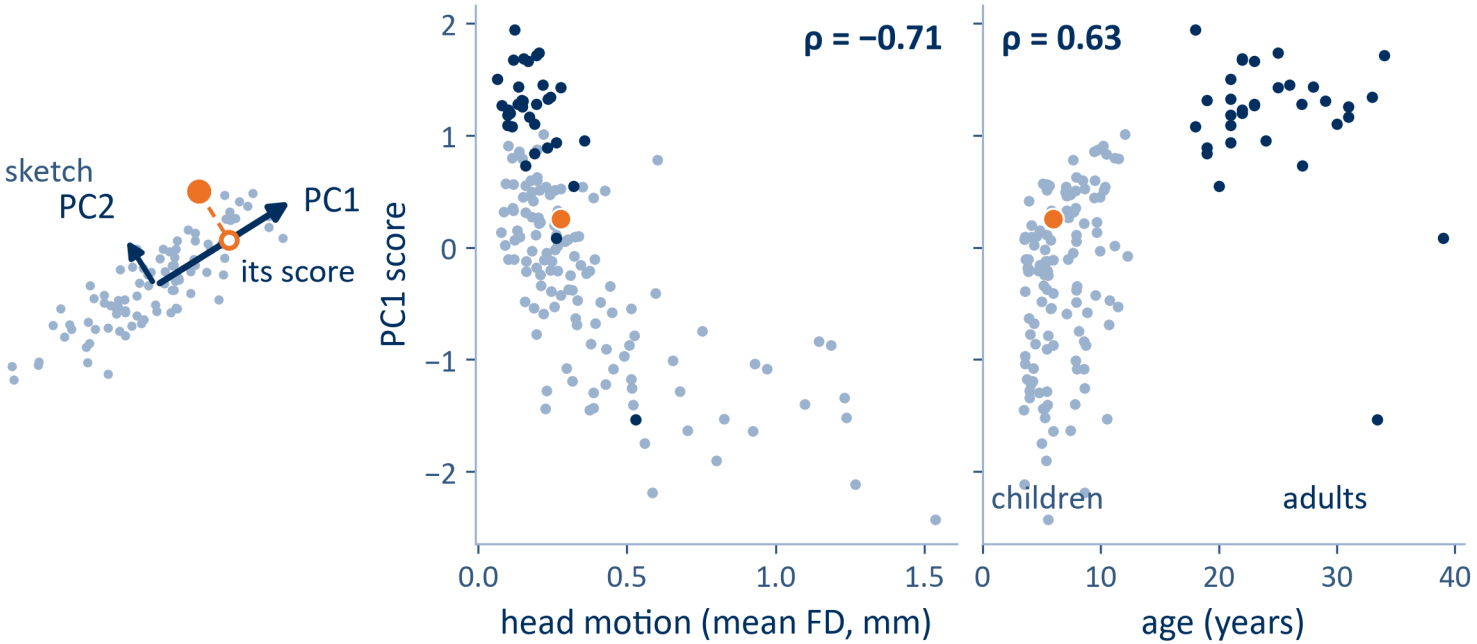
Not: an average miss of 3.5 years means every age is predicted within about 3.5 years.

But: on average adults are predicted 5.8 y too young, children 1.4 y too old (only 33 adults: pulled to the middle).

Do: look at the errors in each group and at the scatter, not only at the average.

Poldrack et al. (2020) · Sabbagh et al. (2020) · Figure: own (figures/make_figs_p4.py) · data: ds000228 (CC0) / OSF 5hju4 (CC BY 4.0)

PCA finds the direction where connectomes differ most: it follows head motion (ρ −0.71) more than age (0.63)



Read: ρ = rank correlation (−1 … 1) · PC1 holds 5% of the variance · its sign is arbitrary

Σ DEFINITION · PCA

PCA: PC1 = the direction in which the people spread most; PC2 = the next, at right angles. No labels used.

$$\mathbf{w}_1 = \arg\max_{\|\mathbf{w}\|=1} \mathrm{Var}(\mathbf{w} \cdot \mathbf{x})$$

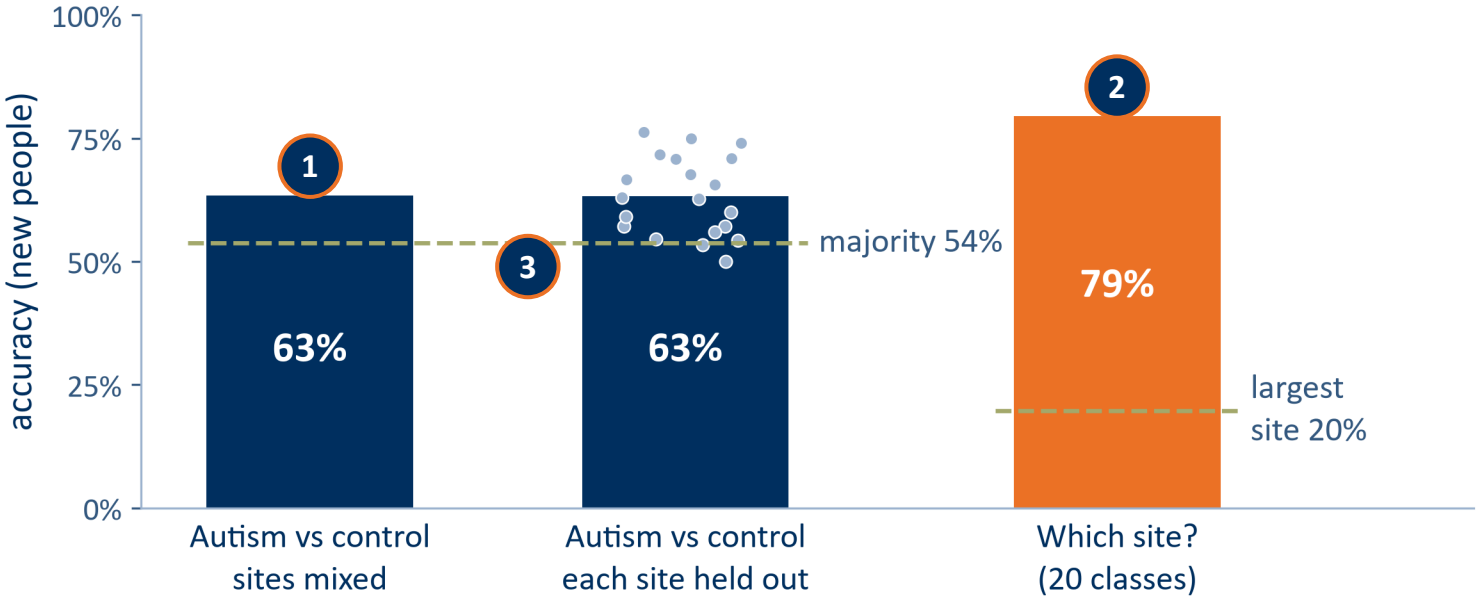
$\mathbf{x}$	a person's 741 numbers
$\mathbf{w}$	a direction (length 1)
$\mathbf{w} \cdot \mathbf{x}$	the person's score
Var	spread over 155 people

EXAMPLE · WITHIN THE 122 CHILDREN Age and motion barely go together (ρ −0.07), yet PC1 still follows motion (−0.71) and age only weakly (0.35).

Pearson (1901) · Power et al. (2012) · Figure: own (figures/make_figs_p4.py) · data: ds000228 (CC0) / OSF 5hju4 (CC BY 4.0)

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Across 20 sites, connectomes give away the scanner site (79% vs 20%) far more than the diagnosis (63% vs 54%)



1. ABIDE: 871 people (403 autistic, 468 controls), 20 sites
2. each scanner leaves a fingerprint in the connectome: the site is recognised 79% of the time
3. held-out sites score 50–76%, and 6 of the 20 do no better than their own majority answer

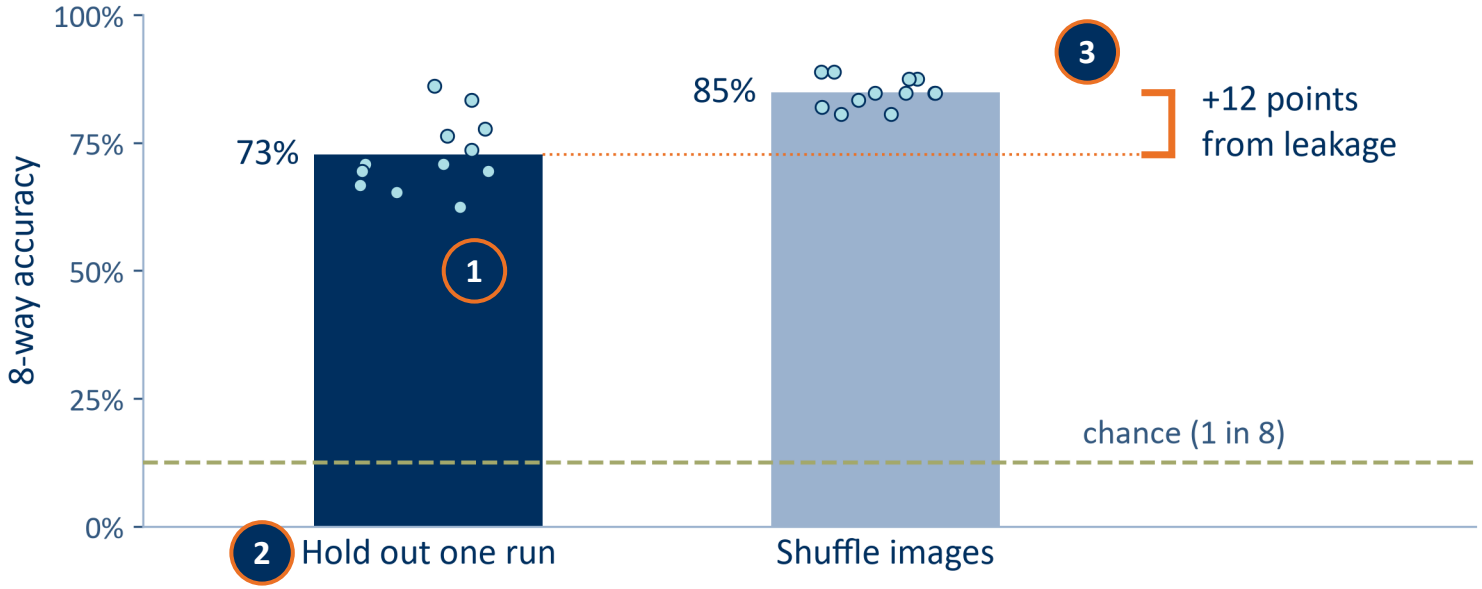
⚠ **PITFALL**

Not: a good score on pooled multi-site data proves a diagnostic marker.
But: the model can learn the scanner, and a new hospital may score far lower.
Do: test on held-out sites and report the spread across them.

Di Martino et al. (2014) · Abraham et al. (2017) · Figure: own (figures/make_figs_p4.py) · data: ABIDE I Preprocessed, PCP (CC BY-NC-SA)

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Within one person, decoding works if whole runs are held out: 73% vs 85% when neighbouring images leak



1. **Decoding:** guess what the person saw from one brain image (one person, 8 categories, 464 voxels)
2. a **run** = one continuous scan of a few minutes; 12 runs, 864 images (one every 2.5 s)
3. BOLD is slow (slide 46): neighbouring images are near-copies, so shuffling puts twins in train and test

KEY IDEA · TRUST CHECK · TIME

Moments close in time are near-copies. Test on separate blocks of time (here, whole runs), and keep filters and z-scoring from reaching into the test block.

Haxby et al. (2001) · Varoquaux et al. (2017) · Figure: own (figures/make_figs_p4.py) · data: Haxby subject 2 via nilearn (CC BY-SA 3.0)

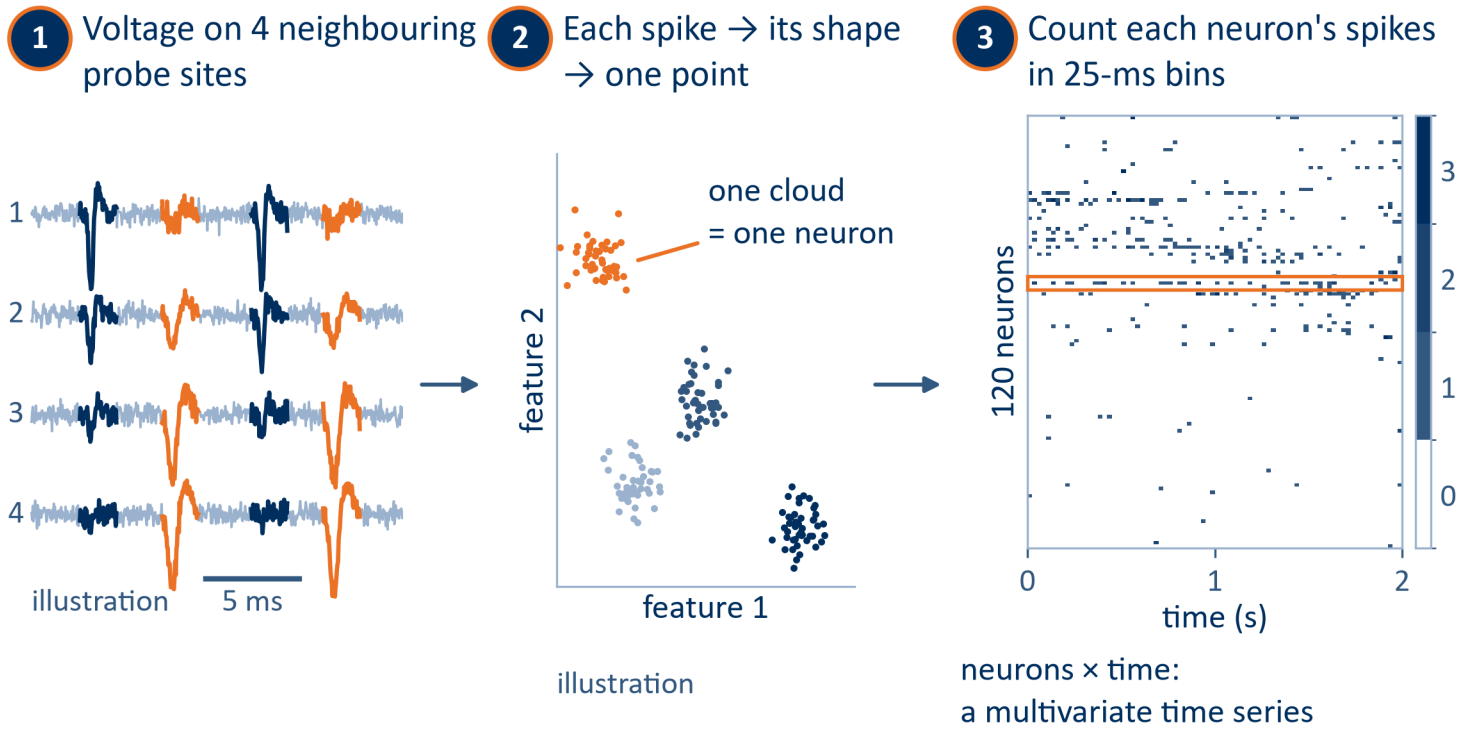
PART 5 OF 5

Spikes: where is the rat?

Can 120 hippocampal neurons tell us where the rat is?

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Spike sorting is clustering: each spike is reduced to one label, its neuron, then counted in 25-ms bins



1. **Probe**: e.g. Neuropixels (slide 4), 384 channels at once, 30,000 samples per second

2. **Sorting = clustering**: no labels, unknown number of neurons; each spike → one neuron (common tool: Kilosort4)

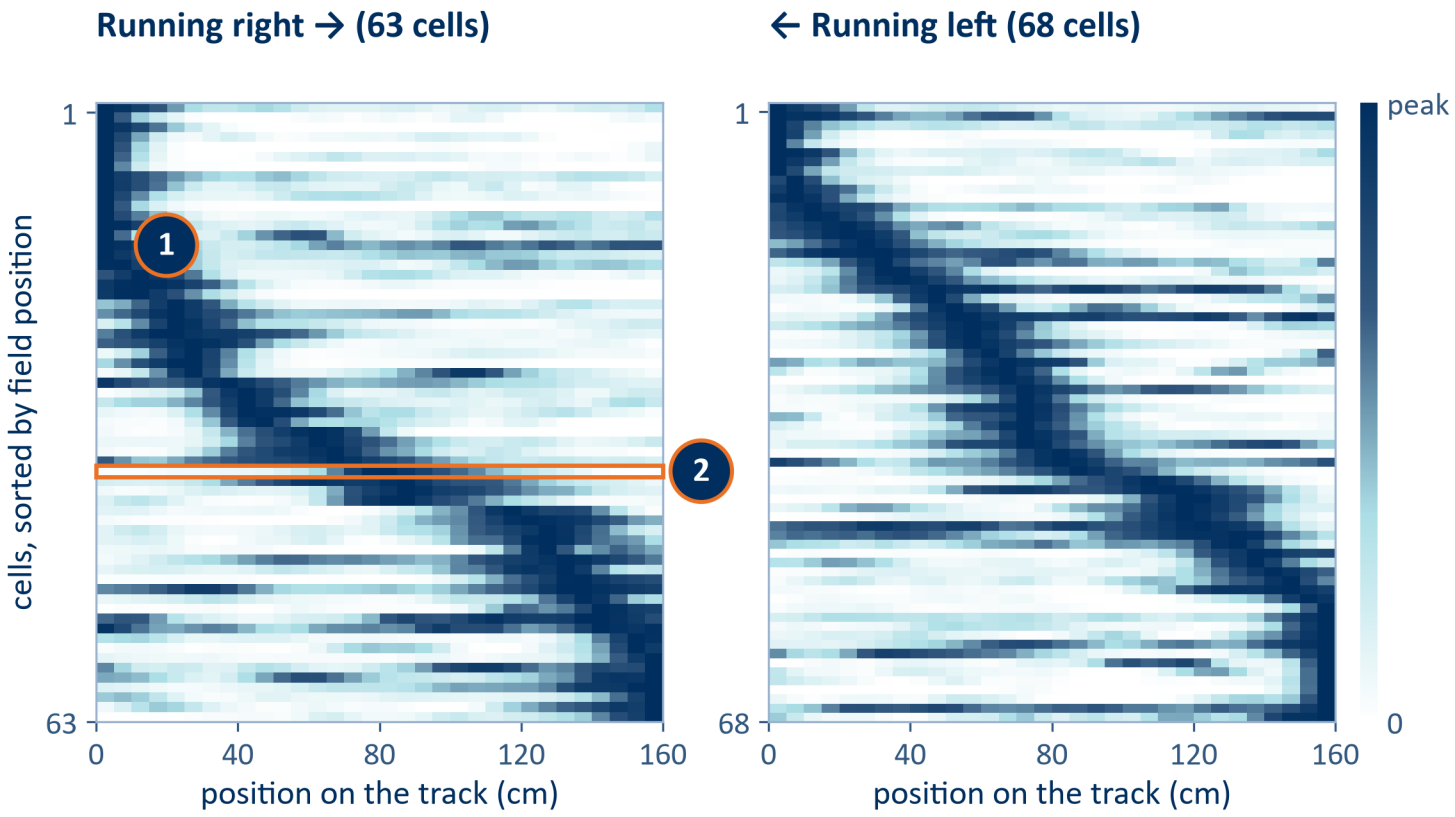
3. **Our matrix**: 120 neurons × 9,563 bins of 25 ms (running only); 96.5% zeros

⚠ **PITFALL · SPIKE SORTING** **Not:** one sorted unit is surely one neuron · **But:** sorting can merge two cells or split one · **Do:** report the sorter and the curation (we use the 120 units the authors released as excitatory cells)

Jun et al. 2017 · Pachitariu et al. 2024 · data: DANDI 000044 (CC BY 4.0) · Figure: own (steps 1–2 simulated)

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Place cells: each hippocampal neuron fires at its own stretch of the 1.6-m track, so together they tile it



Σ DEFINITION · PLACE CELL

A hippocampal neuron that fires mainly when the animal is in one part of space: its place field.

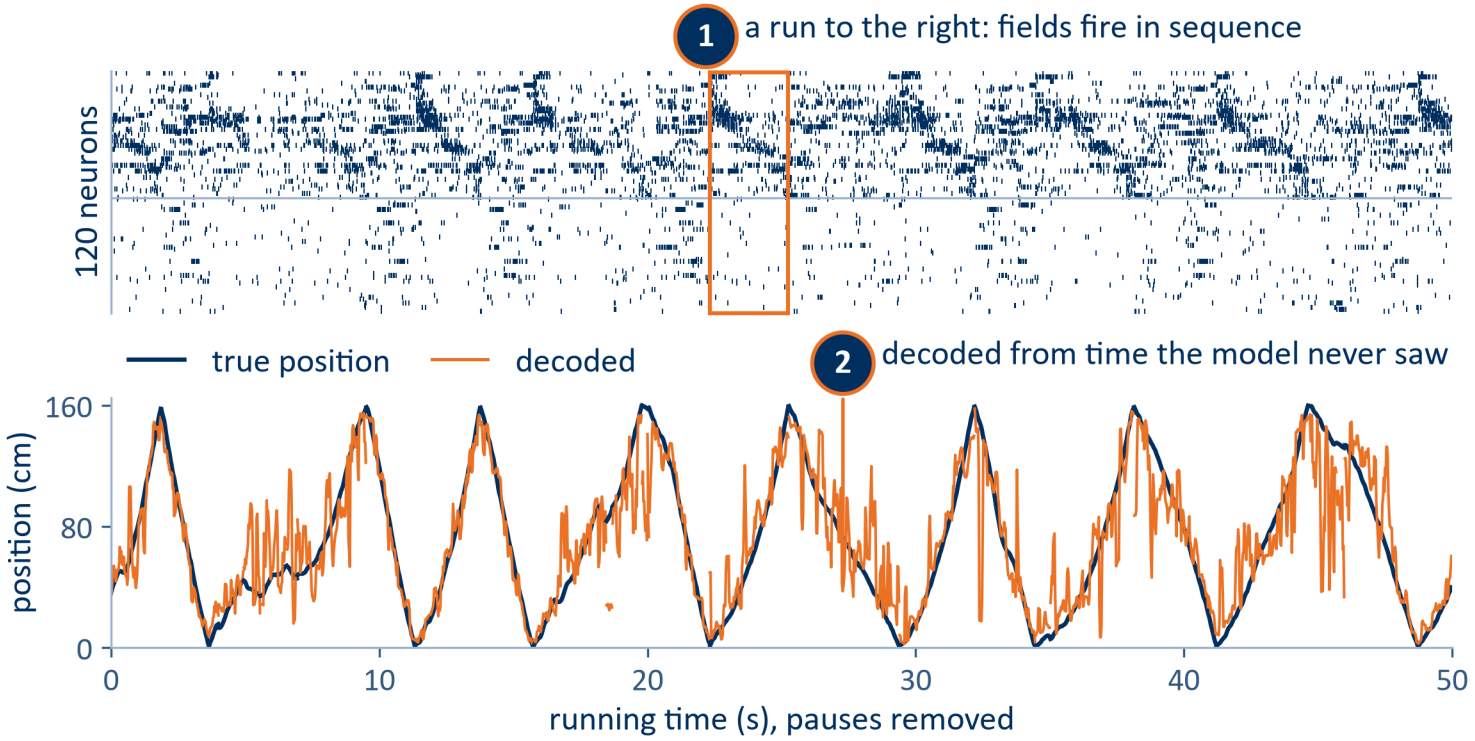
1. Fields tile the whole track
2. One cell (orange): 10 Hz running right, < 1 Hz running left

- The rat runs back and forth on a 1.6-m track: 34 + 39 full laps
- Rate map = spikes ÷ time spent, in 5-cm bins
- 84 of 120 cells have a field; 37 of them in one running direction only

O'Keefe & Dostrovsky 1971 · data: Grosmark & Buzsáki 2016, DANDI 000044 (CC BY 4.0) · Figure: own

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120 neurons place the rat to about 13 cm on unseen stretches of time; shuffled bins inflate R^2 to 0.95



DECODING POSITION, 10 NEIGHBOURS

Tested on	R^2	Error
Unseen time blocks	**0.82**	**13 cm**
Shuffled 25-ms bins	0.95	6 cm
Always the mean	0.00	42 cm

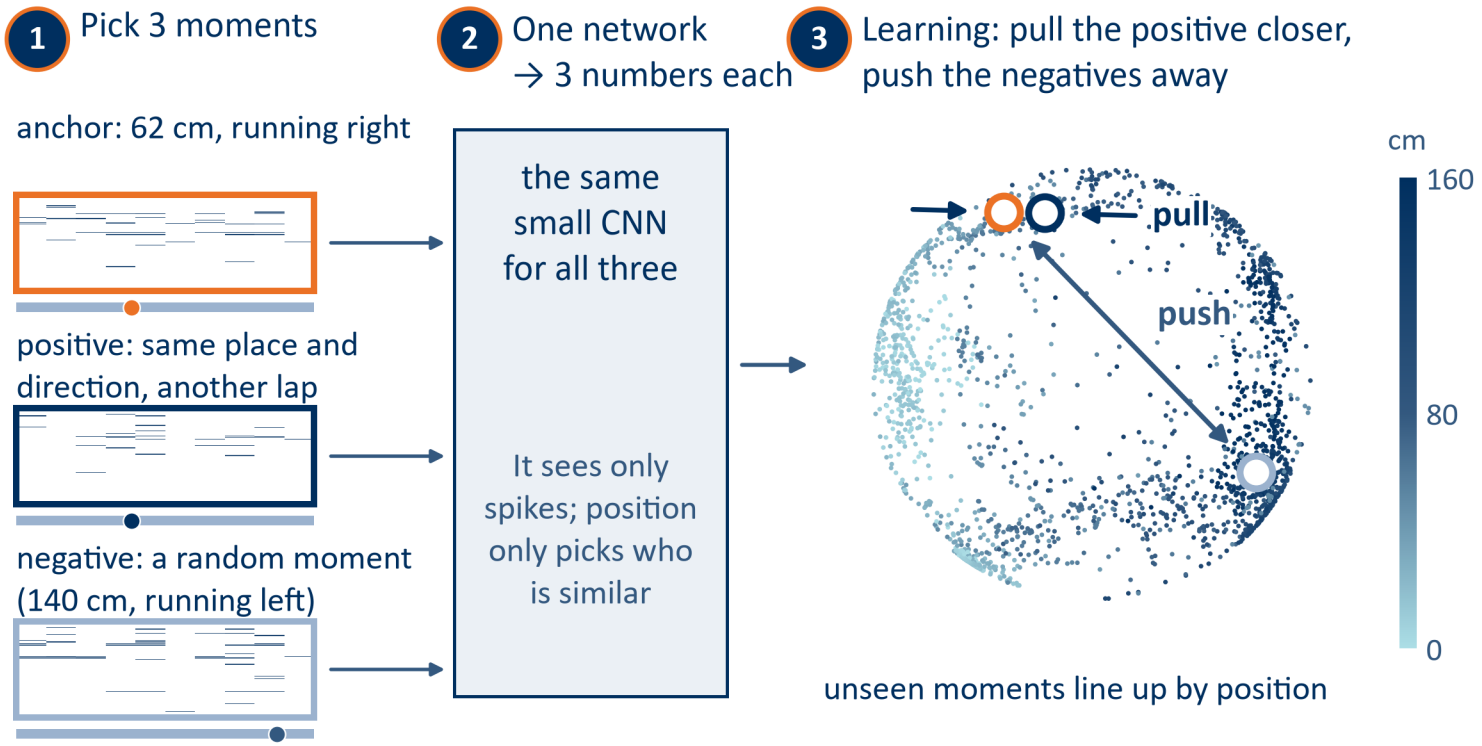
- **Regression: the answer is a number (cm), not a class (slide 9)**
- **kNN**: the 10 most similar training moments vote on the position
- **R^2**: 1 = perfect, 0 = no better than always guessing the mean

⚠ **PITFALL · TIME** **Not: shuffle 25-ms bins into the folds · But: neighbouring bins are near copies (same lap, smoothed counts) · Do: test on contiguous blocks of time, as with fMRI runs (slide 56)**

Glaser et al. 2020 · data: DANDI 000044 (CC BY 4.0) · Figure: own (kNN, 5 time blocks)

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Contrastive learning pulls similar moments together: on unseen time, 3 learned numbers locate the rat better than the raw counts



Σ DEFINITION · CONTRASTIVE LEARNING

Learn by comparison: map each moment to a few numbers so that similar pairs land close and random pairs land far apart.

ONE TEST BLOCK: LAST 20% OF RUNNING

Decoded from	R^2	Error
all 120 neurons	0.73	17 cm
PCA, 3 numbers	0.74	18 cm
contrastive, 3 numbers	**0.87**	**12 cm**
always the mean	−0.01	43 cm

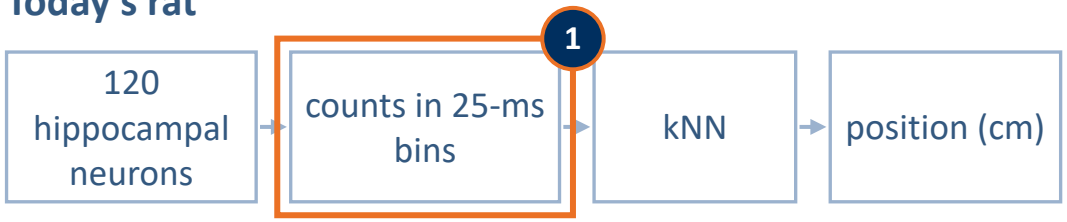
⚠ **PITFALL · SUPERVISED EMBEDDING** **Not:** "the curve is the brain's map" · **But:** here "similar" was defined by position and direction, so the map follows position partly by construction · **Do:** judge it on unseen time, next to a label-free map (PCA)

Schneider et al. 2023 (CEBRA) · Cunningham & Yu 2014 · data: DANDI 000044 · Figure: own

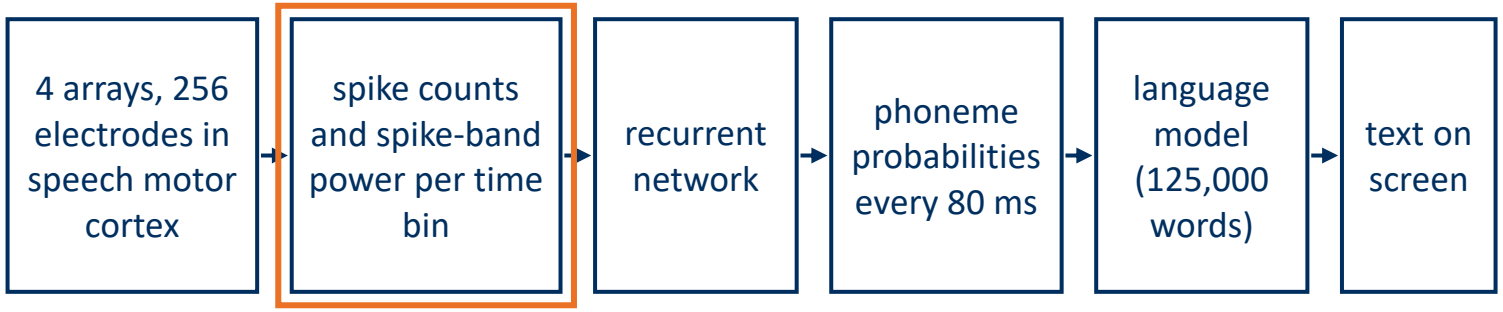
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From rats to patients: decoding speech-motor-cortex spikes gives people with ALS 56–62 words per minute

Today's rat



Speech BCI (e.g. Card et al. 2024)



Study	Result
Willett 2023	62 words/min; 24% word errors with 125,000 words
Card 2024	97.5% of words correct over 8.4 months
Card 2026	~2 years of home use (> 3,800 h); 56 words/min

Natural conversation: about 160 words/min.

1 the same kind of time series: binned spiking activity → a model → a readout

Willett et al. 2023 Nature · Card et al. 2024 NEJM · Card et al. 2026 Nat Med · Figure: own schematic

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HANDS-ON · NOTEBOOK §0–§8

Try it: EEG and fMRI, and how much the evaluation matters

- **Work in pairs**, one laptop. Scan the QR code or open the link (sign in to Google), then run §0 now.
- Before each result: guess, then run. EEG: §1–§5 (Q1) · fMRI: §6–§7 (Q2).
- When I say ***jump***, go to §6 (fMRI).
- We compare your table (§8) together at the end.

github.com/gon-uri/ITBA2026 · notebook

Q1 · Q2 · O4

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Notebook §2–§7 ← SPLIT · slide 29

What the notebook showed: how we evaluated moved the scores more than which features we used

§	Task	What you ran	What you should have seen	Trust check
§2.2	PREDICT	EEG: eyes open or closed, new people: 1 alpha feature → full spectrum (2,560 numbers)	64% → 75% (always the most common answer: 50%)	BASELINE
§2.4	PREDICT	Same model, windows shuffled across people	**91%** (+16 points)	SPLIT
§3	REPRESENT	Map and 2 clusters of the 1,200 windows	islands = people (0.32 vs 0.05 by eyes); 2 clusters ignore the eyes (ARI 0.00)	SPLIT
§4	FEATURES (graph)	Alpha envelope-correlation graph, eyes open vs closed	0.43 vs 0.41, p = 0.60; strongest edges 3.8 cm apart	CONFOUND
§6	PREDICT	fMRI: child or adult from the connectome	“always child” 79% = 50% balanced; connectome 92% balanced	METRIC
§7	PREDICT	fMRI: age from the connectome	off by 3.5 years on average (6.2 for the mean age)	BASELINE

Also in your table: per person 50–100% (UNCERTAINTY) · each person z-scored with all of their windows 92%, impossible for one new patient (SPLIT) · forehead only 75%, without alpha 71% (CONFOUND) · head motion alone 69% balanced, 43% once matched, connectome still 92% (CONFOUND)

data: PhysioNet EEGBCI (Schalk et al. 2004), ODC-By · Richardson et al. 2018 fMRI, CC BY 4.0 · numbers: own runs

Two tasks on three kinds of brain data: each number means something only next to its reference point

	PREDICT classify or regress	**REPRESENT** embed or cluster (compress)
EEG/MEG Q1 · eyes closed?	**75%** eyes open/closed on new people chance 50% · leaky split 91%	**0.32 vs 0.05** windows group by person, not eyes sleep clusters vs stages: ARI 0.26
fMRI Q2 · child or adult?	**92%** child/adult (balanced, connectome features) **3.5 y** age error vs 6.2 y guessing the mean	**ρ −0.71** PCA axis follows head motion age only 0.63
SPIKES Q3 · where is the rat?	**R^2 0.82** rat position on unseen time shuffled bins 0.95	**0.87 vs 0.73** contrastive 3 numbers vs 120 neurons PCA 0.74

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Six trust checks, and today each one changed a number we would otherwise have believed

1 · BASELINE

Does it beat a trivial model?

Eyes: always the same answer 50%; one alpha number 64%; the full spectrum 75%.

slide 25 · notebook §2

2 · SPLIT

Is the test unit the one you will meet in use?

Eyes model: 75% on new people, 91% when their own windows leak into training.

slide 29 · notebook §2

3 · CONFOUND

What else could make this pattern?

Three forehead electrodes match all 64 (75% vs 75%): likely the eyes, not visual cortex.

slides 75 · 40 · 66

4 · UNCERTAINTY

How many units, and how wide is the interval?

With 20 examples, pure noise scores 70% or more in one run in ten.

slide 44

5 · METRIC

Does the score hide a group?

“Always child”: 79% accuracy but 50% balanced accuracy; the connectome reaches 92%.

slide 51

6 · TIME

Did the future, or a near twin, reach training?

Shuffling neighbouring images or bins: fMRI 73% → 85%, rat R^2 0.82 → 0.95.

slides 56 · 74

Varoquaux 2018 · Poldrack, Huckins & Varoquaux 2020 · Brookshire et al. 2024

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What to remember

O1 A brain recording is channels × time: order, sampling rate, rhythms and artefacts carry, or fake, the information.

O2 A model learns by tuning its parameters to lower a loss; judge it on unseen data. Name the task: predict or represent.

O3 Start simple: features or covariances + a linear model. CNNs, contrastive and foundation models learn their own representation, given more data.

O4 No number alone: six trust checks. In the notebook a leaky split added 16 points (75% → 91%); better features only about 10 (64% → 75%).

TAKE IT HOME

github.com/gon-uri/ITBA2026

Resources page

These slides (PDF)

Repository: github.com/gon-uri/ITBA2026 (code MIT; data files keep their licences)

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References

START HERE (TEXTBOOK, REVIEWS, TUTORIALS)

James, Witten, Hastie, Tibshirani & Taylor (2023) An Introduction to Statistical Learning with Applications in Python, Springer, free PDF www.statlearning.com · doi:10.1007/978-3-031-38747-0

Glaser et al. (2020) eNeuro 7:ENEURO.0506-19.2020 · doi:10.1523/ENEURO.0506-19.2020

Cunningham & Yu (2014) Nat Neurosci 17:1500–1509 · doi:10.1038/nn.3776

Lotte et al. (2018) J Neural Eng 15:031005 · doi:10.1088/1741-2552/aab2f2

Bastos & Schoffelen (2016) Front Syst Neurosci 9:175 · doi:10.3389/fnsys.2015.00175

Varoquaux (2018) NeuroImage 180:68–77 · doi:10.1016/j.neuroimage.2017.06.061

Poldrack, Huckins & Varoquaux (2020) JAMA Psychiatry 77:534–540 · doi:10.1001/jamapsychiatry.2019.3671

Brookshire et al. (2024) Front Neurosci 18:1373515 · doi:10.3389/fnins.2024.1373515

DATA USED TODAY

Schalk et al. (2004) IEEE TBME 51:1034–1043 + PhysioNet (Goldberger et al. 2000) · EEG eyes open/closed · doi:10.13026/C28G6P

Kemp et al. (2000) IEEE TBME 47:1185–1194 · Sleep-EDF · doi:10.13026/C2X676

Bagnall et al. (2018) UEA archive, SelfRegulationSCP1 · arXiv:1811.00075

Richardson et al. (2018) Nat Commun 9:1027 · children and adults fMRI · doi:10.1038/s41467-018-03399-2

Haxby et al. (2001) Science 293:2425–2430 · doi:10.1126/science.1063736

Di Martino et al. (2014) Mol Psychiatry 19:659–667 · ABIDE · doi:10.1038/mp.2013.78

Grosmark & Buzsáki (2016) Science 351:1440–1443, DANDI 000044 · rat place cells · doi:10.48324/dandi.000044/0.261005.2101

OWN WORK

Uribarri et al. (2024) DMKD 38:3922–3947 · doi:10.1007/s10618-024-01062-7

Solana, Fransén & Uribarri (2025) AALTD 2024, LNCS pp. 96–114 · doi:10.1007/978-3-031-77066-1_6

Uribarri*, Ekman von Huth* et al. (2026) Int J Med Inform 220:106626 · doi:10.1016/j.ijmedinf.2026.106626

Every source cited on a slide: appendix of this PDF · links and extras: github.com/gon-uri/ITBA2026

Questions?

Gonzalo Uribarri · Stockholm University (DSV)

Slides, notebook and data: github.com/gon-uri/ITBA2026

@ gonzalo.uribarri@dsv.su.se 𝕏 @gonzauri

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Instituto Tecnológico de Buenos Aires

APPENDIX · NOT PRESENTED

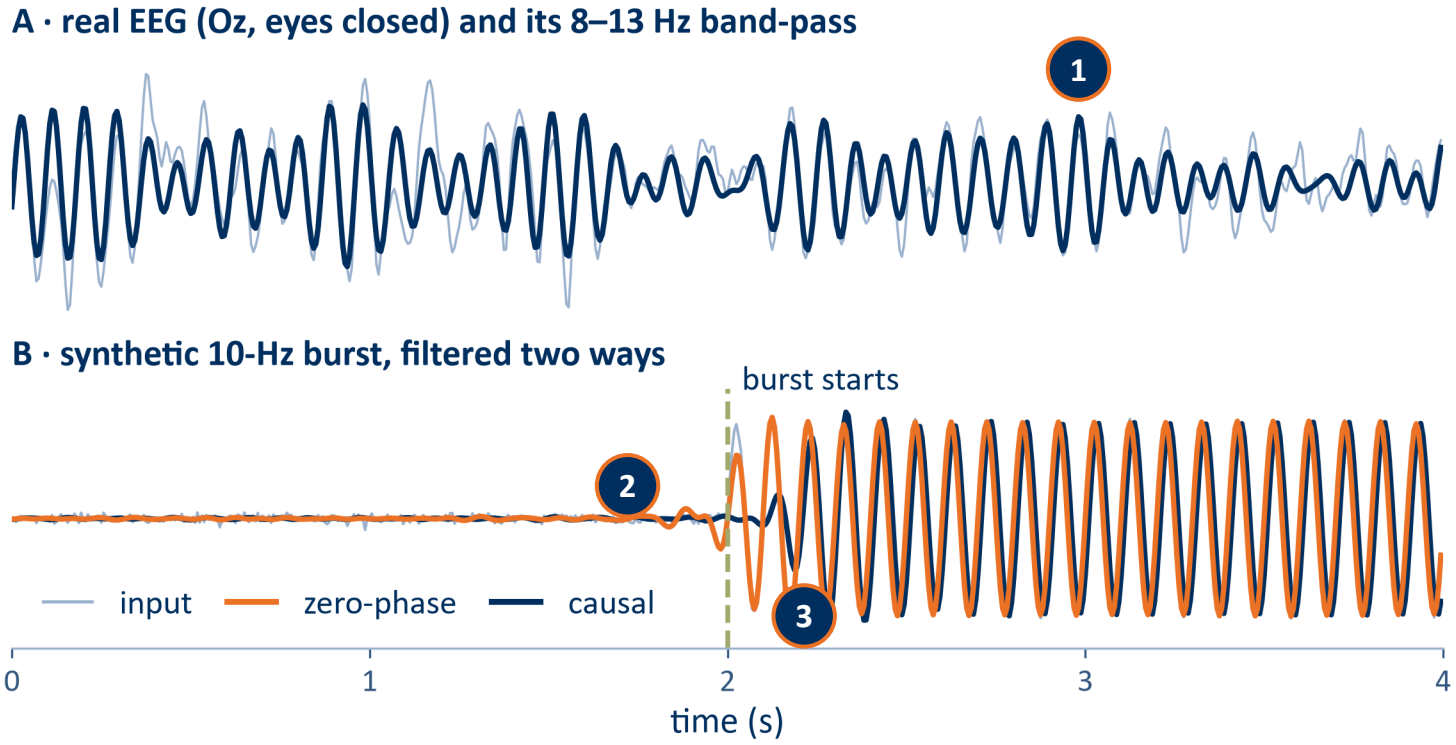
Backup slides and full references

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→ slide 40 · learned filters

Notebook §1

A filter slides a short set of weights along the signal: it keeps some frequencies and removes others



How it works: slide a short set of weights along the signal; each output is a weighted sum of nearby samples: a **convolution**.

1. **Band-pass 8–13 Hz**: keeps alpha, removes the rest.
2. **Zero-phase**: no delay, but can start before the event.
3. **Causal**: past only, so it lags.

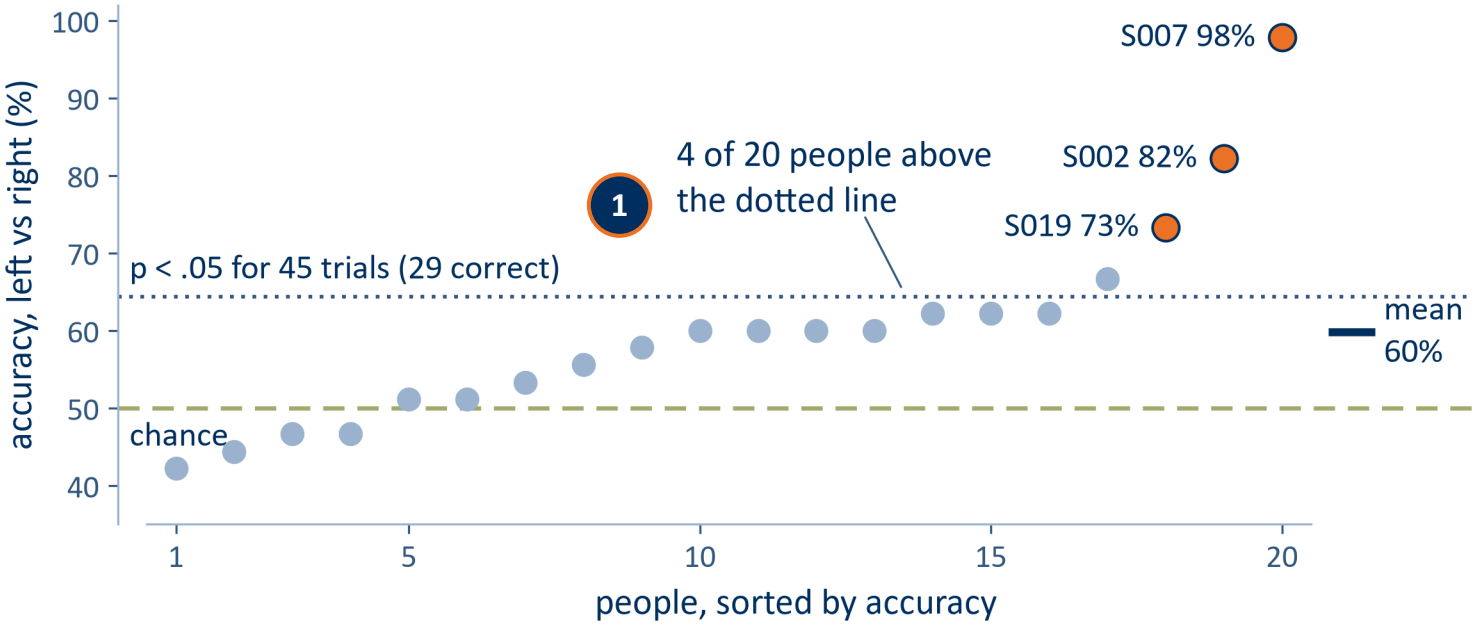
A CNN learns these weights (slide 40).

⚠ **PITFALL** **Not:** filtering is harmless cleaning. **But:** a zero-phase filter lets later samples shape earlier ones. **Do:** filter causally when a model must run in real time (e.g. a BCI).

MNE filtering tutorial (Gramfort et al. 2013) · Data: PhysioNet EEGBCI, ODC-By 1.0 · Figure: own (figures/make_figs_p1.py)

PREDICT EEG/MEG

Spatial filters find the rhythm that imagined movements change, but only 3 of our 20 people reach 70%



1 Imagining a hand movement weakens the 8–30 Hz (mu/beta) rhythm over the opposite motor cortex.

2 **CSP** (common spatial patterns) learns weighted sums of channels (“virtual electrodes”) whose power differs most between left and right.

3 **Their log-power goes to LDA (linear discriminant analysis), a linear readout as on slide 26.**

EEGBCI runs 4/8/12 · 45 trials per person · 5-fold CV within each person · CSP fitted in each fold

Both CSP and the next slide start from the **covariance matrix** of the channels.

⚠ **PITFALL** **Not:** “motor-imagery BCIs work for everyone” · **But:** many people produce no separable rhythm (“BCI inefficiency”) · **Do:** report every person, not only the mean.

Ramoser et al. (2000) · Blankertz et al. (2008) IEEE SPM · Figure: own · data: EEGBCI (Schalk 2004), ODC-By

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PREDICT EEG/MEG FMRI

Method card: covariance → tangent space → linear model

Tangent space Covariance → tangent space → linear model

Barachant et al. 2012, 2013

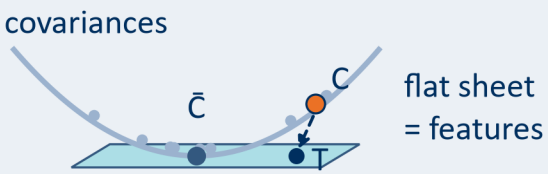
ASKS

Which class does this trial belong to, judged only by how its channels rise and fall together?

HOW IT WORKS

1 Covariance per trial: how each channel pair co-varies
2 Reference = average of the training trials
3 Flatten each matrix at the reference
4 → logistic regression (slide 26)

KEY SKETCH



Flattened at the training average, the curved space lets a straight line work.

READING THE OUTPUT

A class probability per trial. Each weight belongs to a pair of channels. Weights are not brain maps: convert them to patterns before interpreting (Haufe et al. 2014).

⚠ KNOBS & PITFALLS

Fit the reference and scaler inside the CV loop, or the test set leaks in. Short trials or many channels: use shrinkage (pull towards a simpler matrix). New people shift the matrices: re-centre each.

USE IT WHEN / PREFER INSTEAD

Few trials; rhythms that change in strength (motor imagery, SSVEP); fMRI connectomes (slide 49). Prefer ROCKET when the information is in the waveform shape, a CNN or foundation model with plenty of labels.

Congedo, Barachant & Bhatia (2017) · Barachant et al. (2012) IEEE TBME; (2013) · Try: Notebook §B4

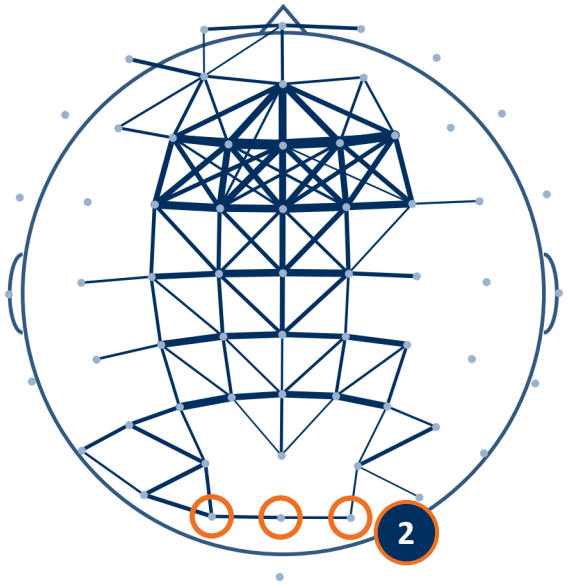
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Electrodes become nodes and co-varying alpha draws the edges, but closing the eyes gives no consistent change

Eyes open · 86 edges



Eyes closed · 116 edges 3



1. **How**: filter 8–12 Hz, take the envelope (how strong alpha is at each moment), correlate every pair of electrodes (+1 = rise and fall together, 0 = unrelated): 2,016 edges
2. Alpha power at the back: 6× higher with eyes closed (25 → 158 $\mu V^2/Hz$)
3. Yet the average edge: 0.43 vs 0.41, higher with eyes closed in only 10 of 20 people

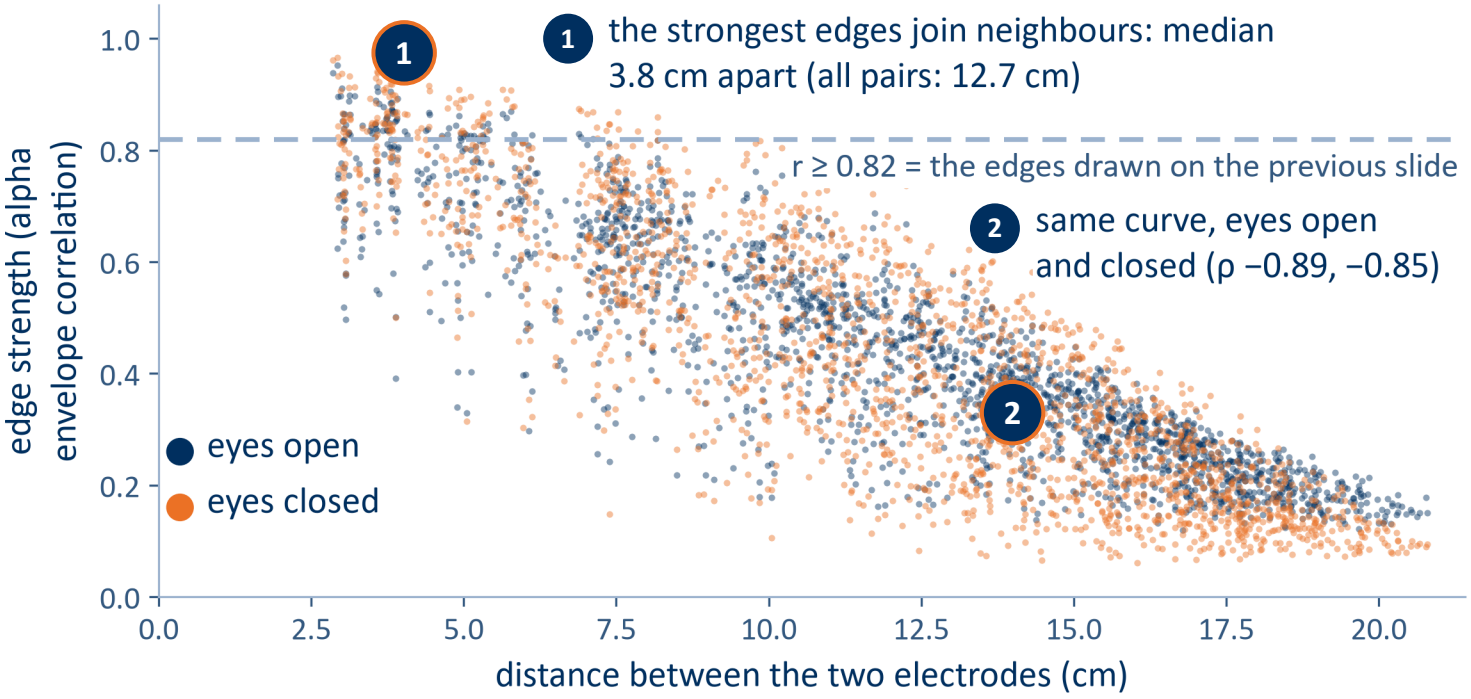
Read: one shared line, r ≥ 0.82 · 81 edges appear in both panels · O1/Oz/O2 ringed

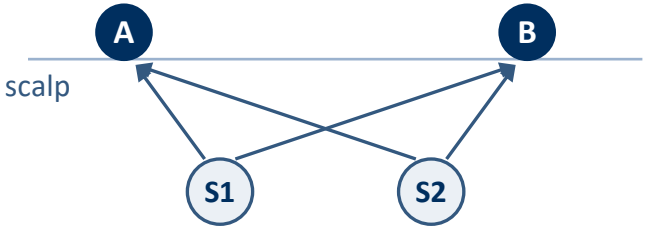
Σ DEFINITION · FUNCTIONAL CONNECTIVITY

Statistical dependence between two recorded signals; by itself it says nothing about wiring.

Hipp et al. 2012 · Figure: own · data: PhysioNet EEGBCI (Schalk et al. 2004), ODC-By 1.0

Neighbouring electrodes record the same sources, so the strongest edges may be physics, not brain coupling





two sources; the head mixes them

- Toy: two independent sources mixed into two electrodes: coherence **0.87**, imaginary coherence **0.01**
- A real coupling with a ≈ 25-ms delay, no mixing: 0.53 and 0.51
- Fixes: lag-based measures (imaginary coherence, wPLI) or source space (helps, does not remove it)

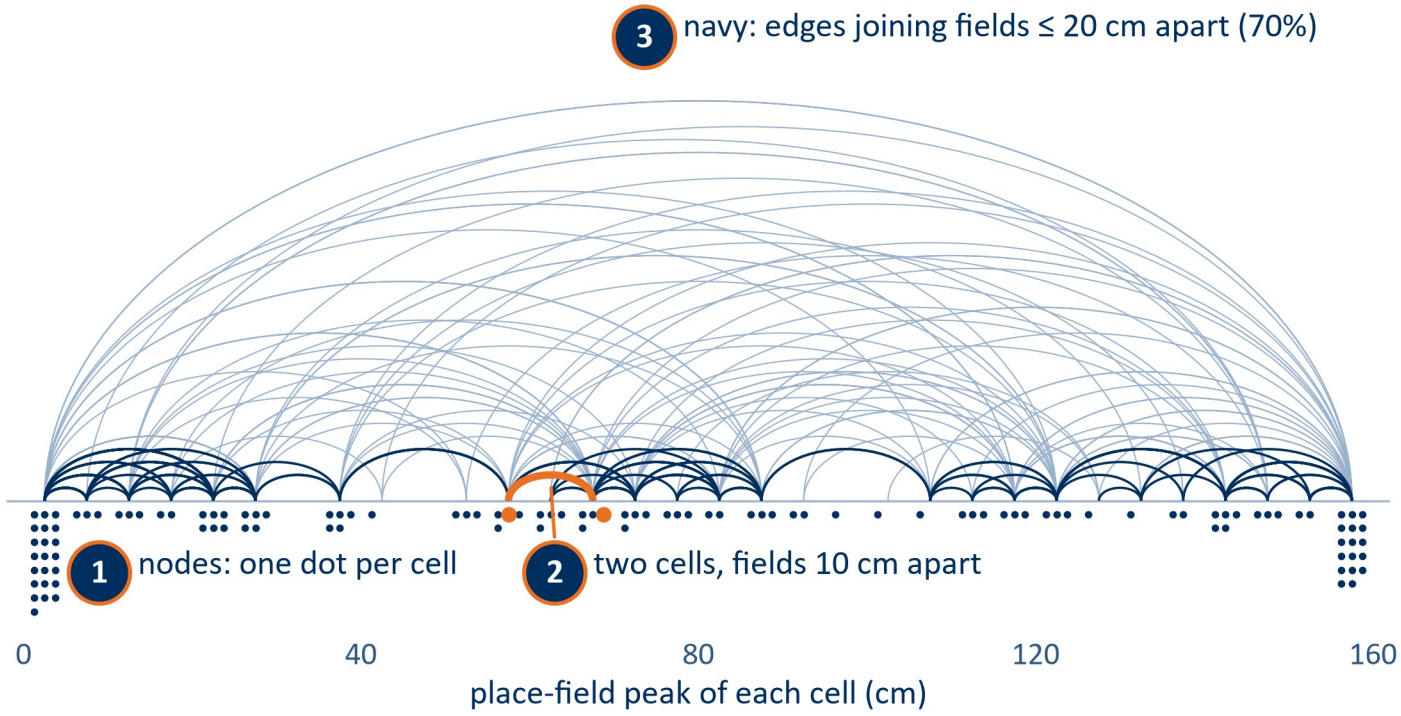
KEY IDEA · TRUST CHECK · CONFOUND

Before believing a pattern, ask what else could produce it: physics, eyes, muscles, motion, site.

Nolte et al. 2004 · Vinck et al. 2011 · Palva et al. 2018 · Figure: own · data: PhysioNet EEGBCI, ODC-By 1.0

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Neurons with nearby place fields fire together: co-activity edges join fields 10 cm apart, random pairs 60 cm



1. **Nodes**: 119 cells active while running
2. **Edges**: correlation of two cells' counts over time; keep the strongest 5% (351 edges)
3. Edges join fields a median 10 cm apart; random pairs 60 cm (random sets of 351 pairs: 50–65 cm)

⚠ **PITFALL · CO-ACTIVITY** **Not: an edge is a synapse · But: two cells co-fire because the rat passes through both fields, a common input as on slide 50 · Do: call it co-activity; a connection needs millisecond-scale evidence (cross-correlograms)**

Perkel et al. 1967 · English et al. 2017 · data: DANDI 000044 (CC BY 4.0) · Figure: own · Try: Notebook §B1.5

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With k = 8, wake splits into three clusters, but agreement with the experts rises only to ARI 0.30

	k = 5	k = 8
Agreement with the scorer, ARI (0 = chance)	0.26	0.30
NMI · purity	0.42 · 64%	0.42 · 72%
Silhouette (higher = tighter groups)	0.32	0.25
Clusters dominated by wake	2	3
Largest share of N3 in one cluster	100%	91%
Largest share of N1 in one cluster	50%	33%

More prototypes split wake further; they do not find N1. The choice of k changes the story, not the data.

Hubert & Arabie 1985 · Rousseeuw 1987 · data: Sleep-EDF (Kemp et al. 2000), ODC-By 1.0 · own run

Every decoder gains from shuffled time bins, and the one that looks up neighbours gains most

Decoder	5 time blocks	Shuffled bins
Position, kNN (10): R^2 (mean error)	0.82 (12.9 cm)	0.95 (6.4 cm)
Position, ridge: R^2 (mean error)	0.80 (16.9 cm)	0.85 (14.7 cm)
Position, always the mean	≈ 0 (42.4 cm)	≈ 0 (42.3 cm)
Direction, logistic regression (balanced)	92%	97%

Direction: majority rate 55%; chance 50%.

Per-block R^2 for kNN on the 5 time blocks: 0.85 · 0.90 · 0.77 · 0.85 · 0.73.

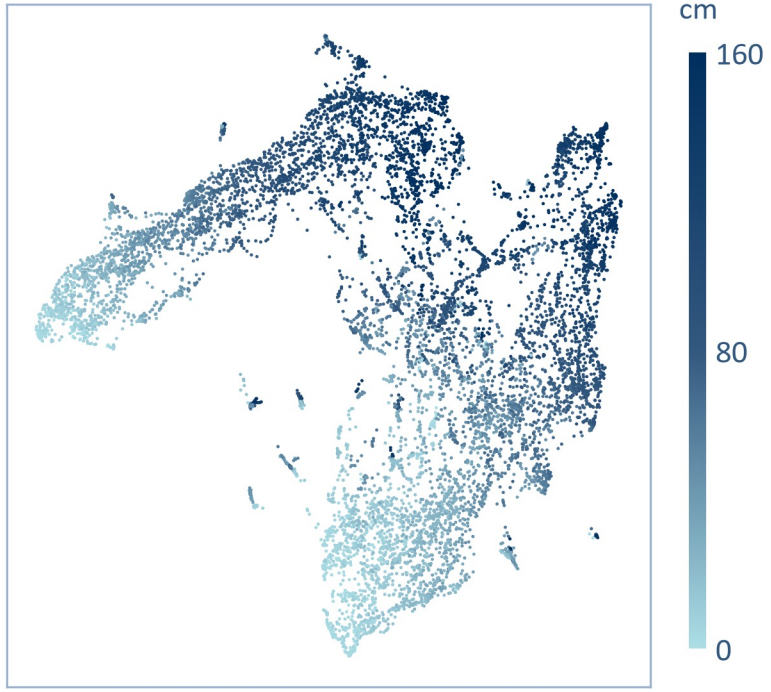
Glaser et al. 2020 · data: DANDI 000044 (CC BY 4.0) · own run (results_spikes.json)

Without labels, PCA and UMAP also order the moments by position; distances between branches mean nothing

PCA (no labels): PC1 vs PC2



UMAP (no labels)

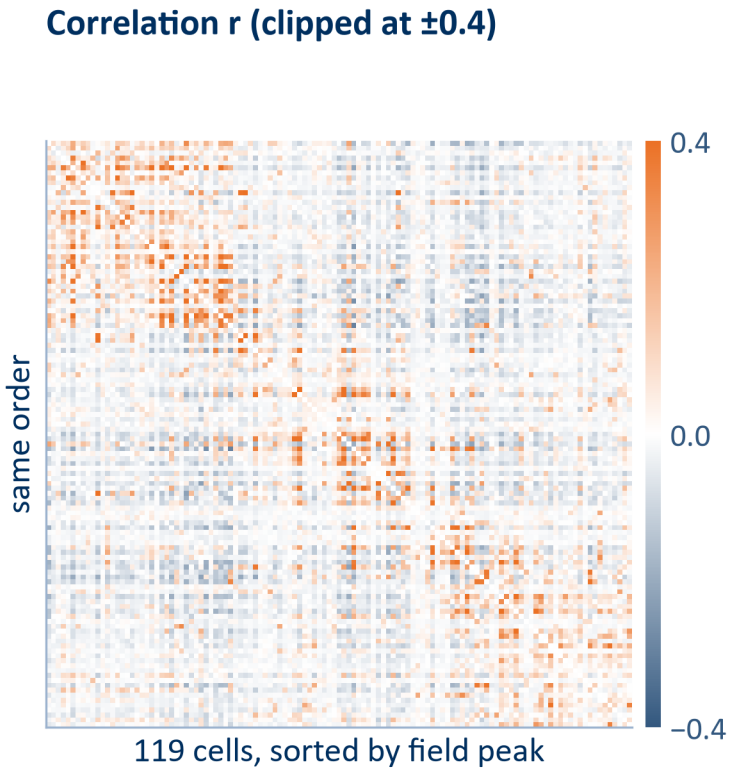


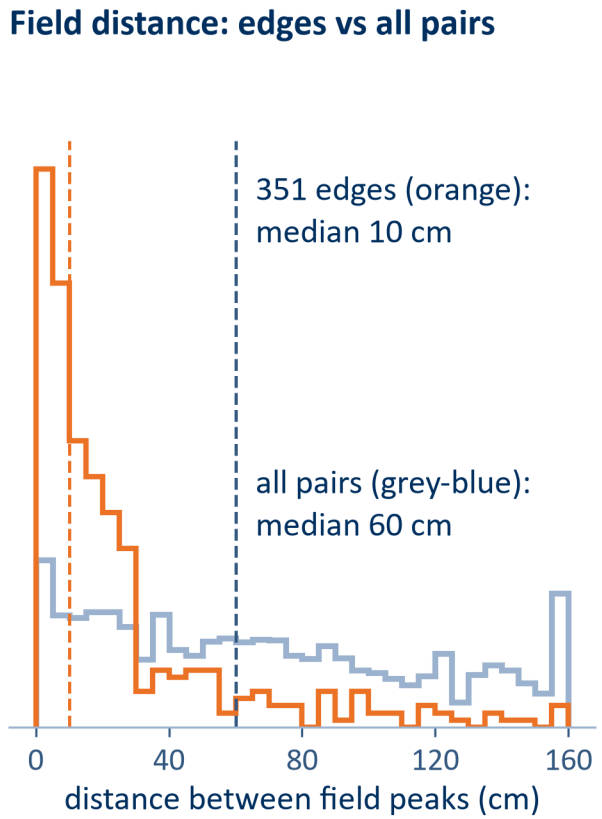
- 3 PCA numbers hold 13.5% of the variance, yet decode position as well as all 120 cells (R^2 0.82 vs 0.82)
- UMAP keeps neighbourhoods only

CEBRA settings (slide 61): offset10 model, 3 numbers, 256 anchors per step, 1,500 steps, cosine, labels = position + direction

McInnes et al. 2018 · Cunningham & Yu 2014 · data: DANDI 000044 (CC BY 4.0) · Figure: own

The co-activity graph holds with place cells only, and the sorted correlation matrix shows it directly





- 71 place cells only: 125 edges, again 10 vs 60 cm
- Over all pairs, correlation falls with field distance (Spearman −0.27)
- Left: cells sorted by field peak; neighbours on the track co-fire near the diagonal

data: DANDI 000044 (CC BY 4.0) · own run (results_spikes.json) · Figure: own

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Brain-to-text decoders in 2023–2026: invasive recordings are far ahead of non-invasive ones

Study	Recording	People	Result
Willett et al. 2023	intracortical arrays, speech motor cortex	1 (ALS)	62 words/min; 23.8% word errors, 125,000 words
Metzger et al. 2023	ECoG, 253 electrodes	1 (brainstem stroke)	78 words/min; 25% word errors, 1,024-word set
Card et al. 2024	4 arrays, 256 electrodes	1 (ALS)	97.5% words correct over 8.4 months
Card et al. 2026	same implant, at home	1	~2 years, > 3,800 h, 56 words/min
Jude et al. 2026	intracortical, attempted finger movements	2	110 characters/min (22 words/min), 1.6% errors
Lévy et al. 2026	MEG vs EEG while typing	35	29% vs 65% character errors

Vocabulary sizes differ (Metzger's 1,024-word set vs 125,000 words), so do not rank the rows.

Willett et al. 2023 · Metzger et al. 2023 · Card et al. 2024; 2026 · Jude et al. 2026 · Lévy et al. 2026

OPTIONAL · NOT PRESENTED

Full references R.1–R.14 (1/12)

OWN RESEARCH

R.1 Uribarri G, Barone F, Ansuini A & Fransén E (2024). Detach-ROCKET: sequential feature selection for time series classification with random convolutional kernels. Data Mining and Knowledge Discovery 38(6):3922–3947. doi:10.1007/s10618-024-01062-7. CC BY 4.0. Code: github.com/gon-uri/detach_rocket.

R.2 Solana A, Fransén E & Uribarri G (2025). Classification of raw MEG/EEG data with Detach-Rocket Ensemble: an improved ROCKET algorithm for multivariate time series analysis. In: Advanced Analytics and Learning on Temporal Data (AALTD 2024, ECML PKDD workshop), LNCS, Springer, pp. 96–114. doi:10.1007/978-3-031-77066-1_6. arXiv:2408.02760.

R.3 Uribarri G*, Ekman von Huth S*, Waldthaler J, Svenningsson P & Fransén E (2026). Deep learning models accurately classify Parkinson's disease from eye-tracking fixation data. International Journal of Medical Informatics 220:106626. doi:10.1016/j.ijmedinf.2026.106626. Open access, CC BY 4.0. *Equal contribution.

DATASETS

R.4 Schalk G, McFarland DJ, Hinterberger T, et al. (2004). BCI2000: a general-purpose brain-computer interface (BCI) system. IEEE Transactions on Biomedical Engineering 51(6):1034–1043. doi:10.1109/TBME.2004.827072.

R.5 Schalk G, McFarland DJ, Hinterberger T, et al. (2009). EEG Motor Movement/Imagery Dataset (v1.0.0). PhysioNet. doi:10.13026/C28G6P. ODC-By 1.0.

R.6 Goldberger AL, Amaral LAN, Glass L, et al. (2000). PhysioBank, PhysioToolkit, and PhysioNet. Circulation 101(23):e215–e220. doi:10.1161/01.CIR.101.23.e215.

R.7 Kemp B, Zwinderman AH, Tuk B, et al. (2000). Analysis of a sleep-dependent neuronal feedback loop: the slow-wave microcontinuity of the EEG. IEEE Transactions on Biomedical Engineering 47(9):1185–1194. doi:10.1109/10.867928.

R.8 Kemp B, Zwinderman AH, Tuk B, et al. (2018). The Sleep-EDF Database [Expanded]. PhysioNet. doi:10.13026/C2X676. ODC-By 1.0.

R.9 Richardson H, Lisandrelli G, Riobueno-Naylor A & Saxe R (2018). Development of the social brain from age three to twelve years. Nature Communications 9:1027. doi:10.1038/s41467-018-03399-2.

R.10 Richardson H, Lisandrelli G, Riobueno-Naylor A & Saxe R. MRI data of 3–12 year old children and adults during viewing of a short animated film. OpenNeuro ds000228 (v1.1.1). doi:10.18112/openneuro.ds000228.v1.1.1. CC0. Preprocessed derivative distributed with nilearn: OSF project 5hju4 (CC BY 4.0).

R.11 Haxby JV, Gobbini MI, Furey ML, et al. (2001). Distributed and overlapping representations of faces and objects in ventral temporal cortex. Science 293(5539):2425–2430. doi:10.1126/science.1063736. Data via PyMVPA/nilearn (CC BY-SA 3.0).

R.12 Di Martino A, Yan CG, Li Q, et al. (2014). The autism brain imaging data exchange: towards a large-scale evaluation of the intrinsic brain architecture in autism. Molecular Psychiatry 19(6):659–667. doi:10.1038/mp.2013.78. CC BY-NC-SA.

R.13 Craddock C, Benhajali Y, Chu C, et al. (2013). The Neuro Bureau Preprocessing Initiative: open sharing of preprocessed neuroimaging data and derivatives. Frontiers in Neuroinformatics, conference abstract (Neuroinformatics 2013). doi:10.3389/conf.fninf.2013.09.00041.

R.14 Grosmark AD & Buzsáki G (2016). Diversity in neural firing dynamics supports both rigid and learned hippocampal sequences. Science 351(6280):1440–1443. doi:10.1126/science.aad1935.

Links and extras: github.com/gon-uri/ITBA2026 · resources page

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OPTIONAL · NOT PRESENTED

Full references R.15–R.32 (2/12)

R.15 Grosmark AD & Buzsáki G. Diversity in neural firing dynamics supports both rigid and learned hippocampal sequences (dataset). DANDI Archive 000044, version 0.261005.2101. doi:10.48324/dandi.000044/0.261005.2101. CC BY 4.0.

R.16 Grosmark AD, Long J & Buzsáki G (2016). Recordings from hippocampal area CA1, PRE, during and POST novel spatial learning. CRCNS.org (hc-11). doi:10.6080/K0862DC5.

R.17 Olivetti E, Kia SM & Avesani P (2014). MEG decoding across subjects. arXiv:1404.4175 (organisers' description of the DecMeg2014 data). Used through Kaggle "DecMeg2014 – Decoding the Human Brain" (2014).

R.18 Miltiadous A, Tzimourta KD, Afrantou T, et al. (2023). A dataset of scalp EEG recordings of Alzheimer's disease, frontotemporal dementia and healthy subjects from routine EEG. Data 8(6):95. doi:10.3390/data8060095. Data: OpenNeuro ds004504, CC0.

R.19 Gramfort A & Hämäläinen MS. MNE sample data. OpenNeuro ds000248 (v1.2.4). doi:10.18112/openneuro.ds000248.v1.2.4. CC0.

R.20 Tangermann M, Müller KR, Aertsen A, et al. (2012). Review of the BCI Competition IV. Frontiers in Neuroscience 6:55. doi:10.3389/fnins.2012.00055. CC BY-ND (numbers only).

R.21 Bagnall A, Dau HA, Lines J, et al. (2018). The UEA multivariate time series classification archive, 2018. arXiv:1811.00075.

OPENING CONTEXT (BRAIN AND RECORDINGS)

R.22 Azevedo FAC, Carvalho LRB, Grinberg LT, et al. (2009). Equal numbers of neuronal and nonneuronal cells make the human brain an isometrically scaled-up primate brain. Journal of Comparative Neurology 513(5):532–541. doi:10.1002/cne.21974.

R.23 Pakkenberg B, Pelvig D, Marner L, et al. (2003). Aging and the human neocortex. Experimental Gerontology 38(1–2):95–99. doi:10.1016/S0531-5565(02)00151-1.

R.24 Herculano-Houzel S, Mota B & Lent R (2006). Cellular scaling rules for rodent brains. PNAS 103(32):12138–12143. doi:10.1073/pnas.0604911103.

R.25 Shapson-Coe A, Januszewski M, Berger DR, et al. (2024). A petavoxel fragment of human cerebral cortex reconstructed at nanoscale resolution. Science 384(6696):eadk4858. doi:10.1126/science.adk4858.

R.26 Beniaguev D, Segev I & London M (2021). Single cortical neurons as deep artificial neural networks. Neuron 109(17):2727–2739.e3. doi:10.1016/j.neuron.2021.07.002.

R.27 Gidon A, Zolnik TA, Fidzinski P, et al. (2020). Dendritic action potentials and computation in human layer 2/3 cortical neurons. Science 367(6473):83–87. doi:10.1126/science.aax6239.

R.28 Adolphs R (2015). The unsolved problems of neuroscience. Trends in Cognitive Sciences 19(4):173–175. doi:10.1016/j.tics.2015.01.007.

R.29 Paulk AC, Kfir Y, Khanna AR, et al. (2022). Large-scale neural recordings with single neuron resolution using Neuropixels probes in human cortex. Nature Neuroscience 25(2):252–263. doi:10.1038/s41593-021-00997-0.

R.30 Manley J, Lu S, Barber K, et al. (2024). Simultaneous, cortex-wide dynamics of up to 1 million neurons reveal unbounded scaling of dimensionality with neuron number. Neuron 112(10):1694–1709.e5. doi:10.1016/j.neuron.2024.02.011.

R.31 International Brain Laboratory, Meshulam L, Angelaki D, et al. (2025). A brain-wide map of neural activity during complex behaviour. Nature 645(8079):177–191. doi:10.1038/s41586-025-09235-0. Open access, CC BY.

R.32 Stevenson IH & Kording KP (2011). How advances in neural recording affect data analysis. Nature Neuroscience 14(2):139–142. doi:10.1038/nn.2731.

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OPTIONAL · NOT PRESENTED

Full references R.33–R.50 (3/12)

R.33 Lee MH, Kwon OY, Kim YJ, et al. (2019). EEG dataset and OpenBMI toolbox for three BCI paradigms: an investigation into BCI illiteracy. GigaScience 8(5):giz002. doi:10.1093/gigascience/giz002. CC BY 4.0.

R.34 Finn ES, Shen X, Scheinost D, et al. (2015). Functional connectome fingerprinting: identifying individuals using patterns of brain connectivity. Nature Neuroscience 18(11):1664–1671. doi:10.1038/nn.4135.

R.35 Marek S, Tervo-Clemmens B, Calabro FJ, et al. (2022). Reproducible brain-wide association studies require thousands of individuals. Nature 603(7902):654–660. doi:10.1038/s41586-022-04492-9.

R.36 Glaser JI, Benjamin AS, Farhoodi R & Kording KP (2019). The roles of supervised machine learning in systems neuroscience. Progress in Neurobiology 175:126–137. doi:10.1016/j.pneurobio.2019.01.008.

R.37 Richards BA, Lillicrap TP, Beaudoin P, et al. (2019). A deep learning framework for neuroscience. Nature Neuroscience 22(11):1761–1770. doi:10.1038/s41593-019-0520-2.

R.38 Mathis A, Mamidanna P, Cury KM, et al. (2018). DeepLabCut: markerless pose estimation of user-defined body parts with deep learning. Nature Neuroscience 21(9):1281–1289. doi:10.1038/s41593-018-0209-y.

R.39 Steinmetz NA, Koch C, Harris KD & Carandini M (2018). Challenges and opportunities for large-scale electrophysiology with Neuropixels probes. Current Opinion in Neurobiology 50:92–100. doi:10.1016/j.conb.2018.01.009. CC BY 4.0 (probe photo, Fig. 1c).

R.40 Steinmetz NA, Aydin C, Lebedeva A, et al. (2021). Neuropixels 2.0: a miniaturized high-density probe for stable, long-term brain recordings. Science 372(6539):eabf4588. doi:10.1126/science.abf4588.

R.41 Niso G, Romero E, Moreau JT, et al. (2023). Wireless EEG: a survey of systems and studies. NeuroImage 269:119774. doi:10.1016/j.neuroimage.2022.119774. Open access.

R.42 Dorkenwald S, Matsliah A, Sterling AR, et al. (2024). Neuronal wiring diagram of an adult brain. Nature 634(8032):124–138. doi:10.1038/s41586-024-07558-y. CC BY 4.0 (Fig. 1a image).

R.43 Schlegel P, Yin Y, Bates AS, et al. (2024). Whole-brain annotation and multi-connectome cell typing of Drosophila. Nature 634(8032):139–152. doi:10.1038/s41586-024-07686-5.

R.44 Bates AS, Phelps JS, Kim M, et al. (2026). Distributed control circuits across a brain-and-cord connectome. Nature 656(8129):957–970. doi:10.1038/s41586-026-10735-w. CC BY 4.0.

R.45 Berg S, Beckett IR, Costa M, et al. (2026). Sexual dimorphism in the complete Drosophila male central nervous system connectome. Cell 189(18):5504–5526.e15. doi:10.1016/j.cell.2026.08.015. CC BY 4.0.

R.46 MICrONS Consortium, Bae JA, Baptiste M, et al. (2025). Functional connectomics spanning multiple areas of mouse visual cortex. Nature 640(8058):435–447. doi:10.1038/s41586-025-08790-w. CC BY.

R.47 Ramón y Cajal S (c. 1900). Purkinje neuron from the human cerebellum, ink drawing. Instituto Cajal (CSIC), Madrid. Public domain, via Wikimedia Commons: https://commons.wikimedia.org/wiki/File:Cajal_-_a_purkinje_neuron_from_the_human_cerebellum.jpg.

R.48 Klapprott M & Debener S (2024). Mobile EEG for the study of cognitive-motor interference during swimming? Frontiers in Human Neuroscience 18:1466853. doi:10.3389/fnhum.2024.1466853. CC BY 4.0.

R.49 Sugden RJ, Pham-Kim-Nghiem-Phu VL, Campbell I, et al. (2023). Remote collection of electrophysiological data with brain wearables: opportunities and challenges. Bioelectronic Medicine 9:12. doi:10.1186/s42234-023-00114-5.

R.50 mBrainTrain. Smarting PRO X specifications (manufacturer web page), https://mbraintrain.com/?p=19501 (accessed 2026-10-23).

Links and extras: github.com/gon-uri/ITBA2026 · resources page

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Full references R.51–R.66 (4/12)

LITERATURE (A–Z)

R.51 Abraham A, Milham MP, Di Martino A, et al. (2017). Deriving reproducible biomarkers from multi-site resting-state data: an autism-based example. NeuroImage 147:736–745. doi:10.1016/j.neuroimage.2016.10.045.

R.52 Ang KK, Chin ZY, Wang C, et al. (2012). Filter bank common spatial pattern algorithm on BCI Competition IV datasets 2a and 2b. Frontiers in Neuroscience 6:39. doi:10.3389/fnins.2012.00039.

R.53 Banville H, Chehab O, Hyvärinen A, et al. (2021). Uncovering the structure of clinical EEG signals with self-supervised learning. Journal of Neural Engineering 18(4):046020. doi:10.1088/1741-2552/abca18.

R.54 Barachant A, Bonnet S, Congedo M, et al. (2012). Multiclass brain–computer interface classification by Riemannian geometry. IEEE Transactions on Biomedical Engineering 59(4):920–928. doi:10.1109/TBME.2011.2172210.

R.55 Barachant A, Bonnet S, Congedo M, et al. (2013). Classification of covariance matrices using a Riemannian-based kernel for BCI applications. Neurocomputing 112:172–178. doi:10.1016/j.neucom.2012.12.039.

R.56 Bastos AM & Schoffelen JM (2016). A tutorial review of functional connectivity analysis methods and their interpretational pitfalls. Frontiers in Systems Neuroscience 9:175. doi:10.3389/fnsys.2015.00175.

R.57 Bellec P, Rosa-Neto P, Lyttelton OC, et al. (2010). Multi-level bootstrap analysis of stable clusters in resting-state fMRI. NeuroImage 51(3):1126–1139. doi:10.1016/j.neuroimage.2010.02.082.

R.58 Bengio Y, Courville A & Vincent P (2013). Representation learning: a review and new perspectives. IEEE Transactions on Pattern Analysis and Machine Intelligence 35(8):1798–1828. doi:10.1109/TPAMI.2013.50.

R.59 Berger H (1929). Über das Elektrenkephalogramm des Menschen. Archiv für Psychiatrie und Nervenkrankheiten 87(1):527–570. doi:10.1007/BF01797193.

R.60 Blankertz B, Tomioka R, Lemm S, et al. (2008). Optimizing spatial filters for robust EEG single-trial analysis. IEEE Signal Processing Magazine 25(1):41–56. doi:10.1109/MSP.2008.4408441.

R.61 Brookshire G, Kasper J, Blauch NM, et al. (2024). Data leakage in deep learning studies of translational EEG. Frontiers in Neuroscience 18:1373515. doi:10.3389/fnins.2024.1373515.

R.62 Card NS, Wairagkar M, Iacobacci C, et al. (2024). An accurate and rapidly calibrating speech neuroprosthesis. New England Journal of Medicine 391(7):609–618. doi:10.1056/NEJMoa2314132.

R.63 Card NS, Singer-Clark T, Peracha H, et al. (2026). Long-term independent use of an intracortical brain–computer interface for speech and cursor control. Nature Medicine 32(7):2504–2510. doi:10.1038/s41591-026-04414-6.

R.64 Chambon S, Galtier MN, Arnal PJ, et al. (2018). A deep learning architecture for temporal sleep stage classification using multivariate and multimodal time series. IEEE Transactions on Neural Systems and Rehabilitation Engineering 26(4):758–769. doi:10.1109/TNSRE.2018.2813138.

R.65 Chen TW, Wardill TJ, Sun Y, et al. (2013). Ultrasensitive fluorescent proteins for imaging neuronal activity. Nature 499(7458):295–300. doi:10.1038/nature12354.

R.66 Chen Z, Peng S, Qin C, et al. (2026). Benchmarking EEG foundation models at scale: lessons from 20,000 evaluations (EEG-Arena). arXiv:2609.32743 (preprint).

Links and extras: github.com/gon-uri/ITBA2026 · resources page

Stockholms universitet

OPTIONAL · NOT PRESENTED

Full references R.67–R.82 (5/12)

R.67 Chevallier S, Carrara I, Aristimunha B, et al. (2024). The largest EEG-based BCI reproducibility study for open science: the MOABB benchmark. arXiv:2404.15319 (preprint).

R.68 Combrisson E & Jerbi K (2015). Exceeding chance level by chance: the caveat of theoretical chance levels in brain signal classification and statistical assessment of decoding accuracy. Journal of Neuroscience Methods 250:126–136. doi:10.1016/j.jneumeth.2015.01.010.

R.69 Congedo M, Barachant A & Bhatia R (2017). Riemannian geometry for EEG-based brain-computer interfaces; a primer and a review. Brain-Computer Interfaces 4(3):155–174. doi:10.1080/2326263X.2017.1297192.

R.70 Cox DR (1958). The regression analysis of binary sequences. Journal of the Royal Statistical Society Series B 20(2):215–232. doi:10.1111/j.2517-6161.1958.tb00292.x.

R.71 Craddock RC, James GA, Holtzheimer PE, Hu XP & Mayberg HS (2012). A whole brain fMRI atlas generated via spatially constrained spectral clustering. Human Brain Mapping 33(8):1914–1928. doi:10.1002/hbm.21333.

R.72 Cunningham JP & Yu BM (2014). Dimensionality reduction for large-scale neural recordings. Nature Neuroscience 17(11):1500–1509. doi:10.1038/nn.3776.

R.73 Dadi K, Rahim M, Abraham A, et al. (2019). Benchmarking functional connectome-based predictive models for resting-state fMRI. NeuroImage 192:115–134. doi:10.1016/j.neuroimage.2019.02.062.

R.74 Decat N, Walter J, Koh ZH, et al. (2022). Beyond traditional sleep scoring: massive feature extraction and data-driven clustering of sleep time series. Sleep Medicine 98:39–52. doi:10.1016/j.sleep.2022.06.013.

R.75 Dempster A, Petitjean F & Webb GI (2020). ROCKET: exceptionally fast and accurate time series classification using random convolutional kernels. Data Mining and Knowledge Discovery 34(5):1454–1495. doi:10.1007/s10618-020-00701-z.

R.76 Dempster A, Schmidt DF & Webb GI (2021). MiniRocket: a very fast (almost) deterministic transform for time series classification. Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining, pp. 248–257. doi:10.1145/3447548.3467231.

R.77 Dempster A, Schmidt DF & Webb GI (2023). Hydra: competing convolutional kernels for fast and accurate time series classification. Data Mining and Knowledge Discovery 37(5):1779–1805. doi:10.1007/s10618-023-00939-3.

R.78 Dempster A, Schmidt DF & Webb GI (2024). quant: a minimalist interval method for time series classification. Data Mining and Knowledge Discovery 38(4):2377–2402. doi:10.1007/s10618-024-01036-9.

R.79 Dong Z, Li R, Wu Y, et al. (2024). Brain-JEPA: brain dynamics foundation model with gradient positioning and spatiotemporal masking. Advances in Neural Information Processing Systems 37, pp. 86048–86073. doi:10.52202/079017-2732.

R.80 Donoghue T, Haller M, Peterson EJ, et al. (2020). Parameterizing neural power spectra into periodic and aperiodic components. Nature Neuroscience 23(12):1655–1665. doi:10.1038/s41593-020-00744-x.

R.81 El Ouahidi Y, Lys J, Thölke P, et al. (2025). REVE: a foundation model for EEG — adapting to any setup with large-scale pretraining on 25,000 subjects. arXiv:2510.21585 (preprint).

R.82 Engemann DA, Kozynets O, Sabbagh D, et al. (2020). Combining magnetoencephalography with magnetic resonance imaging enhances learning of surrogate-biomarkers. eLife 9:e54055. doi:10.7554/eLife.54055.

Links and extras: github.com/gon-uri/ITBA2026 · resources page

Stockholms universitet

OPTIONAL · NOT PRESENTED

Full references R.83–R.99 (6/12)

R.83 English DF, McKenzie S, Evans T, et al. (2017). Pyramidal cell–interneuron circuit architecture and dynamics in hippocampal networks. Neuron 96(2):505–520.e7. doi:10.1016/j.neuron.2017.09.033.

R.84 Esteban O, Markiewicz CJ, Blair RW, et al. (2019). fMRIPrep: a robust preprocessing pipeline for functional MRI. Nature Methods 16(1):111–116. doi:10.1038/s41592-018-0235-4.

R.85 Friedman J, Hastie T & Tibshirani R (2008). Sparse inverse covariance estimation with the graphical lasso. Biostatistics 9(3):432–441. doi:10.1093/biostatistics/kxm045.

R.86 Fulcher BD & Jones NS (2017). hctsa: a computational framework for automated time-series phenotyping using massive feature extraction. Cell Systems 5(5):527–531.e3. doi:10.1016/j.cels.2017.10.001.

R.87 Gemein LAW, Schirrmeister RT, Chrabąszcz P, et al. (2020). Machine-learning-based diagnostics of EEG pathology. NeuroImage 220:117021. doi:10.1016/j.neuroimage.2020.117021.

R.88 Glaser JI, Benjamin AS, Chowdhury RH, et al. (2020). Machine learning for neural decoding. eNeuro 7(4):ENEURO.0506-19.2020. doi:10.1523/ENEURO.0506-19.2020.

R.89 Glover GH, Li TQ & Ress D (2000). Image-based method for retrospective correction of physiological motion effects in fMRI: RETROICOR. Magnetic Resonance in Medicine 44(1):162–167. doi:10.1002/1522-2594(200007)44:1<162::AID-MRM23>3.0.CO;2-E.

R.90 Goodfellow I, Bengio Y & Courville A (2016). Deep Learning. MIT Press. ISBN 978-0-262-03561-3. Free online: https://www.deeplearningbook.org (no DOI).

R.91 Gramfort A, Luessi M, Larson E, et al. (2013). MEG and EEG data analysis with MNE-Python. Frontiers in Neuroscience 7:267. doi:10.3389/fnins.2013.00267.

R.92 Granger CWJ (1969). Investigating causal relations by econometric models and cross-spectral methods. Econometrica 37(3):424–438. doi:10.2307/1912791.

R.93 Grootswagers T, Wardle SG & Carlson TA (2017). Decoding dynamic brain patterns from evoked responses: a tutorial on multivariate pattern analysis applied to time series neuroimaging data. Journal of Cognitive Neuroscience 29(4):677–697. doi:10.1162/jocn_a_01068.

R.94 Hämäläinen M, Hari R, Ilmoniemi RJ, et al. (1993). Magnetoencephalography — theory, instrumentation, and applications to noninvasive studies of the working human brain. Reviews of Modern Physics 65(2):413–497. doi:10.1103/RevModPhys.65.413.

R.95 Han K, Su Y, He L, et al. (2026). Rethinking functional brain connectome analysis: do graph deep learning models help? npj Artificial Intelligence 2:19. doi:10.1038/s44387-025-00067-x.

R.96 Haufe S, Nikulin VV, Müller KR & Nolte G (2013). A critical assessment of connectivity measures for EEG data: a simulation study. NeuroImage 64:120–133. doi:10.1016/j.neuroimage.2012.09.036.

R.97 Haufe S, Meinecke F, Görgen K, et al. (2014). On the interpretation of weight vectors of linear models in multivariate neuroimaging. NeuroImage 87:96–110. doi:10.1016/j.neuroimage.2013.10.067.

R.98 He T, Kong R, Holmes AJ, et al. (2020). Deep neural networks and kernel regression achieve comparable accuracies for functional connectivity prediction of behavior and demographics. NeuroImage 206:116276. doi:10.1016/j.neuroimage.2019.116276.

R.99 Henson RN, Wakeman DG, Litvak V & Friston KJ (2011). A parametric empirical Bayesian framework for the EEG/MEG inverse problem: generative models for multi-subject and multi-modal integration. Frontiers in Human Neuroscience 5:76. doi:10.3389/fnhum.2011.00076.

Links and extras: github.com/gon-uri/ITBA2026 · resources page

Stockholms universitet

OPTIONAL · NOT PRESENTED

Full references R.100–R.117 (7/12)

R.100 Hinton GE & Salakhutdinov RR (2006). Reducing the dimensionality of data with neural networks. Science 313(5786):504–507. doi:10.1126/science.1127647.

R.101 Hipp JF, Hawellek DJ, Corbetta M, et al. (2012). Large-scale cortical correlation structure of spontaneous oscillatory activity. Nature Neuroscience 15(6):884–890. doi:10.1038/nn.3101.

R.102 Hoerl AE & Kennard RW (1970). Ridge regression: biased estimation for nonorthogonal problems. Technometrics 12(1):55–67. doi:10.1080/00401706.1970.10488634.

R.103 Honey CJ, Sporns O, Cammoun L, et al. (2009). Predicting human resting-state functional connectivity from structural connectivity. PNAS 106(6):2035–2040. doi:10.1073/pnas.0811168106.

R.104 Hubert L & Arabie P (1985). Comparing partitions. Journal of Classification 2(1):193–218. doi:10.1007/BF01908075.

R.105 James G, Witten D, Hastie T, Tibshirani R & Taylor J (2023). An Introduction to Statistical Learning with Applications in Python. Springer Texts in Statistics. Springer, Cham. doi:10.1007/978-3-031-38747-0. Free PDF: https://www.statlearning.com.

R.106 Jiang WB, Zhao LM & Lu BL (2024). Large brain model for learning generic representations with tremendous EEG data in BCI (LaBraM). International Conference on Learning Representations (ICLR 2024). arXiv:2405.18765.

R.107 Jiang WB, Wang Y, Lu BL & Li D (2025). NeuroLM: a universal multi-task foundation model for bridging the gap between language and EEG signals. International Conference on Learning Representations (ICLR 2025). arXiv:2409.00101.

R.108 Jolliffe IT & Cadima J (2016). Principal component analysis: a review and recent developments. Philosophical Transactions of the Royal Society A 374(2065):20150202. doi:10.1098/rsta.2015.0202.

R.109 Jude JJ, Levi-Aharoni H, Acosta AJ, et al. (2026). Restoring rapid natural bimanual typing with a neuroprosthesis after paralysis. Nature Neuroscience 29(10):2613–2622. doi:10.1038/s41593-026-02218-y.

R.110 Jun JJ, Steinmetz NA, Siegle JH, et al. (2017). Fully integrated silicon probes for high-density recording of neural activity. Nature 551(7679):232–236. doi:10.1038/nature24636.

R.111 Kampel N, Kiefer CM, Shah NJ, et al. (2023). Neural fingerprinting on MEG time series using MiniRocket. Frontiers in Neuroscience 17:1229371. doi:10.3389/fnins.2023.1229371.

R.112 Kastrati A, Bürki J, Lauer J, et al. (2025). EEG-Bench: a benchmark for EEG foundation models in clinical applications. arXiv:2512.08959 (preprint).

R.113 King JR & Dehaene S (2014). Characterizing the dynamics of mental representations: the temporal generalization method. Trends in Cognitive Sciences 18(4):203–210. doi:10.1016/j.tics.2014.01.002.

R.114 Kontras K, Osselaer T, Mouslech SG, et al. (2026). NeuroAtlas: benchmarking foundation models for clinical EEG and brain-computer interfaces. arXiv:2605.14698 (preprint).

R.115 Kunz EM, Abramovich Krasa B, Kamdar F, et al. (2025). Inner speech in motor cortex and implications for speech neuroprostheses. Cell 188(17):4658–4673.e17. doi:10.1016/j.cell.2025.06.015.

R.116 Lane C, Tripathy M, Murali LK, et al. (2026). Scaling vision transformers for functional MRI with flat maps (Brainmarks). arXiv:2510.13768 (v3).

R.117 Lawhern VJ, Solon AJ, Waytowich NR, et al. (2018). EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces. Journal of Neural Engineering 15(5):056013. doi:10.1088/1741-2552/aace8c.

Links and extras: github.com/gon-uri/ITBA2026 · resources page

Stockholms universitet

OPTIONAL · NOT PRESENTED

Full references R.118–R.134 (8/12)

R.118 LeCun Y, Bengio Y & Hinton G (2015). Deep learning. Nature 521(7553):436–444. doi:10.1038/nature14539.

R.119 Ledoit O & Wolf M (2004). A well-conditioned estimator for large-dimensional covariance matrices. Journal of Multivariate Analysis 88(2):365–411. doi:10.1016/S0047-259X(03)00096-4.

R.120 Lévy J, Zhang M, Pinet S, et al. (2026). Noninvasive decoding of typed sentences from human brain activity. Nature Neuroscience 29(10):2438–2444. doi:10.1038/s41593-026-02303-2.

R.121 Lewicki MS (1998). A review of methods for spike sorting: the detection and classification of neural action potentials. Network: Computation in Neural Systems 9(4):R53–R78. doi:10.1088/0954-898X_9_4_001.

R.122 Li R, Johansen JS, Ahmed H, et al. (2021). The perils and pitfalls of block design for EEG classification experiments. IEEE Transactions on Pattern Analysis and Machine Intelligence 43(1):316–333. doi:10.1109/TPAMI.2020.2973153.

R.123 Lindquist MA, Loh JM, Atlas LY & Wager TD (2009). Modeling the hemodynamic response function in fMRI: efficiency, bias and mis-modeling. NeuroImage 45(1 Suppl):S187–S198. doi:10.1016/j.neuroimage.2008.10.065.

R.124 Liu D, Chen Y, Chen Z, et al. (2026). EEG-FM-Compass: progress, benchmarking and future directions for EEG foundation models. National Science Review nwag466. doi:10.1093/nsr/nwag466. arXiv:2601.17883.

R.125 Lloyd S (1982). Least squares quantization in PCM. IEEE Transactions on Information Theory 28(2):129–137. doi:10.1109/TIT.1982.1056489.

R.126 Logothetis NK (2008). What we can do and what we cannot do with fMRI. Nature 453(7197):869–878. doi:10.1038/nature06976.

R.127 Lotte F, Bougrain L, Cichocki A, et al. (2018). A review of classification algorithms for EEG-based brain–computer interfaces: a 10 year update. Journal of Neural Engineering 15(3):031005. doi:10.1088/1741-2552/aab2f2.

R.128 Lubba CH, Sethi SS, Knaute P, et al. (2019). catch22: CAnonical Time-series CHaracteristics. Data Mining and Knowledge Discovery 33(6):1821–1852. doi:10.1007/s10618-019-00647-x.

R.129 McInnes L, Healy J & Melville J (2018). UMAP: uniform manifold approximation and projection for dimension reduction. arXiv:1802.03426.

R.130 Metzger SL, Littlejohn KT, Silva AB, et al. (2023). A high-performance neuroprosthesis for speech decoding and avatar control. Nature 620(7976):1037–1046. doi:10.1038/s41586-023-06443-4.

R.131 Mhanna R, Achard S, Petersen A & Richiardi J (2026). A comprehensive benchmark of graph neural networks, graph kernels, and classical machine learning approaches on rs-fMRI brain graphs. Medical Imaging with Deep Learning (MIDL 2026), PMLR 315:4479–4495. https://proceedings.mlr.press/v315/mhanna26a.html.

R.132 Middlehurst M, Large J, Flynn M, et al. (2021). HIVE-COTE 2.0: a new meta ensemble for time series classification. Machine Learning 110(11–12):3211–3243. doi:10.1007/s10994-021-06057-9.

R.133 Middlehurst M, Schäfer P & Bagnall A (2024). Bake off redux: a review and experimental evaluation of recent time series classification algorithms. Data Mining and Knowledge Discovery 38(4):1958–2031. doi:10.1007/s10618-024-01022-1.

R.134 Miltiadous A, Gionanidis E, Tzimourta KD, et al. (2023). DICE-Net: a novel convolution-transformer architecture for Alzheimer detection in EEG signals. IEEE Access 11:71840–71858. doi:10.1109/ACCESS.2023.3294618. CC BY-NC-ND 4.0.

Links and extras: github.com/gon-uri/ITBA2026 · resources page

Stockholms universitet

OPTIONAL · NOT PRESENTED

Full references R.135–R.150 (9/12)

R.135 Noble S, Scheinost D & Constable RT (2019). A decade of test-retest reliability of functional connectivity: a systematic review and meta-analysis. NeuroImage 203:116157. doi:10.1016/j.neuroimage.2019.116157.

R.136 Nolte G, Bai O, Wheaton L, et al. (2004). Identifying true brain interaction from EEG data using the imaginary part of coherency. Clinical Neurophysiology 115(10):2292–2307. doi:10.1016/j.clinph.2004.04.029.

R.137 O'Keefe J & Dostrovsky J (1971). The hippocampus as a spatial map. Preliminary evidence from unit activity in the freely-moving rat. Brain Research 34(1):171–175. doi:10.1016/0006-8993(71)90358-1.

R.138 Ortega Caro J, Fonseca AHdO, Averill C, et al. (2024). BrainLM: a foundation model for brain activity recordings. International Conference on Learning Representations (ICLR 2024). bioRxiv doi:10.1101/2023.09.12.557460.

R.139 Pachitariu M, Sridhar S, Pennington J & Stringer C (2024). Spike sorting with Kilosort4. Nature Methods 21(5):914–921. doi:10.1038/s41592-024-02232-7.

R.140 Palva JM, Wang SH, Palva S, et al. (2018). Ghost interactions in MEG/EEG source space: a note of caution on inter-areal coupling measures. NeuroImage 173:632–643. doi:10.1016/j.neuroimage.2018.02.032.

R.141 Pandarinath C, O'Shea DJ, Collins J, et al. (2018). Inferring single-trial neural population dynamics using sequential auto-encoders. Nature Methods 15(10):805–815. doi:10.1038/s41592-018-0109-9.

R.142 Pearson K (1901). On lines and planes of closest fit to systems of points in space. The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science 2(11):559–572. doi:10.1080/14786440109462720.

R.143 Pereira F, Mitchell T & Botvinick M (2009). Machine learning classifiers and fMRI: a tutorial overview. NeuroImage 45(1 Suppl):S199–S209. doi:10.1016/j.neuroimage.2008.11.007.

R.144 Perkel DH, Gerstein GL & Moore GP (1967). Neuronal spike trains and stochastic point processes. Biophysical Journal 7(4):419–440. doi:10.1016/S0006-3495(67)86597-4.

R.145 Poldrack RA, Huckins G & Varoquaux G (2020). Establishment of best practices for evidence for prediction: a review. JAMA Psychiatry 77(5):534–540. doi:10.1001/jamapsychiatry.2019.3671.

R.146 Power JD, Barnes KA, Snyder AZ, et al. (2012). Spurious but systematic correlations in functional connectivity MRI networks arise from subject motion. NeuroImage 59(3):2142–2154. doi:10.1016/j.neuroimage.2011.10.018.

R.147 Quiroga RQ, Nadasdy Z & Ben-Shaul Y (2004). Unsupervised spike detection and sorting with wavelets and superparamagnetic clustering. Neural Computation 16(8):1661–1687. doi:10.1162/089976604774201631.

R.148 Ramoser H, Müller-Gerking J & Pfurtscheller G (2000). Optimal spatial filtering of single trial EEG during imagined hand movement. IEEE Transactions on Rehabilitation Engineering 8(4):441–446. doi:10.1109/86.895946.

R.149 Rousseeuw PJ (1987). Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics 20:53–65. doi:10.1016/0377-0427(87)90125-7.

R.150 Roy Y, Banville H, Albuquerque I, et al. (2019). Deep learning-based electroencephalography analysis: a systematic review. Journal of Neural Engineering 16(5):051001. doi:10.1088/1741-2552/ab260c.

Links and extras: github.com/gon-uri/ITBA2026 · resources page

Stockholms universitet

OPTIONAL · NOT PRESENTED

Full references R.151–R.168 (10/12)

R.151 Ruiz AP, Flynn M, Large J, et al. (2021). The great multivariate time series classification bake off: a review and experimental evaluation of recent algorithmic advances. Data Mining and Knowledge Discovery 35(2):401–449. doi:10.1007/s10618-020-00727-3.

R.152 Rumelhart DE, Hinton GE & Williams RJ (1986). Learning representations by back-propagating errors. Nature 323(6088):533–536. doi:10.1038/323533a0.

R.153 Sabbagh D, Ablin P, Varoquaux G, Gramfort A & Engemann DA (2020). Predictive regression modeling with MEG/EEG: from source power to signals and cognitive states. NeuroImage 222:116893. doi:10.1016/j.neuroimage.2020.116893.

R.154 Satterthwaite TD, Wolf DH, Loughead J, et al. (2012). Impact of in-scanner head motion on multiple measures of functional connectivity: relevance for studies of neurodevelopment in youth. NeuroImage 60(1):623–632. doi:10.1016/j.neuroimage.2011.12.063.

R.155 Schirrmeister RT, Springenberg JT, Fiederer LDJ, et al. (2017). Deep learning with convolutional neural networks for EEG decoding and visualization. Human Brain Mapping 38(11):5391–5420. doi:10.1002/hbm.23730.

R.156 Schneider S, Lee JH & Mathis MW (2023). Learnable latent embeddings for joint behavioural and neural analysis. Nature 617(7960):360–368. doi:10.1038/s41586-023-06031-6.

R.157 Schoffelen JM & Gross J (2009). Source connectivity analysis with MEG and EEG. Human Brain Mapping 30(6):1857–1865. doi:10.1002/hbm.20745.

R.158 Seth AK, Barrett AB & Barnett L (2015). Granger causality analysis in neuroscience and neuroimaging. Journal of Neuroscience 35(8):3293–3297. doi:10.1523/JNEUROSCI.4399-14.2015.

R.159 Shah V, von Weltin E, Lopez S, et al. (2018). The Temple University Hospital seizure detection corpus. Frontiers in Neuroinformatics 12:83. doi:10.3389/fninf.2018.00083.

R.160 Shannon CE (1949). Communication in the presence of noise. Proceedings of the IRE 37(1):10–21. doi:10.1109/JRPROC.1949.232969.

R.161 Smith SM, Miller KL, Salimi-Khorshidi G, et al. (2011). Network modelling methods for FMRI. NeuroImage 54(2):875–891. doi:10.1016/j.neuroimage.2010.08.063.

R.162 Smith SM, Beckmann CF, Andersson J, et al. (2013). Resting-state fMRI in the Human Connectome Project. NeuroImage 80:144–168. doi:10.1016/j.neuroimage.2013.05.039.

R.163 SPM canonical haemodynamic response function (spm_hrf). Wellcome Centre for Human Neuroimaging, SPM documentation, https://www.fil.ion.ucl.ac.uk/spm/docs/wikibooks/Haemodynamic_Response_Function/ (accessed 2026-10-23).

R.164 Stone M (1974). Cross-validatory choice and assessment of statistical predictions. Journal of the Royal Statistical Society Series B 36(2):111–133. doi:10.1111/j.2517-6161.1974.tb00994.x.

R.165 Tan CW, Dempster A, Bergmeir C & Webb GI (2022). MultiRocket: multiple pooling operators and transformations for fast and effective time series classification. Data Mining and Knowledge Discovery 36(5):1623–1646. doi:10.1007/s10618-022-00844-1.

R.166 Tao JX, Ray A, Hawes-Ebersole S & Ebersole JS (2005). Intracranial EEG substrates of scalp EEG interictal spikes. Epilepsia 46(5):669–676. doi:10.1111/j.1528-1167.2005.11404.x.

R.167 Teplan M (2002). Fundamentals of EEG measurement. Measurement Science Review 2(2):1–11 (no DOI).

R.168 Tibshirani R (1996). Regression shrinkage and selection via the lasso. Journal of the Royal Statistical Society Series B 58(1):267–288. doi:10.1111/j.2517-6161.1996.tb02080.x.

Links and extras: github.com/gon-uri/ITBA2026 · resources page

Stockholms universitet

OPTIONAL · NOT PRESENTED

Full references R.169–R.185 (11/12)

R.169 Traut N, Heuer K, Lemaître G, et al. (2022). Insights from an autism imaging biomarker challenge: promises and threats to biomarker discovery. NeuroImage 255:119171. doi:10.1016/j.neuroimage.2022.119171.

R.170 van den Oord A, Li Y & Vinyals O (2018). Representation learning with contrastive predictive coding. arXiv:1807.03748.

R.171 van der Maaten L & Hinton G (2008). Visualizing data using t-SNE. Journal of Machine Learning Research 9:2579–2605. https://www.jmlr.org/papers/v9/vandermaaten08a.html.

R.172 Varoquaux G, Baronnet F, Kleinschmidt A, et al. (2010). Detection of brain functional-connectivity difference in post-stroke patients using group-level covariance modeling. In: MICCAI 2010, LNCS, pp. 200–208. doi:10.1007/978-3-642-15705-9_25.

R.173 Varoquaux G, Gramfort A, Poline JB & Thirion B (2010). Brain covariance selection: better individual functional connectivity models using population prior. Advances in Neural Information Processing Systems 23 (NeurIPS 2010). arXiv:1008.5071.

R.174 Varoquaux G, Gramfort A, Pedregosa F, et al. (2011). Multi-subject dictionary learning to segment an atlas of brain spontaneous activity. In: Information Processing in Medical Imaging (IPMI 2011), LNCS, pp. 562–573. doi:10.1007/978-3-642-22092-0_46.

R.175 Varoquaux G & Craddock RC (2013). Learning and comparing functional connectomes across subjects. NeuroImage 80:405–415. doi:10.1016/j.neuroimage.2013.04.007.

R.176 Varoquaux G, Raamana PR, Engemann DA, et al. (2017). Assessing and tuning brain decoders: cross-validation, caveats, and guidelines. NeuroImage 145:166–179. doi:10.1016/j.neuroimage.2016.10.038.

R.177 Varoquaux G (2018). Cross-validation failure: small sample sizes lead to large error bars. NeuroImage 180:68–77. doi:10.1016/j.neuroimage.2017.06.061.

R.178 Vinck M, Oostenveld R, van Wingerden M, et al. (2011). An improved index of phase-synchronization for electrophysiological data in the presence of volume-conduction, noise and sample-size bias. NeuroImage 55(4):1548–1565. doi:10.1016/j.neuroimage.2011.01.055.

R.179 Wattenberg M, Viégas F & Johnson I (2016). How to use t-SNE effectively. Distill 1(10). doi:10.23915/distill.00002.

R.180 Welch P (1967). The use of fast Fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms. IEEE Transactions on Audio and Electroacoustics 15(2):70–73. doi:10.1109/TAU.1967.1161901.

R.181 Whitham EM, Pope KJ, Fitzgibbon SP, et al. (2007). Scalp electrical recording during paralysis: quantitative evidence that EEG frequencies above 20 Hz are contaminated by EMG. Clinical Neurophysiology 118(8):1877–1888. doi:10.1016/j.clinph.2007.04.027.

R.182 Willett FR, Kunz EM, Fan C, et al. (2023). A high-performance speech neuroprosthesis. Nature 620(7976):1031–1036. doi:10.1038/s41586-023-06377-x.

R.183 Yuval-Greenberg S, Tomer O, Keren AS, et al. (2008). Transient induced gamma-band response in EEG as a manifestation of miniature saccades. Neuron 58(3):429–441. doi:10.1016/j.neuron.2008.03.027.

R.184 Zanini P, Congedo M, Jutten C, et al. (2018). Transfer learning: a Riemannian geometry framework with applications to brain–computer interfaces. IEEE Transactions on Biomedical Engineering 65(5):1107–1116. doi:10.1109/TBME.2017.2742541.

R.185 Zare M (2026). A negative-control protocol for clinical EEG foundation-model benchmarks: dataset identity and external-cohort stress testing. arXiv:2607.24519 (preprint).

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OPTIONAL · NOT PRESENTED

Full references R.186–R.191 (12/12)

SOFTWARE AND TUTORIALS

R.186 Pedregosa F, Varoquaux G, Gramfort A, et al. (2011). Scikit-learn: machine learning in Python. Journal of Machine Learning Research 12:2825–2830.

R.187 Abraham A, Pedregosa F, Eickenberg M, et al. (2014). Machine learning for neuroimaging with scikit-learn (nilearn). Frontiers in Neuroinformatics 8:14. doi:10.3389/fninf.2014.00014.

R.188 MNE-Python (Gramfort et al. 2013, R.91) and its tutorials "Background on filtering" and "Repairing artifacts with ICA", https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html.

R.189 CEBRA 0.6.1 (Schneider et al. 2023, R.156), Apache-2.0, https://cebra.ai.

R.190 Middlehurst M, Ismail-Fawaz A, Guillaume A, et al. (2024). aeon: a Python toolkit for learning from time series. Journal of Machine Learning Research 25(289):1–10. https://jmlr.org/papers/v25/23-1444.html. Version 1.6.0 used for the ROCKET-family models and the UEA loaders.

R.191 detach_rocket (BSD-3), https://github.com/gon-uri/detach_rocket (see R.1).

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